

AmiCORE Apheresis System

Operator's Manual
SW v. 2.0

REF 6R8800

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Chapter 1 - Introduction

Section 1.1 **About this Manual**

This AmiCORE Operator's Manual ("this manual") includes:

- **Table of Contents**
- **Chapter 1: Introduction** – How to use this manual; the intended use of the AmiCORE Apheresis System; donor requirements and care; adverse effects; performance; blood component quality; and warnings and cautions.
- **Chapter 2: System Overview** – General overview; how to use the touch screen and the bar code scanner.
- **Chapter 3: System Considerations** – Daily system set up and shut down.
- **Chapter 4: Procedures** – Instructions for performing apheresis procedures with the AmiCORE Apheresis System.
- **Chapter 5: Troubleshooting**
- **Chapter 6: Maintenance and Cleaning**
- **Chapter 7: System Specifications**
- **Appendix A** – Nomograms; manual calculations for blood loss; residual red cell volume; and density of plasma.
- **Glossary of Graphics** – Defines all touch screen graphics.

Document Purpose

This manual provides the requirements and steps to successfully collect platelets using the AmiCORE Apheresis System. This manual assumes that operators are trained in the use of the AmiCORE Apheresis System, and have:

- Basic venipuncture skills.
- Familiarity with the institution's applicable guidelines and standard operating procedures (SOPs), including aseptic technique.

Conventions

Actual screens and summary information may vary from the examples shown in this manual.

This manual uses italics for the names of tabs, buttons, icons, and prompts, for example:

- Touch the *Check button*.
- The *Hang Containers prompt* on the screen prompts the operator to perform hang containers.

This manual uses symbols to identify warnings, cautions, and notes:

Symbol	Meaning
	Warning. Used in this manual to alert personnel of potentially hazardous conditions that may cause serious adverse reactions or personal injury.
	Caution. Used in this manual to alert personnel of a potentially hazardous situation which, if not avoided, may result in minor or moderate injury to the donor or damage to the instrument or other property. It may also alert against unsafe use.
	Note. Used in this manual to provide additional information about the content.

Operator Qualification and Training

For operator qualifications and manufacturer's training programs, contact your local customer service representative.

Section 1.2 Intended Use and Indication for Use

Intended Use

The AmiCORE Apheresis System is an automated blood cell separator intended for use in the collection of blood components.

Indications for Use

The AmiCORE Apheresis System is an automated blood cell separator indicated for the collection of blood components. AmiCORE is indicated to collect:

Platelets Pheresis, Leukocytes Reduced (single or double units)

Apheresis Kits

AmiCORE apheresis kits are the only apheresis kits that are to be used with the AmiCORE Apheresis System.

Products

Platelets Pheresis, Leukocytes Reduced may be stored in the platelet storage containers with an appropriate volume of storage fluid for up to 5 days at 20°C to 24°C with continuous gentle agitation.

Section 1.3 Adverse Effects

This section discusses possible adverse effects donors may experience during apheresis procedures.

Adverse donor reactions are similar to those sometimes experienced during blood collection procedures. These may include bruising or hematoma formation at the venipuncture site, hyperventilation, light-headedness, hypovolemia or vasovagal reactions such as pallor, sweating, hypotension, nausea, vomiting, and fainting.

Reactions unique to apheresis collection procedures may also occur. Cooling of saline or donor blood may be sufficient to induce chills. Infusion of anticoagulants containing citrate may lead to donor symptoms of moderate hypocalcemia due to chelation of calcium by unmetabolized citrate. Such donor reactions are usually manifested by a “tingling” feeling, often around the mouth or fingers. Allergic symptoms including skin redness, itching, hives, etc., may be seen. Other manifestations may include muscle discomfort, muscle twitching or spasms, or the presence

of an unusual taste or smell sensation. Treatment of the donor's blood should be temporarily slowed or halted if such symptoms occur.

Symptoms of severe hypocalcemia, rarely seen in donors undergoing large volume leukapheresis, include tetany, seizure, cardiac arrhythmia, and death.

Complications such as blood loss, hemolysis, air embolism, and blood clotting may be associated with improper operating conditions.



Warning:

The physician or operator responsible for the operation of the separator should refer to the package insert accompanying each drug product used during blood processing procedures, for full drug product information.

Section 1.4 Product Claims and Specifications

WBC Counts

The AmiCORE Apheresis System provides processed leukoreduced platelet products.

Platelets Pheresis, Leukocytes Reduced:

- < 5×10^6 95% of the time with 95% confidence*
- < 1×10^6 90% of the time

* For collections under 9.0×10^{11} this value is per collection. For collections greater than or equal to 9.0×10^{11} this value is per platelet dose.

Yield and Volume

The platelet yield range per storage container is 1.5×10^{11} to 4.7×10^{11} platelets. Volume and yield requirements are specified in Appendix A.5, Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC.



Note:

Concentration of platelets/mL = platelet yield/volume.

Section 1.5 Contraindications

The AmiCORE Apheresis System is contraindicated in those cases where adequate anticoagulation cannot be achieved.

Section 1.6 Donor Requirements and Care

Donor Requirements

Donors should be selected based on the requirements of the appropriate state, federal, or national health-regulating agencies as well as institutional standard operating procedures (SOPs) for the individual blood collection center.

Donor Care

Follow institutional guidelines and SOPs for venipuncture site and donor care.



Note:

The donor should be informed of the basic procedure and understand the potential adverse reactions associated with apheresis.

Donor and Equipment Positioning

Position the venous access point at least 24 inches (61 cm) above the floor. Position the donor chair or bed lower than the top panel of the separator, to enhance gravity flow of saline.

Section 1.7 Warnings and Cautions

Warnings and cautions in this manual emphasize important information and define specific hazards that can cause serious adverse reactions or personal injury during normal operation. They are presented here, along with additional warnings that must be followed, to bring your attention to their existence and highlight their importance.

Warnings

Before operating the AmiCORE Apheresis System, you should review the following:



Warning:

Do not use the AmiCORE Apheresis System in the presence of explosive gases (e.g., anaesthetics).



Warning:

Do not connect a donor if the instrument displays a red banner. This indicates Demo Mode which may have disabled safety systems.



Warning:

Use only the power cord supplied with the AmiCORE instrument, and plug into a polarized, grounded, three-hole outlet.



Warning:

Do not use the power cord if it is frayed or damaged.



Warning:

Do not reuse a disposable kit. Reuse of a disposable kit may lead to a donor being exposed to a potential biohazard from the blood of another donor.



Warning:

Do not perform maintenance tasks while a procedure is in progress.



Warning:

Only authorized service representatives should access the inside of the instrument.



Warning:

If the instrument is turned off during a procedure, immediately isolate the donor from the instrument by clamping the donor line and disconnecting the donor. Do not turn the instrument on with the donor connected.



Warning:

An improperly installed disposable kit can cause kinks or damage the kit, which can lead to hemolysis.



Warning:

Do not touch containers on the scales after install check or fluid prime. Doing so can result in errors with monitoring blood loss.



Warning:

The separator has been designed and verified not to cause hemolysis of red blood cells. End the procedure without fluid return if ongoing (i.e., non-transient) hemolysis or red blood cells are observed in the plasma container or plasma lines.



Warning:

Continuously monitor the donor line for air bubbles during manual reinfusion of disposable kit contents or saline, and immediately clamp the line if air is observed.



Warning:

Inspect the disposable kit for air.



Warning:

Use only the anticoagulant type and concentration specified in this manual to prevent citrate toxicity, hemolysis, and abnormal storage pH.



Warning:

Use only the saline type and concentration specified in this manual.



Warning:

Damaged or unapproved apheresis needles may result in hemolysis or cell activation.



Warning:

Always seal the pathway to the whole blood sampling site to ensure air from the site does not enter the donor line.



Warning:

Do not actuate the centrifuge door lock override and open the centrifuge door while the centrifuge is spinning.



Warning:

Inspect for air bubbles between the disposable kit and apheresis needle before performing manual reinfusion of disposable kit contents or saline. Do not perform manual reinfusion if air is present.



Warning:

The operator must be trained before operating the system.



Warning:

Do not interchange the AC and saline lines. Ensure that the red spike is connected to the AC container.



Warning:

Connect or disconnect the donor only when prompted by the instrument. Doing so at any other time may result in the infusion of air, anticoagulant, or other fluids to the donor.



Warning:

Do not press the CASSETTE BYPASS button while the donor is connected. Unloading the cassette while the donor is connected may result in the infusion of air, anticoagulant, or other fluids to the donor.



Warning:

If the touch screen becomes blank, unreadable, or nonresponsive during a procedure, press the *STOP button* and disconnect the donor.



Warning:

Only 17 gauge thin-walled apheresis needles may be used.



Warning:

The operator is responsible for confirming that the entered procedure data displayed on the touch screen is correct.



Warning:

Clean and disinfect blood spills immediately. Treat all spills as potential biohazards. Precautions should be taken at all times to prevent exposure to and transmission of blood borne pathogens.



Warning:

Consideration should be given to the donor blood sample volume, as this may lead to excessive blood loss.



Warning:

The air detector and outside of the tubing must be kept dry during disposable set installation and the procedure.

Cautions

Following is a list of all cautions. These cautions are intended to inform you of conditions that may result in minor or moderate injury to the donor or damage to the instrument or other property. However, it is also important that they be adhered to in their procedural context.



Caution:

Accessories and disposables used with the instrument must be approved by the instrument manufacturer. See Appendix A.2 for a list of approved accessory equipment.



Caution:

Instrument performance may be affected by external electromagnetic fields generated by non-EMC compliant medical equipment and other electromagnetic sources (such as small hand-held radio transceivers).



Caution:

The AmiCORE Apheresis System is not intended for use as portable equipment.



Caution:

The power cord, pressure cuff, and any other accessories must be kept away from walkways to prevent tripping.



Caution:

The instrument must be used and stored in the proper operating environment (temperature, humidity, altitude, and surface incline requirements).



Caution:

Follow institutional SOPs for phlebotomy technique. Use proper phlebotomy technique to avoid damaging donor veins.



Caution:

Do not block the vent on the front or rear of the instrument. Blocking the vents may cause the instrument to overheat.



Caution:

When maintaining the instrument, use only replacement components, cables, and accessories authorized by the instrument manufacturer and be sure to replace the cassette base and all shields, covers, and screws in their exact locations as described in the AmiCORE Service Manual. Failure to do so may result in increased emissions or decreased immunity of the instrument.



Caution:

Use aseptic technique throughout the procedure.



Caution:

Ensure the proper connection of all solution containers so that there is no leakage at the connection sites.



Caution:

Although the system is automated, apheresis procedures must be monitored.



Caution:

Do not touch the screen with sharp objects.



Caution:

Do not use sharp objects to open the disposable kit packaging.



Caution:

Improperly performed venipuncture may lead to hematoma formation. Operator should monitor venipuncture site for any adverse effects.



Caution:

Allow sanitizing agent to dry before installing a disposable kit.



Caution:

Turn the main power switch (lower back of the AmiCORE instrument) to OFF and unplug from the power source before replacing the power cord.



Caution:

Do not pull the power cord wire. Grasp plug at the power source to remove.



Caution:

The instrument must be plugged in and both power switches turned ON at least once every 6 months for 24 hours. Failure to do so may require battery replacement.



Caution:

Do not attempt to clear the needle, draw line, return line, or resolve the venipuncture problem by infusing saline to the donor.



Caution:

Leave the protective caps on the AC and saline spikes until the solution containers have been hung in place and the operator is ready to make the connections.



Caution:

Do not operate the instrument unless the air filter and air inlet cover are installed.



Caution:

Do not use disposable kits or accessories that have exceeded their expiration date.



Caution:

Do not use solutions if they appear cloudy.



Caution:

Do not use the disposable kit if sterility protectors are damaged, loose, or not intact, or if parts of the disposable kit are damaged or severely kinked.



Caution:

Properly load the donor draw and return lines into the proper clamps and air detector.



Caution:

Properly load the centrifuge pack into the centrifuge.



Caution:

The disposable kit contains DEHP.



Caution:

Equipment connected to the instrument's communication ports must comply with all appropriate UL/CSA/IEC standards for the equipment type.



Caution:

The antenna for the wireless transmitter must be installed to provide a separation distance of at least 20 cm from all persons, and must not be co-located or operating in conjunction with any other antenna or transmitter.



Caution:

Cold solutions can cause considerable discomfort for donors, and overly warm solutions can cause hemolysis.



Caution:

Stripping the tubing may activate platelets.



Caution:

Do not use the centrifuge door lock override while the system is powered.



Caution:

Do not use the separator if there is condensation on it. The condensation poses no safety hazard; however, the separator might be damaged.



Caution:

Avoid contact with the moving pumps, cassette base, and the centrifuge rotor as it decelerates.



Caution:

End the procedure without fluid return in instances where set integrity is compromised in order to prevent blood loss and biohazard.



Caution:

Kinks in the kit tubing may cause container volumes during the procedure to be inaccurate.



Caution:

Keep hair, fingers, jewelry, etc. away from pumps, clamps, or centrifuge components to avoid the possibility of pinching or becoming caught in moving parts of the instrument.



Caution:

Avoid changing parameters which affect Whole Blood to Process or Target Platelet Yield during collection. Doing so may result in an actual yield or storage volume that differs from the target.



Caution:

Fluid spills in the cassette area should be thoroughly cleaned, centrifuge pressure sensors

should be free to move, and donor pressure sensor hydrophobic filters should be replaced if wetted.



Caution:

Treat all used disposable kits as potential biohazards.



Caution:

If the instrument has been moved or relocated, perform weigh scale checks before starting a procedure to ensure scale accuracy.



Caution:

Unless the touch screen indicates otherwise, the operator must use the STOP *button* when preparing the instrument to shut off. Failure to do so may damage the instrument.



Caution:

If the STOP *button* or audio check fails, contact a qualified service representative.

Chapter 2 - System Overview

This chapter contains an overview of the AmiCORE Apheresis System, as well as information on using the touch screen and the bar code scanner (optional).

Section 2.1 **AmiCORE Apheresis System Overview**

The AmiCORE Apheresis System uses a continuous-flow centrifuge to separate whole blood into its individual blood components. Whole blood is drawn from the donor and mixed with anticoagulant (AC). The blood and AC are pumped into the cassette, transferred to the centrifuge, and spun at high speed.

Platelets are temporarily collected within the centrifuge pack. Plasma is collected in a container (to be used as a storage media for the platelets). The remaining blood components are returned to the donor. At the end of the procedure, the platelets and plasma are transferred to a long-term storage container.

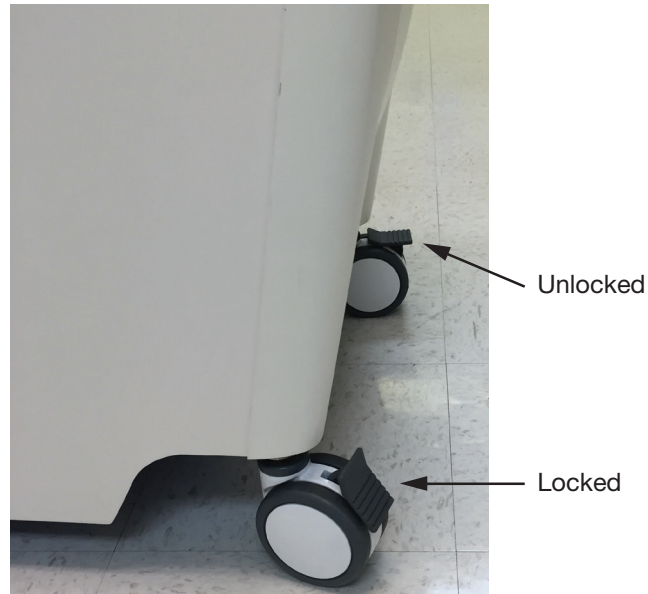
Section 2.2 **AmiCORE Apheresis System Setup**

Position the AmiCORE Apheresis System in the desired location, on a flat surface. Press down on the back of the wheel locks to lock the system in place.



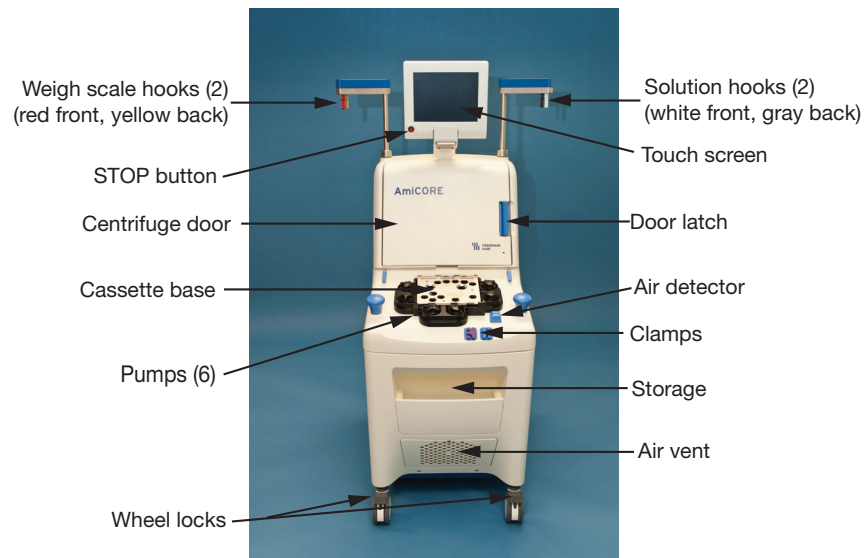
Note:

Ensure that the wheels are properly locked into place prior to beginning a procedure to minimize movement of the instrument.

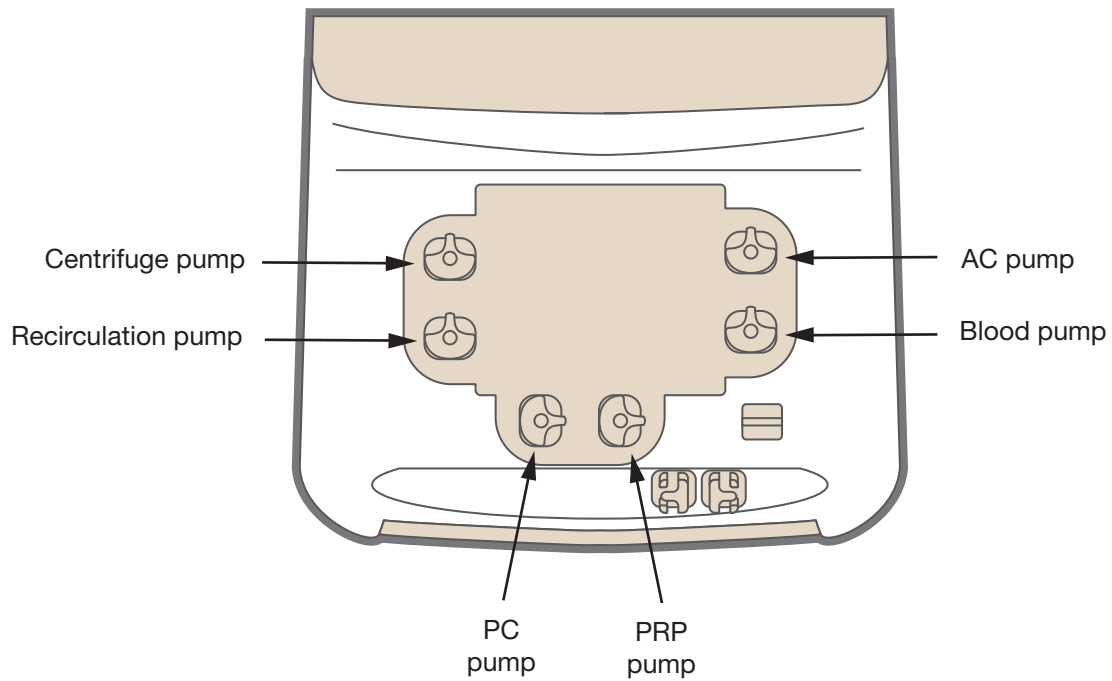


2.1 Wheels and Wheel Locks

Section 2.3 Instrument Components

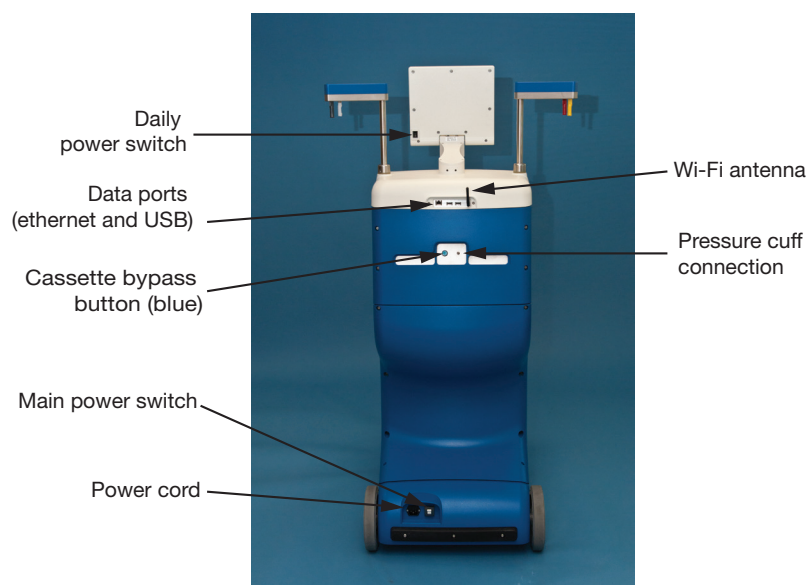


2.2 Front of Instrument



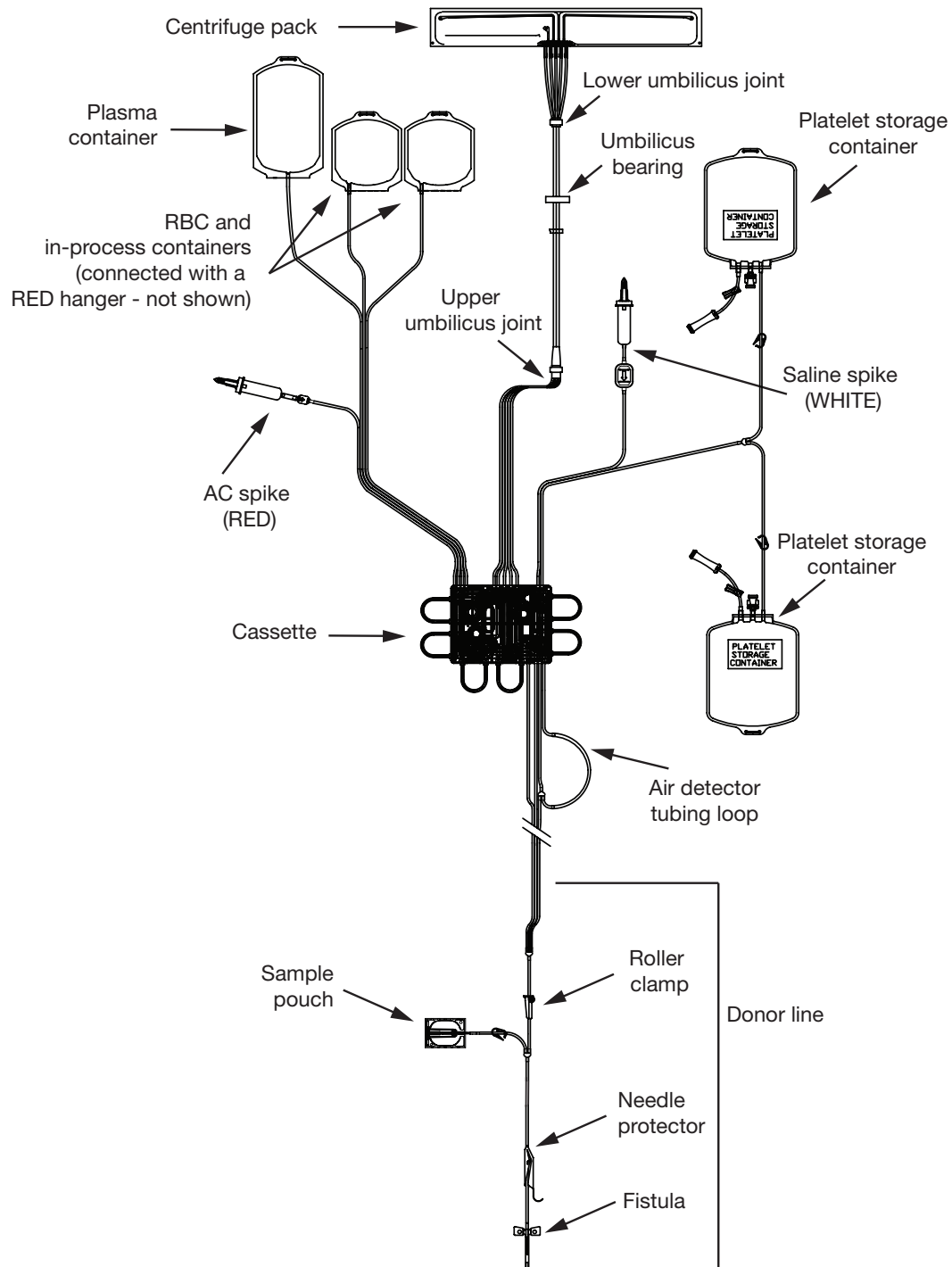
2.3 Pumps

- **AC** – Pumps the anticoagulant to be mixed with the drawn blood.
- **Blood** – Draws and returns blood from/to the donor.
- **Centrifuge** – Pumps blood into the centrifuge to be processed.
- **PC** – Pumps plasma and platelet concentrate into the platelet container.
- **PRP** – Pumps platelet rich plasma from the separation chamber to the collection chamber of the centrifuge pack.
- **Recirculation** – Pumps plasma into the centrifuge.



2.4 Back of Instrument

Section 2.4 Disposable Kit Components



2.5 Typical Disposable Kit

Section 2.5 Touch Screen

The touch screen enables the operator to control the procedure, gather status information, and handle alerts. The screen displays prompts to enter information or perform other tasks.

Using the Touch Screen

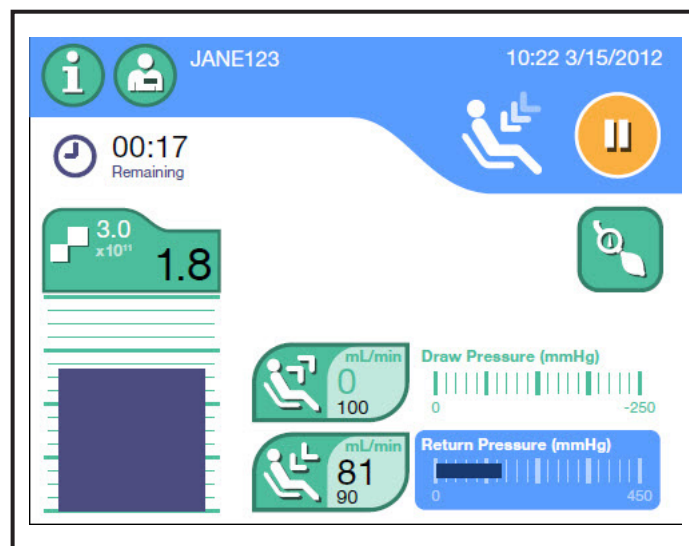
This section describes typical interactive elements that may appear on the touch screen, and how the operator uses them to perform tasks. Interactive elements include screens, overlays, tabs, pages, buttons, and keypads.



Note: Touch screen entries should conform to the instrument's Administrative Settings. Visual indicators (typically orange text or highlights) notify the operator of an entry that is outside of the administrative settings range.

Screens

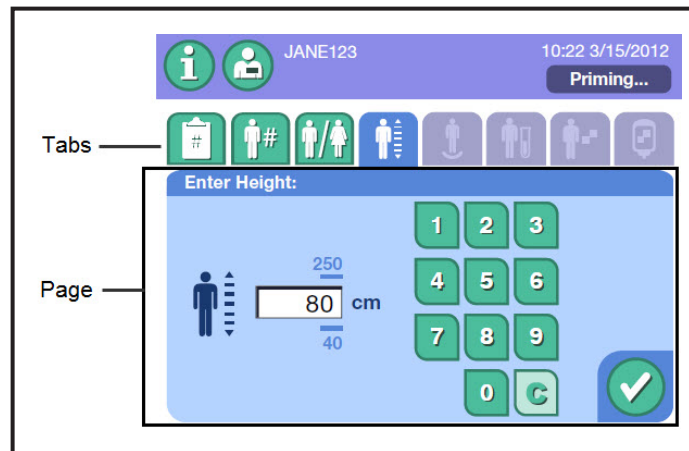
Screens contain elements such as buttons, tabs, prompts, and status information.



2.6 Typical Screen (Return Cycle)

Tabs

Tabs appear on screens and provide access to additional information and controls associated with the screen. Touching a tab opens a page.



2.7 Typical Page (Donor Height)

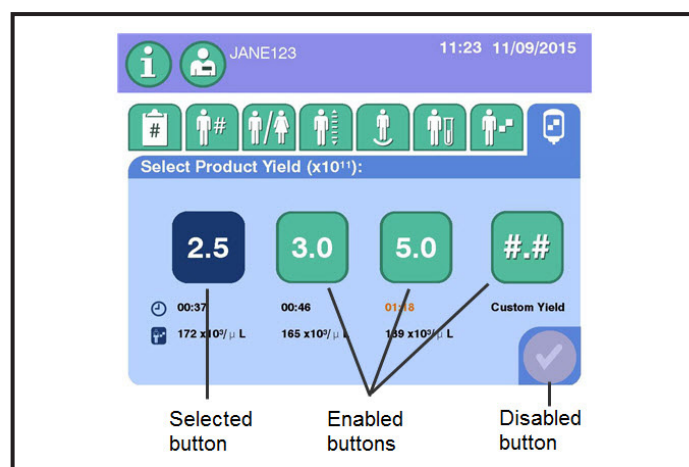
Pages

Pages contain sets of similar elements, such as donor and procedure data entry fields. A page remains available until the associated screen is closed or a different tab is selected.

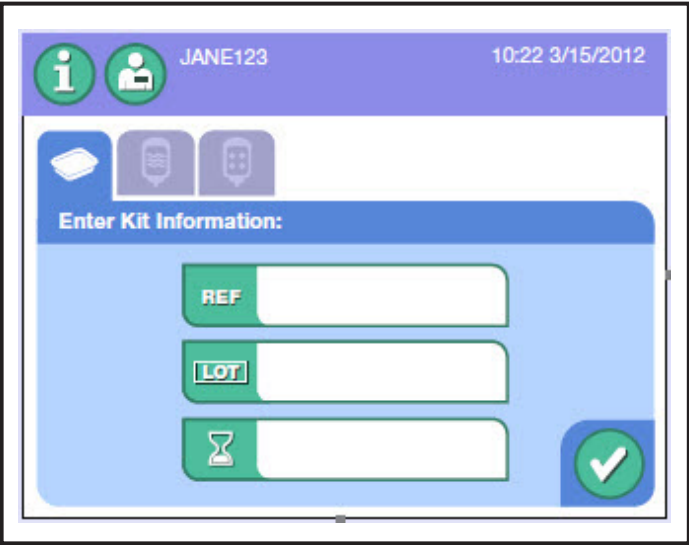
Buttons

There are various types of buttons to help enter information, navigate sets of data, and confirm selections. Buttons change color based on current button status. Grayed-out buttons are disabled, green buttons are enabled, and blue buttons are selected.

Buttons can be disabled due to administrator configuration, procedure settings, or status. Touching the *Check button* typically saves the data and closes the active screen or overlay.



2.8 Typical Buttons



2.9 Typical Buttons

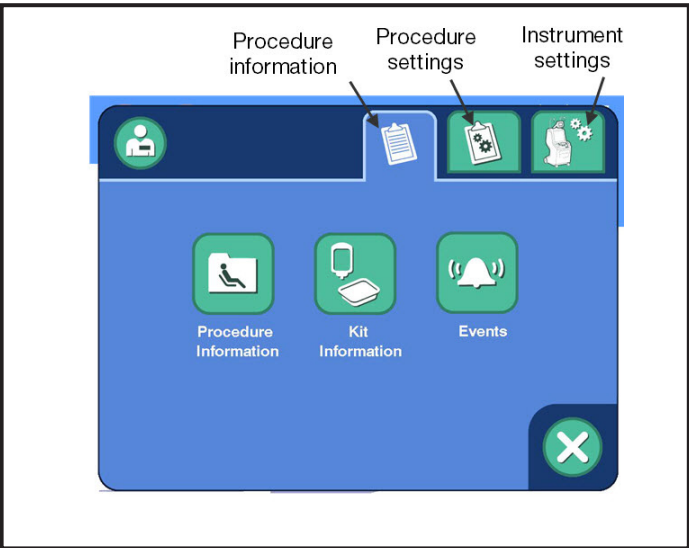
For a list of all buttons, see the Graphics Glossary.



Information
Button

Information Button

The *Information* button enables the operator to view information about the current procedure and perform tasks, such as add saline or AC containers, or adjust hematocrit, pre-count, and target volume settings. Touch the *Information* button to view information or make adjustments.



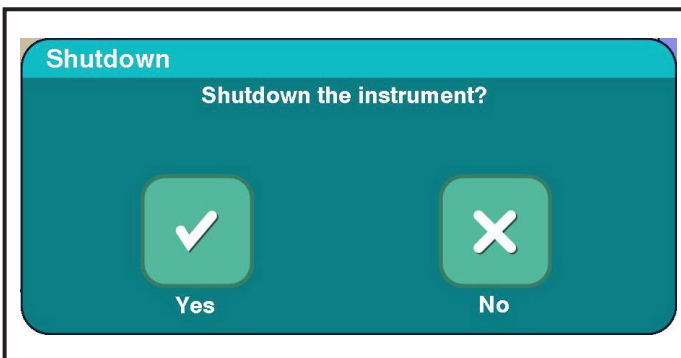
2.10 Typical Information Main Page

See the table on the following page for more information about options on the *Procedure Settings* tab.

Parameter	Range	Default Value	Description/Additional Information
<i>AC Ratio</i>	7:1 to 12:1	10:1	Ratio of whole blood to anticoagulant added by the instrument.
<i>Max Cycle Volume</i>	150 to 300 mL	250 mL	Maximum volume of whole blood to be processed during the draw cycles.
<i>Citrate Infusion Rate</i>	0.50 to 1.50 mg/kg/min	1.25 mg/kg/min	Maximum rate at which citrate is returned to the donor.
<i>Target Draw Rate</i>	40 to 150 mL/min	110 mL/min	Maximum whole blood draw rate.
<i>Target Return Rate</i>	30 to 150 mL/min	150 mL/min	Maximum return rate.

Overlays

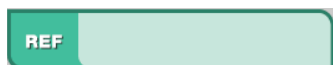
There are various types of overlays, such as confirmation overlays, data entry overlays, and keypad overlays - see “Keypads.”



2.11 Typical Confirmation Overlay

To close a confirmation overlay, touch a response button, such as the *Yes button*.

Data entry overlays enable the operator to view, enter, or change information. Touch the appropriate buttons to enter or change information. If applicable, a keypad opens. The following steps describe using the data entry overlay for entry of disposable kit data. Other overlays require similar actions from the operator.



When the Disposable Kit Data Entry overlay opens, touch the Code Number field to display the applicable keypad.

Code Number Field

The screenshot shows a touch screen interface for data entry. At the top, there is a status bar with a green 'i' icon, a green person icon with 'L', the time '11:22', the date '21/04/2015', and a 'Priming.' button. Below this is a row of eight icons: a clipboard with '#', a person with '#', a person with a slash, a person with a vertical line, a person with a curved line, a person with a vertical line, a person with a vertical line, and a person with a vertical line. Below the icons is a text field labeled 'Enter Procedure ID:' containing the letter 'L'. To the right of the text field are two green buttons: a 'C' button and a back arrow button. Below the text field is a numeric keypad with numbers 1-0, a QWERTY keyboard layout, and a green checkmark button at the bottom right.

2.12 Typical Data Entry Keypad

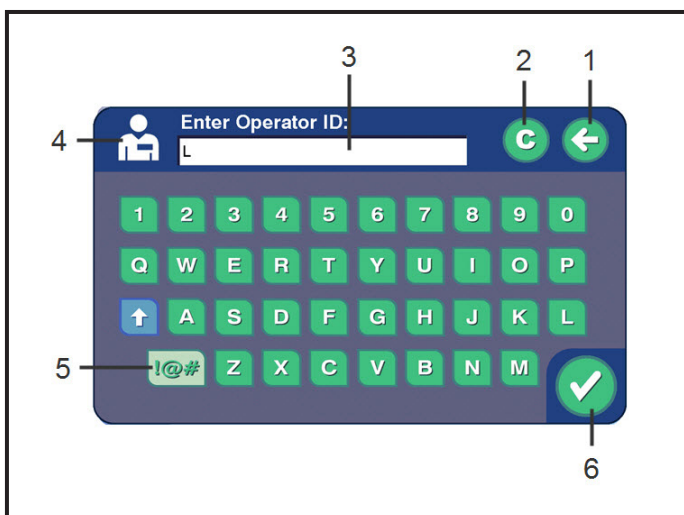
The screenshot shows a touch screen interface for date entry. It has a dark blue background. At the top, it says 'Enter AC Expiration:'. Below this are two input fields: a blue box labeled 'MM' and a green box labeled 'YYYY'. To the left of the 'MM' box is a small icon of a clock. To the right of the 'YYYY' box is a small icon of a clock. To the right of the input fields are three green buttons: an up arrow, a 'C' button, and a down arrow. At the bottom right is a large green checkmark button.

2.13 Typical Use-by Date Overlay

Touch the applicable buttons to modify the date, if necessary. Touch the *Check* button to save the entry and close the overlay.

Keypads

The AmiCORE Apheresis System has several keypads for operators to manually enter information in the system. This section provides examples of several keypads, and actions are similar for all keypads. Some keypads have an option to select from several preset values. Preset values are determined by administrative settings.



2.14 Typical Keypad

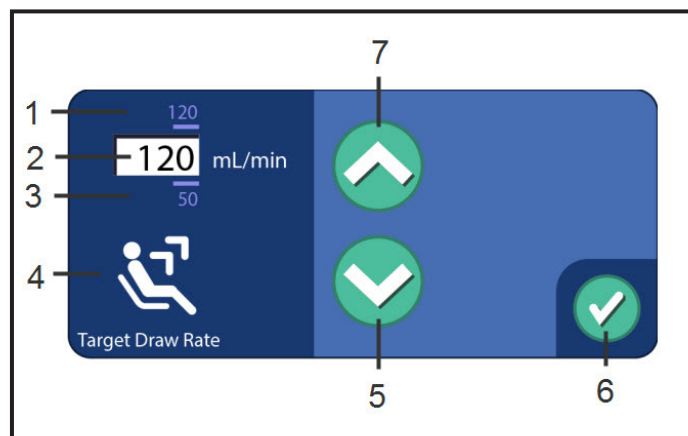
1. **Delete Button** – deletes the last character from the Data Entry field
2. **Clear Button** – deletes all entries from the Data Entry field
3. **Data Entry Field** – displays keypad entries
4. **Icon (Varies)** – identifies which information is being entered, such as Operator ID
5. **Special Character Button** – toggles between special characters and the alphanumeric keypad
6. **Check Button** – saves the entry and closes the overlay



2.15 Typical Platelet Target Volume Keypad

1. **Maximum Yield** – displays the maximum possible platelet target yield

2. **Data Entry Field** – displays the keypad entry to set the platelet target yield
3. **Minimum Yield** – displays the minimum platelet target yield
4. **Time Remaining** – displays time remaining to achieve target yield
5. **Post-Count** – displays donor's platelet count after collection
6. **WB to Process** – displays volume of whole blood to process
7. **Platelet Storage Fluid Button** – provides suggested storage fluid volume for the entered target platelet yield
8. **Check Button** – saves the entry and closes the overlay
9. **Clear Button** – deletes all entries from the Data Entry field
10. **Preset View Button** – returns to the *Platelet Yield Preset buttons*



2.16 Typical Target Draw Rate Keypad

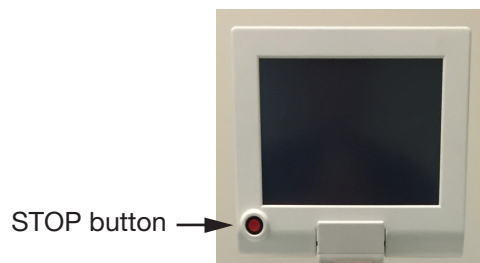
1. **Maximum Draw Rate** – displays the maximum possible draw rate
2. **Data Entry Field** – displays the keypad entry to set the draw rate
3. **Minimum Draw Rate** – displays the minimum draw rate
4. **Icon** – displays parameter name
5. **Decrease Value Button** – decreases Draw Rate value
6. **Check Button** – saves the entry and closes the overlay
7. **Increase Value Button** – increase Draw Rate value

Section 2.6 Overview of the STOP Button

The STOP *button* (red button located below the touch screen) has two functions:

1. Stop and end a procedure
2. Safely power down the instrument

Pressing the STOP *button* stops all pumps and closes all clamps to isolate the donor, and displays an overlay on the touch screen.



2.17 Typical STOP Button

See Chapter 4 for additional information about the STOP *button*.

Section 2.7 Using a Bar Code Scanner (Optional)

The bar code scanner connects to a USB port on the back of the instrument. An institution can use bar codes to collect and track information about a donor, operator, procedure, disposable kits, and ancillary items used during a procedure.

- If a bar code scan is accepted, the instrument emits a short, high-pitched beep (success tone).
- If the bar code scan fails, the instrument emits two lower-pitched beeps in succession (invalid tone).
- The necessary field must be open on the touch screen for the bar code information to be scanned.



Note:

Hold the scanner close to smaller bar codes, and farther away from large bar codes to obtain a proper read.



Note: If bar codes are positioned close together, and it is difficult to scan the intended bar code, use an object (e.g., paper) to cover the undesired bar code.



Note: If the bar code is highly reflective (e.g., laminated), you may need to tilt the scanner at an angle so the bar code can be scanned.

1. Point the scanner's light beam over the center of the bar code.
When the scanner accepts the bar code, the audible success tone is emitted.



2.18 Typical Linear Bar Code



2.19 Typical 2D Bar Code

If the scanner fails to emit a light beam, if an audible beeps does not sound, or if a bar code does not appear in the entry field, contact a qualified service representative or your local customer service representative.

2. Verify that the correct information is displayed, then touch the *Check button*.

If the bar code scanner fails to read properly, use the keypad. Enter all characters displayed under the bar code.

Section 2.8 Physical Specifications

Following are the AmiCORE Apheresis System's temperature and humidity specifications.

Temperature:	The AmiCORE Apheresis System and apheresis kit are designed to work properly when their temperatures are between 15 degrees C (59 degrees F) and 30 degrees C (86 degrees F).
Humidity:	Do not use the AmiCORE Apheresis System if there is condensation on it. The condensation poses no safety hazard; however, the separator might be damaged.

Section 2.9 China Hazardous Substance Table

Unit Name	Hazardous Substance					
	Lead (Pb)	Mercury (Hg)	Cadmium (Cd)	Hexavalent Chromium (Cr6+)	Polybrominated Biphenyls (PBB)	Polybrominated Diphenyl Ethers (PBDE)
Printed Circuit Board Assemblies	X	O	O	O	O	O
Wire and Cable Assemblies	O	O	O	O	O	O
Housing Assemblies and Hardware	X	O	O	O	O	O
Electro-mechanical Components	O	O	O	O	O	O
This table is prepared in accordance with the provisions of SJ/T 11364. O: Indicates that said hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement of GB/T 26572. X: Indicates that said hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement of GB/T 26572.						

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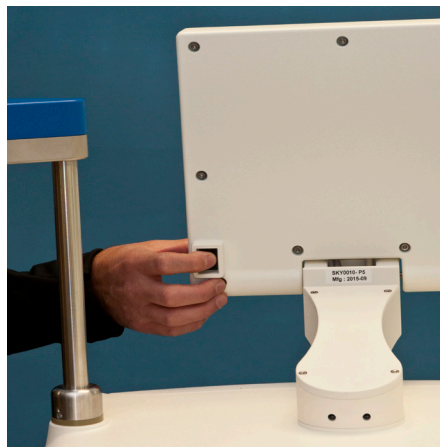
Chapter 3 - System Considerations

Section 3.1 Daily Instrument Set-Up

This section describes daily set-up tasks.

System Start Up

The instrument has two power switches: the daily power switch (back of the touch screen) and the main power switch (lower back of the instrument).



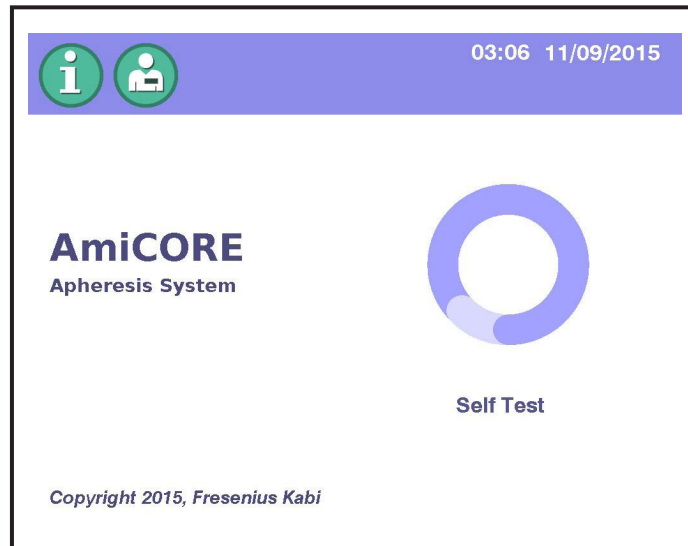
3.1 Typical Daily Power Switch



3.2 Typical Main Power Switch

The main power switch controls power to the entire instrument and is used when there is a need to unplug the instrument. The daily power switch brings power to system components that are necessary to perform a procedure, such as the touch screen.

1. Ensure that the main power switch is set to ON. Toggle the daily power switch to ON. The touch screen displays the AmiCORE Logo screen with the animated *Processing icon* while the instrument performs self tests.



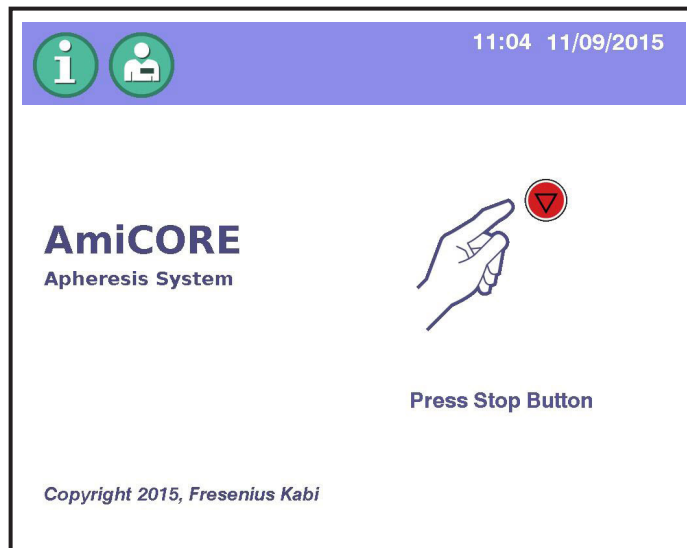
3.3 AmiCORE Logo Screen with Processing Icon (Self Test)

2. Verify the date and time displayed at the top of the screen. If the date or time is incorrect, contact your Administrator.



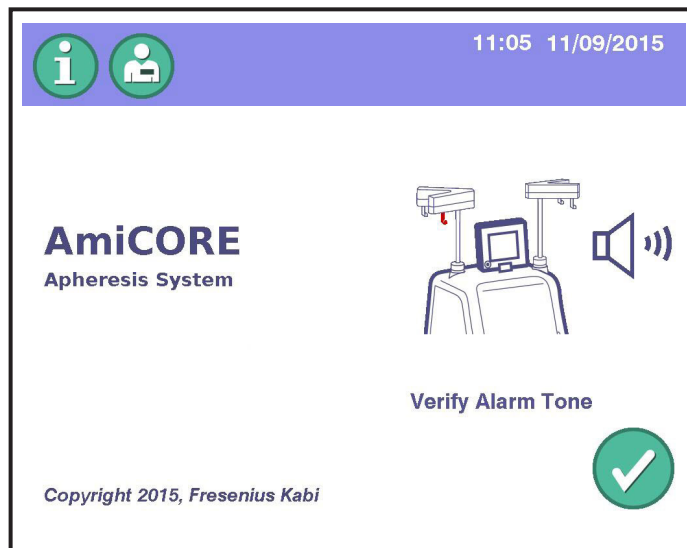
Note: If a red banner appears above the copyright on the AmiCORE Logo screen, contact a qualified service representative.

3. The system is ready to proceed when the prompt to test the *STOP button* appears.



3.4 AmiCORE Logo Screen with STOP Button Test Prompt

4. Press the *STOP button*. If the test is successful, the *Verify Audio Signal* prompt appears.



3.5 AmiCORE Logo Screen with Verify Audio Signal Prompt

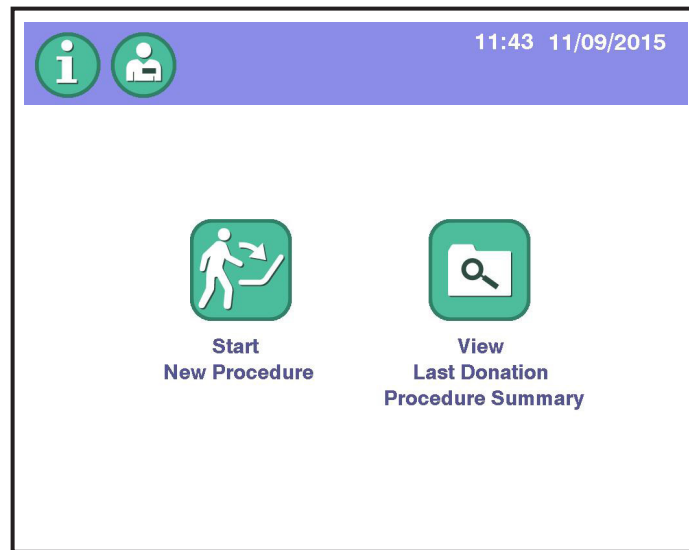
5. Check that an audio tone sounds.



Caution:

If the *STOP button* or audio check fails, contact a qualified service representative.

6. Touch the *Check button*. The Home screen is displayed.



3.6 Typical Home Screen

Section 3.2 Powering the Instrument Off

This section describes how to shut down and turn off the instrument at the end of the day. To power the instrument off during a procedure, see “Using the STOP Button” in Chapter 4.

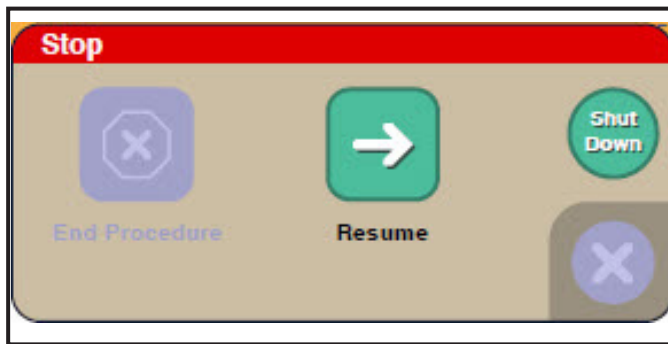
The AmiCORE instrument has two power switches. Follow instructions for which switch to use.



Caution:

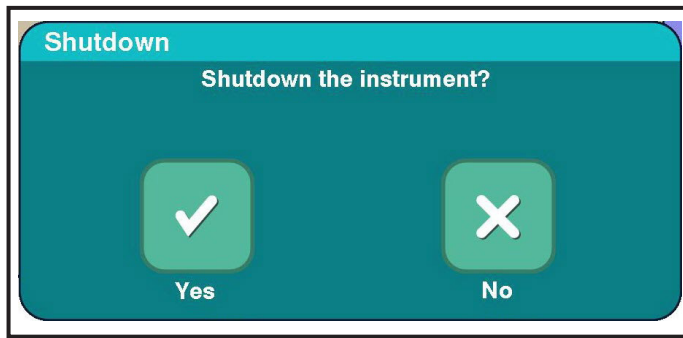
Unless the touch screen indicates otherwise, the operator must use the STOP *button* when preparing to turn off the instrument. Failure to do so may damage the instrument.

1. Press the STOP *button* located next to the touch screen. A Stop overlay appears.



3.7 Stop Overlay

2. Touch the *Shut Down* button. A confirmation overlay appears.



3.8 Typical Confirmation Overlay, Shut Down

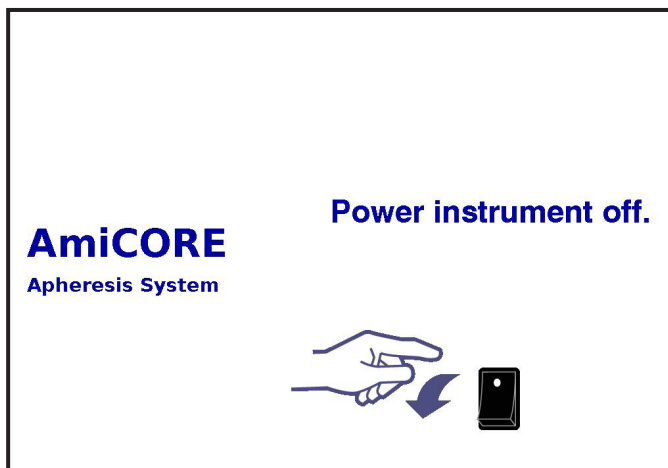
3. Touch the *Yes* button. A Shut Down screen appears.
4. When the Power Instrument Off screen appears, toggle the daily power switch (back of the touch screen) to the OFF position.



*Shut Down
Button*



Yes Button



3.9 Power Off Screen



Note:

Ensure that the main power switch (lower back of the instrument) is in the OFF position before unplugging the power cord. Failure to do so may cause the backup battery to discharge.



Note:

Ensure that the instrument is not turned on again within 3 seconds of being turned off. Doing so may cause the instrument to reboot improperly.



Note:

In a power outage scenario, if the instrument is shut down, the operator will not be able to resume the current procedure.



Note:

If the instrument is powered off due to a power outage or non-recoverable alert, the procedure record may not be completely accurate.

Chapter 4 - Procedures

This chapter describes how to perform a procedure with the AmiCORE Apheresis System.

Section 4.1 Platelet Procedure Overview

After system start up, the operator installs the apheresis kit and solution containers. Kit priming then begins.

During priming, the operator enters procedure and donor information. When priming is complete, venous access is established and the apheresis process begins.

The donor's whole blood is mixed with AC solution and pumped into the separation chamber. Some plasma is recirculated into the separation chamber to maintain the blood entering the chamber at an ideal hematocrit.

The primary stage of separation takes place in the separation chamber. Centrifugal force moves packed red blood cells (PRBCs) along the outer wall of the separation chamber, and platelet-rich plasma (PRP) along the inner wall.

The PRBCs then exit the separation chamber via the PRBC line. The PRP is pumped from the separation chamber into the collection chamber, where the secondary stage of separation takes place.

As PRP enters the collection chamber, platelets are packed by centrifugal force against the outer wall of the chamber. The platelets remain in the chamber while the plasma exits through the plasma line. Any plasma not collected in the plasma container to use as storage fluid is returned to the donor.

RBCs may be visible for short periods of time in the PRP line, but the separator routes these RBCs back into the separation chamber. RBCs should never be visible in the plasma line or the plasma containers.

During reinfusion, the remaining uncollected blood components in the kit are returned to the donor.

Following reinfusion, the centrifuge pack is removed from the centrifuge. A volume of plasma is then added to the collection chamber for platelet resuspension.

After resuspension, the platelets are transferred from the collection chamber into the storage container with plasma.

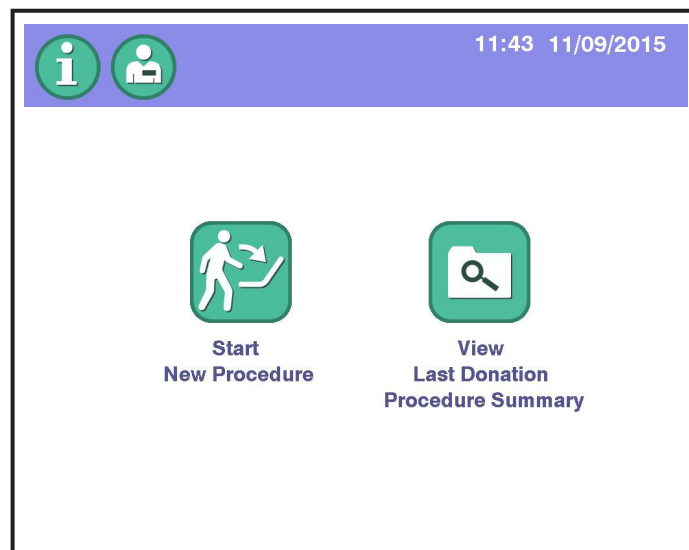
Section 4.2 Collection

The AmiCORE Apheresis System steps the operator through the tasks needed to set up and perform a procedure.

See Chapter 2 for instructions on working with the touch screen.

Starting a New Procedure

1. Touch the *Start New Procedure* button.

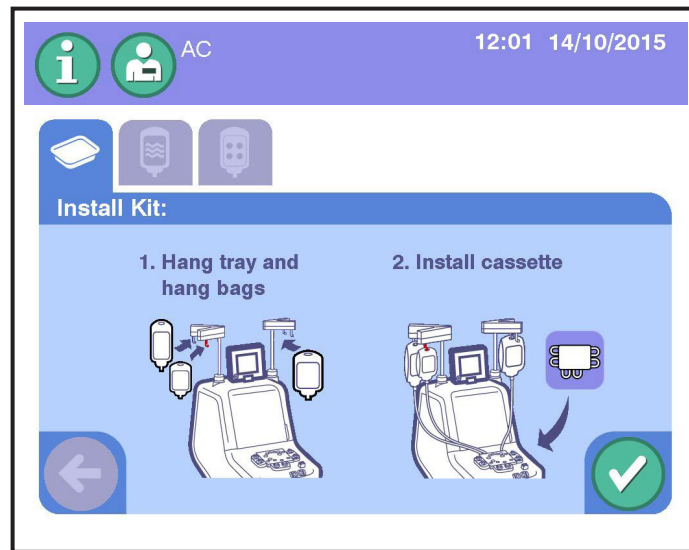


4.1 Typical Home Screen

2. When prompted, use keypads to enter the operator ID.
3. Enter disposable kit information.

Installing a Disposable Kit

When the Install Kit page is displayed, proceed with the kit installation.



4.2 Typical Install Kit Screen

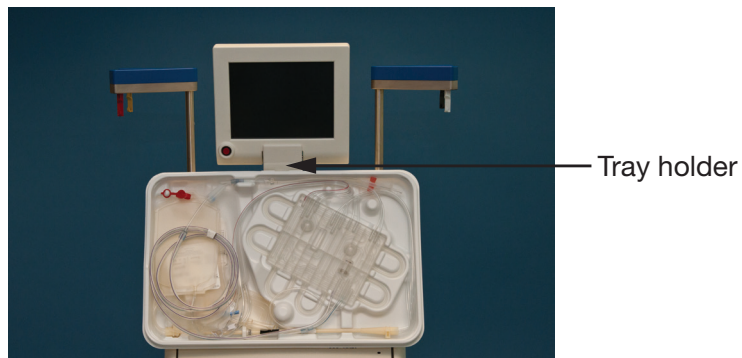
1. Remove the plastic covering from the kit tray.



Note:

The plastic covering on the apheresis kit serves as a dust cover for the kit and keeps the contents of the kit intact. It is not a sterility barrier and it does not define the use by date of the kit.

2. With the centrifuge compartment door closed, snap the kit tray onto the kit tray holder under the touch screen.



4.3 Typical Kit Tray On the Tray Holder

Installing Donor Line and Containers

1. Starting from the left side of the tray, remove the donor line and place it around the right blue handle.
2. Remove the red hanger (with RBC container and in-process container) and hang it on the left front (red) weigh scale hook.

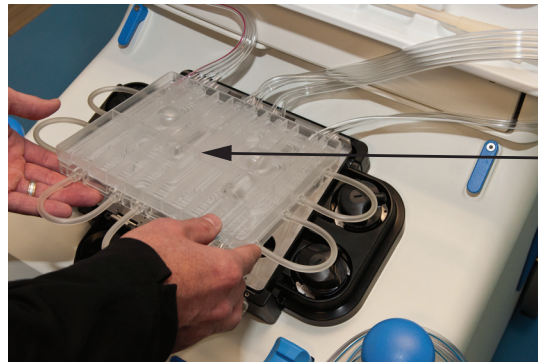


Note: The words “bag” and “container” are used interchangeably throughout this manual.

3. Remove the plasma container and hang it on the left back (yellow) weigh scale hook.
4. Remove the platelet storage containers and hang them on the right back (gray) hook.

Installing the Cassette

1. Remove the cassette from the tray. Slide the cassette into the metal guides at the back of the base and then lower the cassette onto the base.



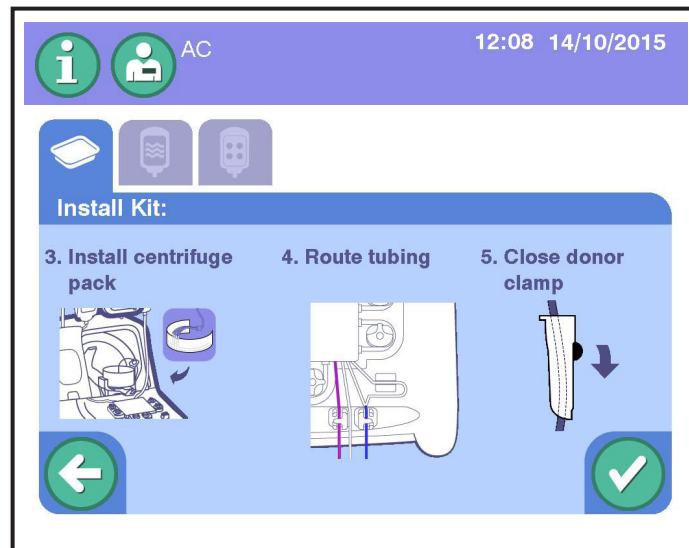
4.4 Typical Cassette Installation

2. Slide the tubing from the top left of the cassette into the left tubing guide. Slide the tubing from the top right of the cassette into the right tubing guide.



Note: Not using the tubing guides may result in damage to the tubing.

3. Touch the *Check button*. The system prompts you to install the centrifuge pack.



4.5 Typical Install Centrifuge Pack and Route Tubing Prompts

Installing the Centrifuge Pack

1. Remove the umbilicus and discard the tray per institutional SOPs.
2. To unlatch the centrifuge compartment door, squeeze the door handle and open the door completely.



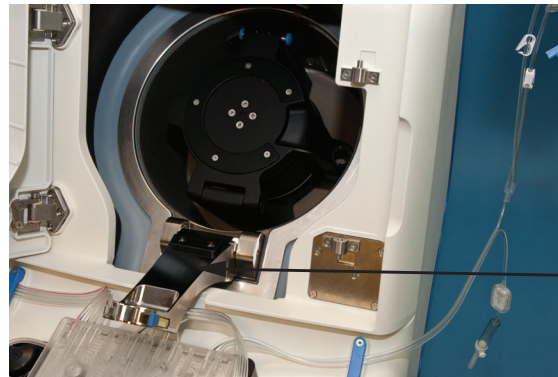
4.6 Unlatching the Centrifuge Compartment Door

3. Pull the zero omega arm toward you.



Note:

Place the centrifuge lines to the side to ensure that the lines are not pinched by the zero omega arm.



Zero omega arm

4.7 Zero Omega Arm

4. Rotate the centrifuge until the blue centrifuge release buttons are at the 12 o'clock position.



Centrifuge

4.8 Blue Centrifuge Release Buttons at 12 o'clock Position

5. Press the blue centrifuge release buttons while pulling the centrifuge lid toward you until it rests open.

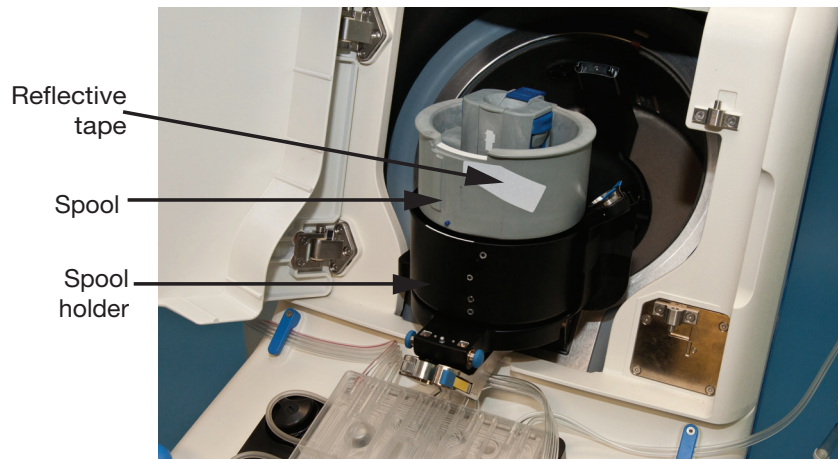


Caution:

The correct spool and spool holder must be used to achieve intended results.

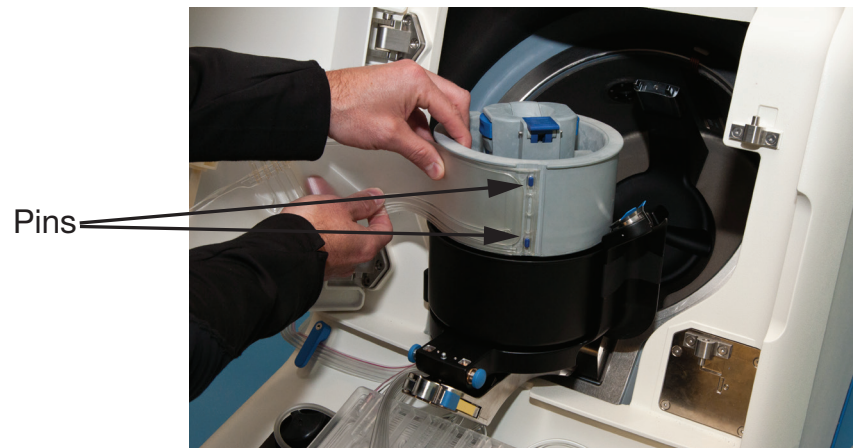
6. Compress the blue spool release wings and pull the spool up until it clicks and stops. Do not compress the gray spool release buttons located below the blue spool release wings or the spool will come off the spool holder.

7. Locate the separation chamber and position it over the spool's reflective tape.



4.9 Spool and Spool Holder

8. Secure the end of the separation chamber to the two pins that are on one side of the spool.



4.10 Double Spool Pins

9. Wrap the centrifuge pack half-way around the spool and fit the single pin into the corresponding hole in the pack.



4.11 Single Spool Pin

10. Finish wrapping the pack around the spool and secure the end to the two pins that are on the side of the spool where you first attached the pack.



4.12 Double Spool Pins

11. Check that all five centrifuge pack holes are secured to a spool pin. Make sure the edges of the centrifuge pack are smooth and even.



Note:

The centrifuge pack will only fit correctly on the spool one way. If the holes on the centrifuge pack do not line up with the pins on the spool, then the pack is not positioned correctly.



Caution:

Do not proceed with centrifuge installation until the centrifuge pack is positioned correctly on the spool.

12. Compress the blue spool release wings and push the spool back into the spool holder. Make sure the centrifuge pack remains smooth.
13. Grasp the upper part of the spool, without touching the spool release wings or buttons, and gently pull on the spool. This ensures that the spool is properly secured in the spool holder.



Note: If you can pull the spool out of the spool holder, the spool was not secured and must be pushed back into the spool holder as described above.



Caution: Do not proceed with centrifuge installation until the spool is secured in the spool holder.

To Install the Lower Umbilicus Joint

1. Press the rear portion of the blue button in the center of the spool while slowly opening the lower umbilicus joint holder.



4.13 Opening the Lower Umbilicus Joint Holder

2. Slide the lower umbilicus joint into the lower umbilicus joint holder.



4.14 Lower Umbilicus Joint Holder

3. To help ensure no kinks are formed, place a finger behind the lines coming from the lower umbilicus joint while pushing the lower umbilicus joint holder back into place.



4.15 Avoid Kinks in Lines

4. Latch the lower umbilicus joint holder by pushing the front portion of the blue button back into place.



4.16 Lower Umbilicus Joint Holder and Button

5. Ensure that lines are not pinched or kinked. Make sure the lower umbilicus joint holder and the button are flush and level with the top of the spool.

6. Tug gently on the umbilicus to test that the lower umbilicus joint holder has been latched correctly.



4.17 Umbilicus Joint Holder Correctly Installed



Note: If pulling on the umbilicus opens the lower umbilicus joint holder, the locking mechanism was not engaged and must be closed again as previously described.



Caution: Failure to properly latch the lower umbilicus joint holder may result in the umbilicus breaking during centrifuge spin up. Do not proceed with centrifuge installation until the umbilicus joint holder is appropriately locked in place.

To Install the Umbilicus Bearing

1. Locate the umbilicus bearing holder found on the outside rim of the centrifuge.
2. Push the umbilicus bearing into the umbilicus bearing holder until it snaps into place.

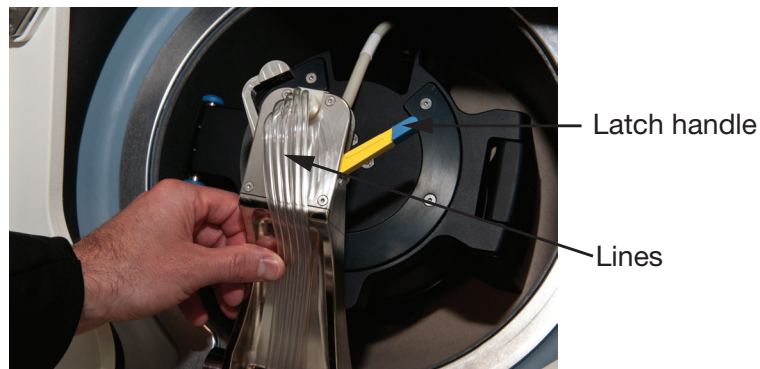


4.18 Umbilicus Bearing in Bearing Holder

3. Check that the bearing is completely seated and flush in the bearing holder.
4. Close the centrifuge by lowering it into place. The centrifuge should click when it latches into place.
5. Return the zero omega arm to its upright locked position.

Installing the Upper Umbilicus Joint

1. Rotate the latch handle outwards and slide the upper umbilicus joint into the zero omega arm and smooth out the lines.



4.19 Upper Umbilicus Joint in Zero Omega Arm

2. Rotate the latch handle inward until the yellow portion is no longer visible.

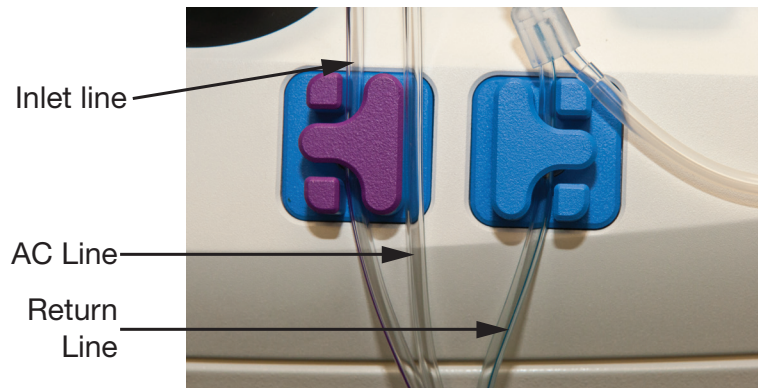


4.20 Zero Omega Arm Latch Closure

3. Place the centrifuge tubing lines in the zero omega tubing guide.
4. Close the centrifuge compartment door. Make sure that the lines are not kinked, twisted, or caught in the door.

Routing the Kit Tubing

1. Thread the purple (inlet) line through the purple clamp.



4.21 Typical Kit Tubing Placement

2. Guide the clear (AC) line between the two clamps.
3. Thread the blue (return) line through the blue clamp.
4. Carefully thread the tubing loop (smaller diameter) into the air detector. Gently slide the tubing from side to side until it is secure in the air detector.



4.22 Insert Tubing Loop into Air Detector

5. Close the roller clamp on the donor line.



Note: Ensure kit tubing is not pinched or kinked before starting kit install check.

6. Touch the *Check button*. The cassette base is automatically lowered and the pumps engage the pump tubing. The system performs the kit install check.



Note: Ensure the pump tubing is correctly installed in the pumps.

Installing Solution Containers

Install Platelet Storage Container

1. Close the slide clamp on the sample pack located on the platelet storage container.
2. If the disposable kit contains two platelet storage containers, do the following:
 - a. Close the clamp to one of the platelet storage containers, ensuring that the clamp on the other platelet storage container is open.
 - b. Close the slide clamps on both sample packs located on the platelet storage container.

Install Saline Container



4.23 Typical Install Saline Solution Prompt



Note:

A minimum of 500 mL of saline is required at the start of the procedure.

1. Enter saline information.
2. Confirm saline volume, toggle to change the volume if needed.
3. Hang the saline container on the right front (white) hook.

4. Spike the saline container with the WHITE spike on the white-striped tubing.



4.24 Typical Saline Spike (WHITE)



Note:

The sequence of hanging and spiking the container may be performed in reverse order, based on institutional SOPs or operator preference. The sequence does not affect disposable kit priming.

5. If a rigid saline container was spiked, open the vent on the saline spike before the saline drip chamber is primed.
6. Prime the saline drip chamber until approximately half full.
7. Touch the *Check button*.

Install AC Container

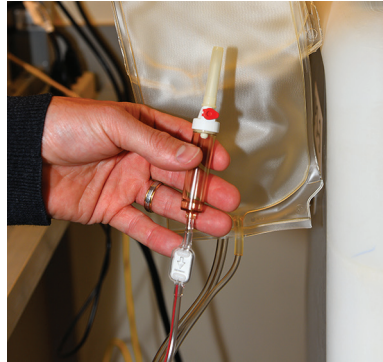


Note:

A minimum of 250 mL of AC is required at the start of the procedure.

1. Enter AC information.
2. Confirm AC volume, toggle to change the volume if needed.
3. Hang the AC container on the left front (red) hook.

4. Spike the AC container with the RED spike on the red-striped tubing.



4.25 Typical AC Spike (RED)



Note:

The sequence of hanging and spiking the container may be performed in reverse order, based on institutional SOPs or operator preference. The sequence does not affect disposable kit priming.

5. If a rigid AC container was spiked, open the vent on the AC spike before the AC drip chamber is primed.
6. Prime the AC drip chamber until approximately half full.
7. Touch the *Check button*.

Entering Donor and Procedure Parameters

During system prime, the touch screen displays a series of tabs that step the operator through entering procedure and donor information.

4.26 Typical Parameter Entry

See the table on the following page for additional information about parameters.

1. After entering the values for a parameter, touch the *Check button* to save the setting and advance to the next tab.



Note: The system displays an orange indicator if you have entered a value that is outside the range of the separator's operation. Review and make any necessary changes.

Preset values	2.5	3.0	5.0	#.#
Est. collection time	00:29	00:33	00:58	Custom Yield
Est. platelet / post count of donor	221 x10 ⁹ /μL	215 x10 ⁹ /μL	196 x10 ⁹ /μL	

4.27 Typical Product Yield Entry

- After all procedure parameters are entered, the system displays a Setup Confirmation screen. Verify that all entries are correct.

4.28 Typical Setup Confirmation Screen

- If changes are needed, touch the applicable parameter tab. When all parameter entries are correct, touch the *Check button* on the Setup Confirmation screen to confirm the entered information.



Warning:

Do not connect the donor until the touch screen instructs you to do so.

Parameter/ Icon	Range	Suggested Value	Description/Additional Information
Operator ID 	Alphanumeric	N/A	Person operating the instrument.
REF	Alphanumeric	N/A	Kit/solution code identification number; entry will be checked by the system.
LOT	Alphanumeric	N/A	Manufacturing LOT identification number.
Expiry	Month / Date	N/A	Use by date of the kit/solution; entry will be checked by the system.
Procedure ID 	Alphanumeric	N/A	Procedure identification number.

Parameter/ Icon	Range	Suggested Value	Description/Additional Information
<i>Donor ID</i> 	Alphanumeric	N/A	Donor identification number.
<i>Gender</i> 	Male / Female	No default	Gender of donor.
<i>Height</i> 	100–254 cm	No default	Height of donor.
<i>Weight</i> 	40–300 kg	No default	Weight of donor.
<i>Hematocrit</i> 	30–55%	No default	Hematocrit of donor.
<i>Pre-count</i> 	150– 600 x 10 ³ /μL	No default	Platelet pre-count of donor.
<i>Product Yield</i> 	1.5– 12.0 x 10 ¹¹	No default	Platelet product yield.
<i>Storage Volume</i> 	0–890 mL	No default	Volume of plasma that will be used to store the platelet product.

Parameter/ Icon	Range	Suggested Value	Description/Additional Information
<i>WB to Process</i> 	N/A	No default	Volume of whole blood to process (in mL).

Venipuncture

Venipuncture When Collecting Blood Samples

1. Place the pressure cuff on the donor's arm and position the cuff for donor comfort.



Note: Ensure that the pressure cuff tubing is free of restriction (e.g., kinks) when the cuff is attached to the donor.

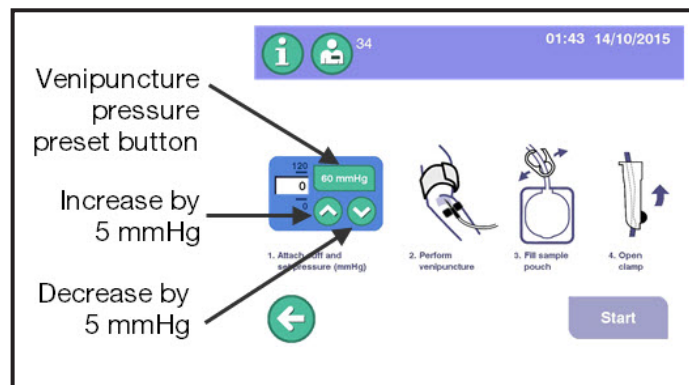


Note: The centrifuge will remain spinning at full speed if the Venipuncture and Sampling screen is displayed, otherwise the centrifuge will spin for 10 minutes before stopping.



Note: If the centrifuge is stopped when the Venipuncture and Sampling screen is displayed, the centrifuge will spin up to full speed. The *Start* button will be enabled once the centrifuge is up to full speed.

2. On the Venipuncture and Sampling screen, use the buttons in the blue area to inflate the pressure cuff to the desired level. To set the pressure to 60 mmHg, touch the *60 mmHg Preset button*. Use the arrows to increase/decrease the pressure level in 5 mmHg increments.



4.29 Typical Cuff Pressure Screen for Venipuncture

Parameter/ Icon	Range	Suggested Value	Description/Additional Information
<i>Venipuncture Cuff Pressure</i>	30–120 mmHg	60 mmHg	Cuff pressure value used to perform venipuncture.

3. Ensure the MasterGuard Anti-Stick Needle Protector is positioned away from the needle wings so it does not interfere with the venipuncture procedure.
4. Perform venipuncture per institutional SOPs.
5. Lower the cuff pressure to the desired setting to fill the blood sampling pouch.
6. Allow the blood sampling pouch to fill.
7. Close the blood sampling pouch clamp.
8. Place two hermetic seals on the blood sampling pouch line near the Y-junction. Once hermetic seals have been placed, you may begin collection. The sampling pouch may be removed from the kit by adding a third hermetic seal and cutting between two of the seals. Leave two seals on the line leading to the kit.



Warning:

If the Sampling Barrel system is used to fill a blood collection tube before two hermetic seals are made, the closed system has been opened and platelets must be stored no longer than 24 hours.



Note:

Fill the blood collection tubes as soon as possible to avoid clotting.

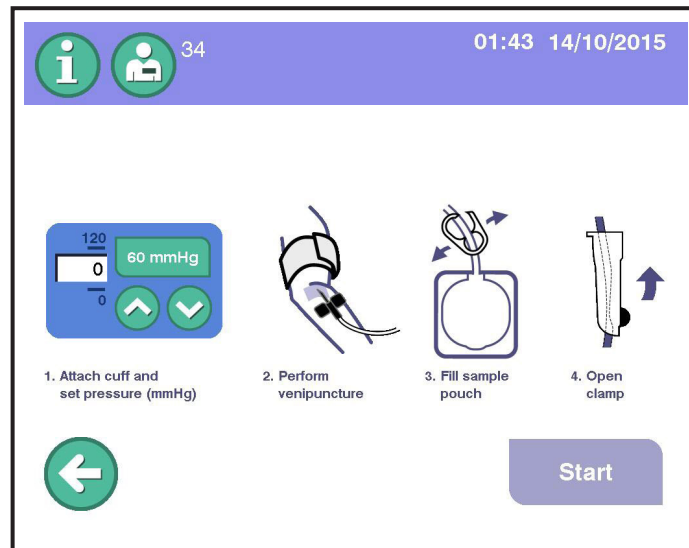
9. Position the sample pouch to prevent air from entering the blood collection tube.
10. Open the barrel holder cover.
11. Center and push the blood collection tube into the barrel holder, penetrating the tube stopper with the multiple-sample needle.
12. Fill the blood collection tube.
13. Remove the blood collection tube from the barrel holder and mix the tube well.

Repeat steps 11–13 as needed for additional blood collection tubes.

Venipuncture without Blood Sampling

Follow this procedure when no blood sample will be taken.

1. Place the pressure cuff on the donor's arm and position the cuff for donor comfort.
2. Inflate the pressure cuff to the desired level.

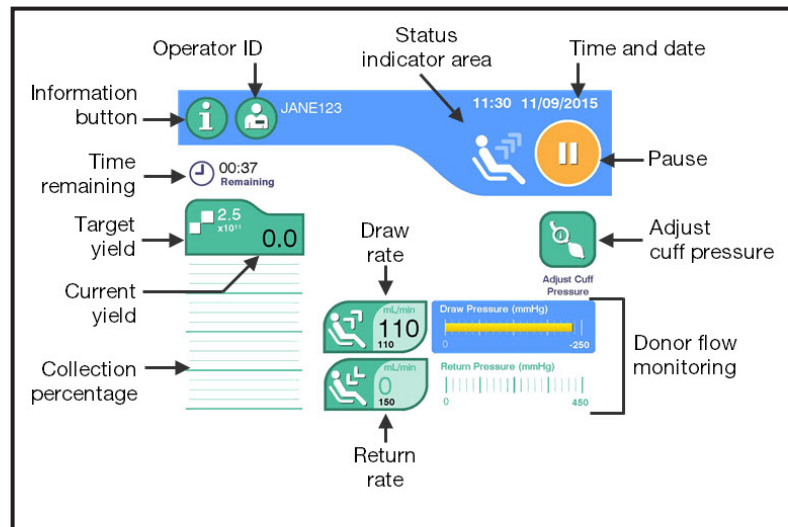


4.30 Typical Cuff Pressure Screen for Venipuncture

3. If available, ensure the MasterGuard Anti-Stick Needle Protector is positioned away from the needle wings so it does not interfere with the venipuncture procedure.
4. Perform venipuncture per institutional SOPs.
5. Place two hermetic seals on the blood sampling pouch line near the Y-junction. If desired, the sampling pouch may be removed from the kit by adding an additional hermetic seal and cutting between two of the seals. Leave two seals on the line leading to the kit. Once hermetic seals have been placed, you may begin platelet collection.

Platelet Collection

1. Open the donor line roller clamp.
2. Touch the *Start button*. The first draw cycle begins.
3. Follow your institutional SOPs for donor monitoring throughout the procedure.



4.31 Typical Collection Screen

To change a parameter, select the desired button and make the adjustment on the touch screen.

Parameter	Range	Default Value	Description/Additional Information
<i>Product Yield</i>	1.5–12.0 x 10 ¹¹	No default	Platelet product yield.
<i>Target Draw Rate</i>	40–150 mL/min	110 mL/min	Maximum whole blood draw rate.
<i>Target Return Rate</i>	30–150 mL/min	150 mL/min	Maximum return rate.
<i>Cuff Pressure</i>	30–120 mmHg	40 mmHg	Cuff pressure value used during collection.

The separator alerts the operator with a single tone audible alert before the first return cycle begins.

An audible tone also sounds near the end of the procedure to alert the operator to change the platelet yield, if desired.

Intelligent Flow Control (IFC) is a feature of the AmiCORE Apheresis System that automatically responds to variations in flow rate during the draw phase of a procedure. If the donor's vein appears unable to maintain the programmed draw rate, IFC temporarily halts the draw and may increase the pressure cuff or decrease the draw rate. These steps are repeated until the problem is resolved. If an occlusion occurs, an audible alarm will sound to notify the operator.

The flow rate is continuously monitored and represented graphically on the Collection screen with a bar graph. The color of the donor flow bar indicates the level of flow.

- **Blue** – Good flow to/from the donor.
- **Yellow** – Possible flow restrictions to/from the donor.
- **Orange** – Occlusion imminent. Alert will be triggered.

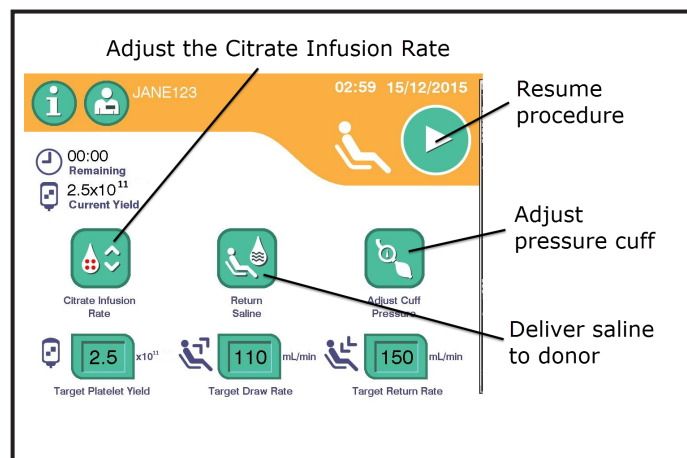
To end the procedure before it is finished and reinfuse the donor's red blood cells, see the section later in this chapter entitled "Using the STOP Button."

When the separator determines that platelet collection is complete, the system performs the reinfusion phase. Once reinfusion is complete, the donor may be disconnected and the platelet product can be transferred into the storage container.

(Optional) To Pause a Procedure

Pause a procedure to make adjustments or add a new AC container.

1. Touch the *Pause button*. The touch screen displays options for actions that the operator can take while the procedure is paused (e.g., resume the procedure, adjust citrate infusion rate, administer saline, adjust cuff pressure).



4.32 Typical Procedure Paused Screen

2. Select the desired button and make adjustments on the screen.



Note:

Some options for *Pause* may not be available depending on the instrument status during a procedure.

Parameter	Range	Default Value	Description/Additional Information
<i>Citrate Infusion Rate</i>	0.50–1.50 mg/kg/min	1.25 mg/kg/min	Maximum rate at which citrate is returned to the donor.
<i>Saline</i>	N/A	N/A	Administer saline to the donor via gravity flow.
<i>Cuff Pressure</i>	30–120 mmHg	40 mmHg	Cuff pressure value used during pause.

3. Touch the *Resume Procedure button*.

Adding a New Saline or AC Container

During the procedure, if the operator notices that the saline or AC container volume is low, the operator should add a new container. To add a new AC container, the procedure must be paused.



Information Button



Kit Information Button



Add Container Button

1. Touch the *Information button* in the top left corner.
2. Touch the *Kit Information button*. On the Kit Information overlay, touch the tab for the container to be added.
3. Touch the *Add Container button* for “Add Saline/Add AC”. Confirm the information, if required.
4. When the Install Container overlay is displayed, remove the old container from the instrument and remove the spike. Confirm the volume of the new container, and hang and spike the new container.



Note:

The same Install Container overlay is displayed when the *Add Container button* is touched on the Alert screen.

5. When the new container is installed, touch the *Check button*.

Table of Collection Indicator Icons

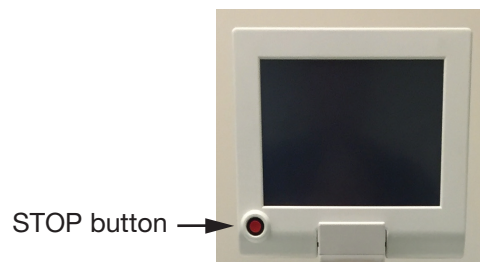
The following table describes icons that may appear during collection to indicate the instrument is performing tasks that do not need operator action. The icons appear at the top of the Collection screen or in a small overlay in the middle of the Collection screen.

Icon	Description/Additional Information
	Scale disturbance. The instrument briefly stops collection and then resumes.

Icon	Description/Additional Information
	Pump ramping issue. The instrument briefly stops all pumps and then resumes collection.
	This overlay is displayed after a new saline container is added during collection. The instrument pumps saline into the kit to prime the saline line.
	This overlay is displayed after an air purge is completed or the operator has returned saline to the donor. The instrument will draw a small blood volume from the donor to prime the return line.
	High pressure has been observed in the centrifuge. The centrifuge will spin down and volume is moved in order to clear the blockage.
	Draw occlusion. If intelligent flow control is enabled, the instrument will decrease the draw rate and increase the cuff pressure to improve flow from the donor.

Using the STOP Button

The STOP *button* is the red button located below the touch screen.



4.33 Typical STOP Button

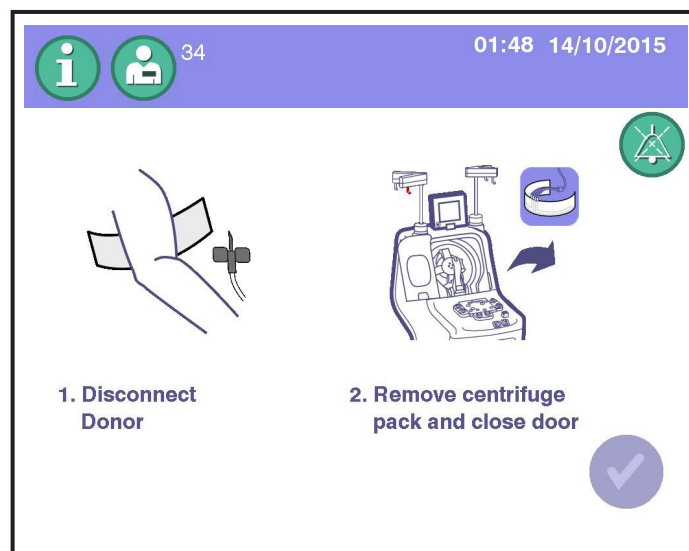
Pressing the STOP *button* while a donor is connected automatically stops all pumps and closes all clamps. After the STOP *button* is pressed, the system displays overlays with options based on the current status of the

procedure. Select and confirm the options displayed on the overlays.
For example:

- Resume the procedure
- End the procedure with fluid return and with product transfer
- End the procedure with no fluid return or product transfer
- Shut down the instrument

Disconnect the Donor

To Disconnect the Line from the Donor



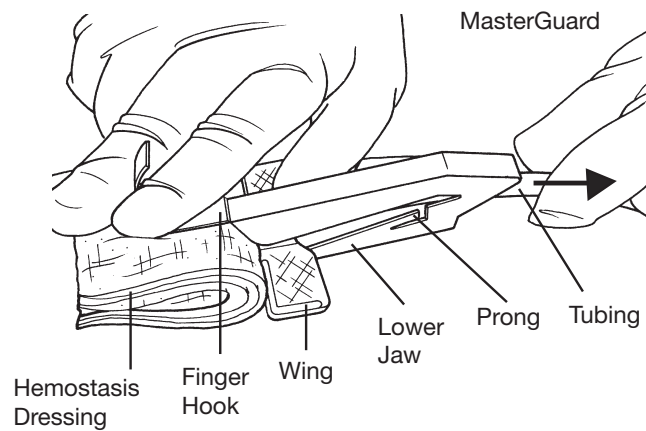
4.34 Typical Disconnect Donor Screen

1. When prompted to disconnect the donor, close the roller clamp on the donor line.
2. Follow your institutional SOPs for needle removal, needle disposal, and donor care.

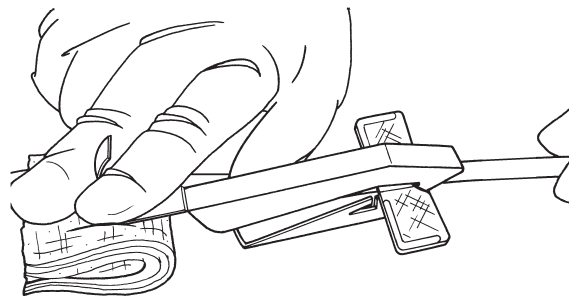
(Optional) Removal of Apheresis Needle with MasterGuard Anti-Stick Protector

1. Upon completion of the procedure, place a hemostasis dressing over the venipuncture site according to your institutional SOPs. Remove tape.

2. With the curved finger hook pointed up, slide the protector forward until the front of the lower jaw is positioned under the wings.



3. Place a finger inside the finger hook, while keeping the hemostasis dressing over the venipuncture site.
4. Pull the tubing until the needle has exited and the wings are secured behind the locking prongs.



5. An audible or tactile “click” will assure the protector is locked in place. Only at this time should the finger hook be released while continuing to maintain pressure on the venipuncture site per your institutional SOPs.
6. Dispose of the needle per institutional SOPs.
7. Follow your institutional SOPs for donor care.

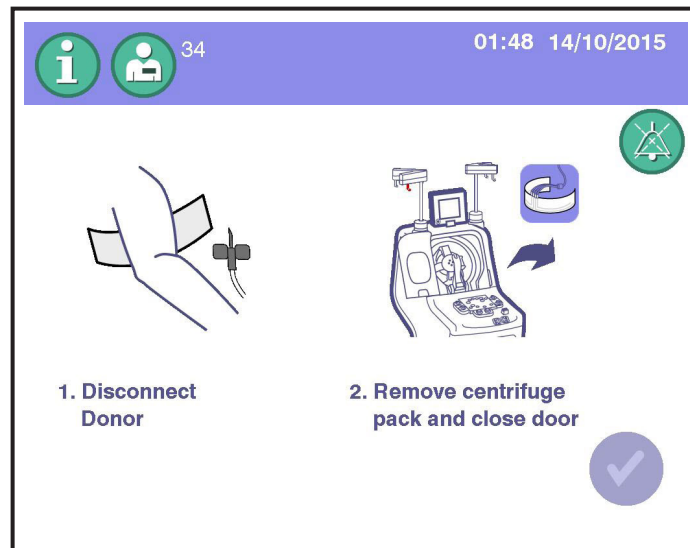


Warning:

The apheresis needle will be covered by the MasterGuard Anti-Stick Needle Protector only if the protector is held in place while the needle is removed by pulling the tubing. After use, visually confirm that the protector is locked over the needle and the wings are securely held behind the locking prongs.

Product Transfer

To Remove the Centrifuge Pack from the Centrifuge



4.35 Typical Remove Centrifuge Pack Screen

1. Check that the clamp to the storage container is open.
2. Open the centrifuge compartment door completely.
3. Unlatch the upper umbilicus joint holder.
4. Remove the upper umbilicus joint from the holder.
5. Pull the zero omega arm to its open position.
6. Rotate the centrifuge until the blue centrifuge release buttons are in the 12 o'clock position.
7. Press the blue centrifuge release buttons to open the centrifuge.
8. Remove the umbilicus bearing from the bearing holder.
9. Press the rear portion of the blue button, and open the lower umbilicus joint holder.
10. Pull the lower umbilicus joint out of the lower umbilicus joint holder.
11. Push the holder back in place and latch it by pushing the blue button into place.
12. Press the blue spool release wings and pull the spool up until it clicks and stops. Do not press the gray spool release buttons located below the blue spool release wings or the spool will come off the spool holder.

13. Remove the centrifuge pack from the spool pins and remove the pack from the spool.
14. Press the blue spool release wings and lower the spool back into the spool holder.
15. Close the centrifuge.
16. Return the zero omega arm to its original position.
17. Close the centrifuge compartment door. Check that no lines are caught in the centrifuge door.



Note: Ensure that the centrifuge compartment door is closed.

18. Touch the *Check button* on the Remove Centrifuge Pack screen to transition to product transfer.

To Resuspend the Platelets

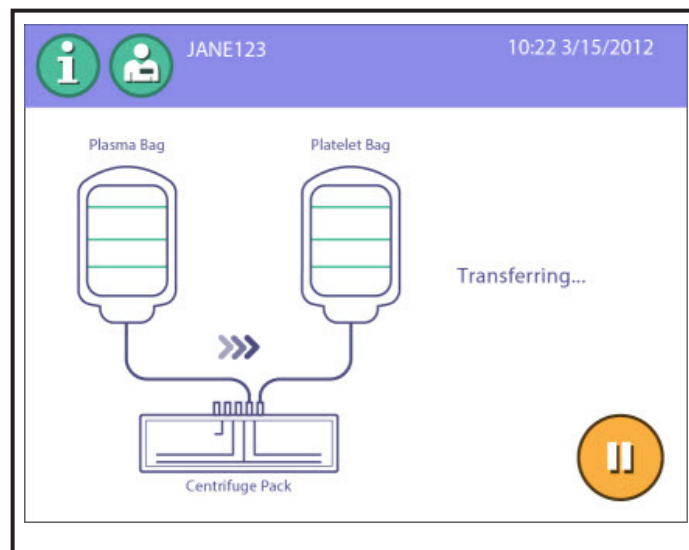
1. Hold the collection chamber taut, grasping one end in each hand. Use a brisk shaking motion to agitate the collection chamber vigorously for at least thirty seconds or until the platelets are completely resuspended.
2. Hang the centrifuge pack on the right front hook using one of the two collection chamber holes located on the end of the pack. Make sure the platelets are completely resuspended.
3. Touch the *Transfer button*. The AmiCORE Apheresis System will transfer the platelets and the appropriate volume of storage fluid to the storage container.



Transfer Button



Note: Transfer can be paused and platelets resuspended in the pack any time and then resumed.

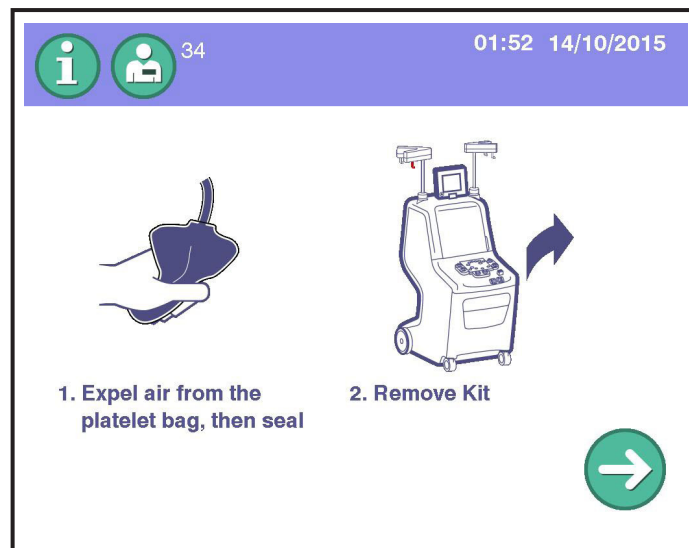


4.36 Typical Product Transferring Screen

Product Handling

To Remove the Disposable Kit from the Instrument

1. Close the clamp to the platelet storage container, but do not yet seal the line.
2. Seal the donor lines, the saline line, and the AC line.
3. Touch the *Check button*. The base automatically unloads the cassette.



4.37 Typical Product Transfer Complete Screen

To Express Air from the Storage Container

Excess air should be expressed from the platelet storage container to avoid adversely affecting the pH of the product and to reduce potential foaming.

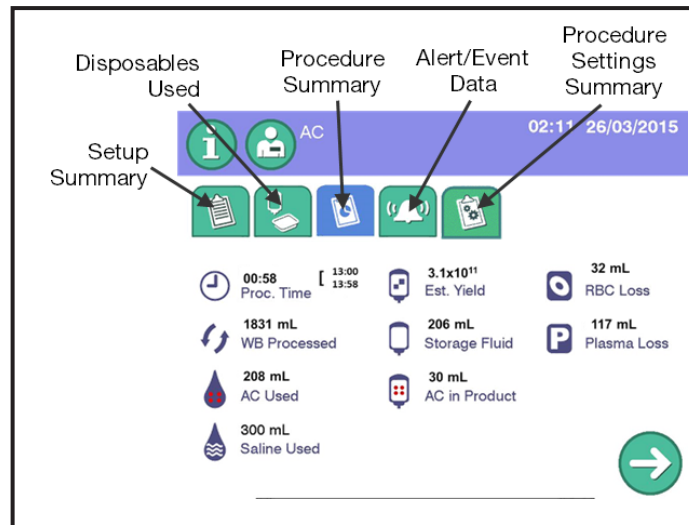
1. Check that the air has been expressed from the sample pack and that the slide clamp is closed.
2. Roll the filled storage container from the bottom up to push the air toward the ports.
3. Carefully open the clamp on the filled storage container and purge the air into the line leading to the cassette until the platelets reach the bottom of the ports.
4. Close the clamp on the filled storage container.

To Remove the Storage Container from the Kit

1. Make three hermetic seals directly on the line leading to the storage container close to the cassette. Ensure that adequate line length remains associated with the storage container to permit subsequent sterile connection needs.
2. Cut between two seals, leaving two seals on the line leading to the storage container.

Section 4.3 Procedure Results

The Procedure Results screen appears at the end of every procedure. The operator should record all procedure information as required by institutional SOPs.



4.38 Typical Procedure Results Screen



Note: If there is any question as to the accuracy of the volumes provided on the Procedure Results screen for the platelet product, the product should be weighed to determine the actual product volume.

Collection Time (min) – The elapsed time from the beginning of the first donor collection phase through the beginning of the reinfusion phase.

Collection Start (time stamp) – The time when the operator selected the *Begin Collection* button.

Collection End (time stamp) – The time when the procedure began reinfusion.

WB Processed (mL) – The approximate volume of anticoagulated whole blood processed during the collection.

AC Used (mL) – The approximate volume of AC used during the procedure.

Saline Used (mL) – The approximate volume of saline used during the procedure.



Note: This volume does not include saline that is manually infused.

Platelet Yield – The approximate yield of platelets in the platelet storage container.

Storage Fluid (mL) – The approximate volume of storage fluid transferred to the platelet storage container at the end of product transfer.

AC in Product (mL) – The approximate volume of AC in the platelet product.

RBC Loss (mL) – The estimated absolute volume of red blood cells removed from the donor.

Plasma Loss (mL) – The estimated plasma volume removed from the donor excluding storage fluid.

Sampling Platelet Products

Prior to sampling, observe the apheresis platelet product by holding the container up to a light source while gently swirling. An opalescent shimmer or swirl should be present. If platelet aggregates are seen, the apheresis platelet product should be rested at room temperature for 2 to 4 hours until platelet aggregates dissipate. If aggregation has not dissipated after 2 to 4 hours, it is suggested that the apheresis platelet product be transferred to a platelet incubator with continuous agitation. Platelet aggregation may take up to 24 hours to dissipate. Final disposition of the platelet product should be determined by institutional SOPs.

Obtain initial sample for platelet counts using your institutional SOPs. A sample may be obtained from the sample pack.

Collection tubes (e.g., EDTA tubes) should be filled according to the manufacturer's directions for use. Collecting an adequate sample volume will maintain an appropriate anticoagulant to sample ratio and may result in a more accurate count.

Product sample packs are available as an ancillary product for collection of additional platelet samples.



Note: The weight of the plastic tubing is approximately 0.3 grams per inch and will contain approximately 0.2 mL of fluid per inch.

(Optional) To Collect a Sample Using the Product Sample Packs

1. Check that the apheresis platelet product is in one storage container and ensure that the clamp is closed.
2. Ensure the apheresis platelet product is well mixed by gently agitating the storage container. An optional method for gentle agitation is to hold the container in the palm of the hands and use a circular motion

to ensure that the apheresis platelet product flows from one end of the container to the other.

3. Position container port side down, open the slide clamp on the platelet sample pack line and allow the sample pack to fill completely with platelets.
4. Squeeze the sample pack several times (approximately 3 times) to ensure the sample is representative of the product.
5. Close the clamp.
6. Make three hermetic seals on the platelet sample pack line.
7. Cut the line between seals leaving two seals on the line leading to the storage container.
8. Immediately transfer the platelet sample into an appropriate laboratory tube. Mix the tube following manufacturer's directions or institutional SOPs. Tubes may be placed on a tube rocker, if available.
9. Obtain a platelet count following institutional SOPs.

Splitting Platelet Products

A maximum of 4.7×10^{11} platelets in the appropriate storage fluid volume may be stored in a single platelet storage container. The AmiCORE separator requires storage in two containers when the expected collection is greater than 4.7×10^{11} platelets. Platelets Pheresis, Leukocytes Reduced may be stored in the platelet storage containers with an appropriate volume of storage fluid for up to 5 days at 20°C to 24°C with continuous gentle agitation.

The platelet product may be split by weighing the product or using a gravity method. The method utilized should be validated.



Note: Once the splitting process starts, it should continue with as little interruption as possible.

Double Platelet Products

(Optional) To Split the Fluid Volume Equally in the Platelet Storage Container when a Double Product is Collected (Weighing Method)



Note: Scale should be tared with the appropriately configured empty container prior to weighing and splitting the product.

1. Check that the apheresis platelet product is in one storage container and ensure that the clamp is closed.

2. Place the container with the apheresis platelet product on the tared scale and determine the product net weight.



Note:

Care should be taken to ensure that the positioning of the extended lines does not affect the weighing process.

3. Divide the product net weight by two to obtain the desired target product net weight per container.
4. Ensure the apheresis platelet product is well mixed by gently agitating the storage container. An optional method for gentle agitation is to hold the container in the palm of the hands and use a circular motion to ensure that the apheresis platelet product flows from one end of the container to the other.
5. Place the empty container on the tared scale and open the clamp.
6. Open the clamp on the container with the apheresis platelet product. Use the clamp to control the transfer of the platelets.
7. Transfer the platelet product per institutional SOPs until the target product net weight (± 10 gram) is reached.
8. Close the clamps.
9. Check that the containers demonstrate acceptable split weights, if not adjust the weight.
10. Remove the container from the scale and hold the two containers port side up by the attached line.
11. Open the clamps on both storage containers to allow the fluid in the lines to drain.



Caution:

The apheresis platelets should be removed from the line connecting the storage containers prior to storage. The tubing material does not support platelet viability for extended periods of time.

12. Close the clamps on the storage containers.
13. Store the platelet containers using standard blood banking conditions in accordance with regulatory guidelines and institutional SOPs.

(Optional) To Split the Fluid Volume Equally in the Platelet Storage Container when a Double Product is Collected (Gravity Method)

1. Check that the apheresis platelet product is in one storage container and ensure that the clamp is closed.

2. Ensure the apheresis platelet product is well mixed by gently agitating the storage container. An optional method for gentle agitation is to hold the container in the palm of the hands and use a circular motion to ensure that the apheresis platelet product flows from one end of the container to the other.
3. Open the clamps on both storage containers.
4. Check that the line connecting the containers is filled with fluid.
5. Hang the storage containers at the same head height with the port side down until the fluid levels are approximately equal.



Note: Hanging the two containers on one hook may result in uneven distribution of the product.

6. Close the clamps on both storage containers.
7. Invert the storage containers so that they are in their upright (ports up) position and hold the two containers by the attached line.
8. Open the clamps on both storage containers to allow the fluid in the lines to drain.



Caution: The apheresis platelets should be removed from the line connecting the storage containers prior to storage. The tubing material does not support platelet viability for extended periods of time.

9. Close the clamps on the storage containers.
10. Weigh each container and verify that the weights are within 10 grams of each other. If not, adjust the weight.
11. Store the platelet containers using standard blood banking conditions in accordance with regulatory guidelines and institutional SOPs.

Kit Disposal

The kit is biohazardous waste and should be disposed of according to institutional guidelines.

1. Disconnect the needle from the kit.
2. Dispose of the needle and kit per institutional SOPs.



Note: (Optional) Recycle the kit tray by placing it in a recycling container.

3. Once the kit has been removed, ensure the procedure results have been appropriately recorded per institutional SOPs.



Note: Only results from the last procedure where a donor was connected are available for operator access. The last donation procedure summary can be accessed from the Home screen.



*View Last Donation
Procedure Summary
Button*

4. Touch the *Check button* to return to the main menu.

Chapter 5 - Troubleshooting

Section 5.1 Introduction

This chapter describes how to troubleshoot the instrument. Refer to institutional standard operating procedures (SOPs).

Performance

The quantity and quality of blood components collected by the AmiCORE Apheresis System depends primarily on:

- Volume of whole blood processed and rate of processing
- Donor hematocrit and platelet count
- Handling and storage of products
- Instruments and techniques used to measure results

If you see a change in the separator's performance, please call your local customer service representative for assistance.

Product Quality

When product quality does not meet expectations a failure investigation should be initiated. The following sections indicate conditions when leukoreduction, red blood cell content, product yield, and product volume may not meet the expected results.

Leukoreduction

Improper sampling methods and WBC counting techniques may provide results that erroneously indicate that leukoreduction has not been achieved.

The following list itemizes conditions when platelet products should be counted for residual white blood cells (WBCs):

- Manual product transfer was performed for any reason.
- A greater than usual amount of red blood cells (RBCs) are noted in the centrifuge pack when it is removed from the centrifuge. The presence of a small amount of RBCs, such as a spot approximately

1 inch (2.5 cm) in diameter, near the entrance to the collection chamber is normal.

- The centrifuge spins down during the course of the procedure.
- The clamps to the storage containers are inadvertently left open when the kit is removed from the separator, resulting in blood components remaining in the kit being transferred to the storage containers.
- Anytime RBCs are noted in the platelet containers.
- The instrument flags the product for RBC contamination.
- Anytime lipemic plasma is noted in the plasma or platelet containers.

Red Blood Cell Content Evaluation

Follow applicable regulatory requirements when evaluating RBC content. RBC content evaluation should be performed under the following circumstances:

- Any time a greater than usual amount of red blood cells (RBCs) are noted when the centrifuge pack is removed from the centrifuge. The presence of a small amount of RBCs, such as a spot approximately 1 inch (2.5 cm) in diameter, near the entrance to the collection chamber is normal.
- Any time RBCs are seen in the platelet container.
- Any time the instrument flags the product for RBC contamination.

Why Actual and Target Yields May Vary

- If the platelet product is not adequately resuspended during the product transfer phase, the actual platelet yield may be lower than the requested yield.
- Inaccurate donor information was used for the procedure parameters.
- The *WB to Process* or *Yield* parameters were changed for any reason during collection.
- Variations in platelet sampling methods and counting techniques.
- The more inlet line occlusions that occur during platelet collection, the greater the possibility that the actual yield is lower than the target yield.
- The more alerts that occur during platelet collection, the greater the possibility that the actual yield is lower than the target yield.

- The more times processing is stopped during platelet collection, the greater the possibility that the actual yield is lower than the target yield.
- If the donor exhibits lipemic plasma, the actual platelet yield may be lower than the target yield.
- If the procedure is ended early for any reason, the actual platelet yield may be lower than the target yield.

Storage Volume

The actual storage volume should be within $\pm 10\%$ of the target volume.

The following conditions may result in an actual product volume that is different than the target volume:

- If a procedure is ended early for any reason, the volume of plasma in the platelet product may be lower than the target volume.
- If the plasma container is disturbed during product transfer, the volume of plasma in the platelet product may be different than the target volume.
- If the platelet yield is changed late in the procedure, the volume of plasma in the platelet product may be lower than the target volume.
- If product transfer-specific alerts occurred during product transfer, the volume of plasma in the platelet product may be lower than the target volume.
- Platelet under-collections due to lipemic plasma.



Note:

The volume of storage fluid in the platelet storage containers should be confirmed, by weighing the products, any time there is any question about the presence of adequate storage fluid to support the platelet product for proper storage. Refer to Appendix A.5 - Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC.

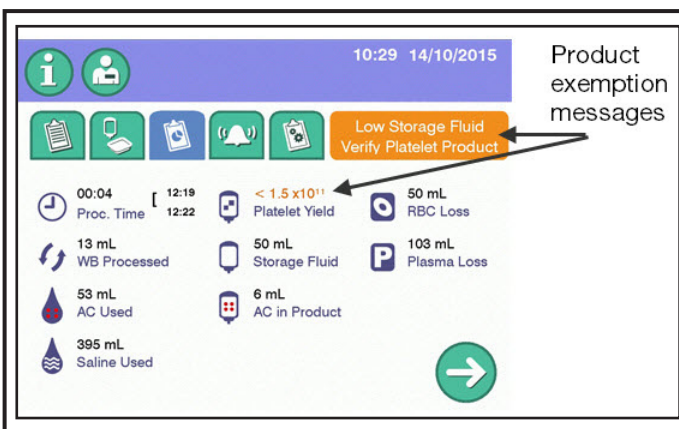


Note:

If there is any question as to the accuracy of the volumes provided on the Procedure Results screen for the platelet product, the product should be weighed to determine the actual product volume.

Section 5.2 Product Exemption Messages

In some instances, one or more product exemption messages will be displayed on the Procedure Summary screen. Whenever product exemption messages are displayed, the operator should check for product quality. Also, the data obtained from procedures with one or more of these messages should be excluded from validation and quality control data. The messages appear on an orange background or in orange text.



5.1 Typical Exemption Messages on Procedure Summary Screen

Low Storage Fluid – The storage fluid transferred into the platelet container is less than the minimum required to store the collected platelets for the full storage duration. Testing should be performed to check plasma volume for product acceptability per institutional guidelines. The Low Storage Fluid message may also be displayed if a product transfer-specific alert occurred during product transfer.

Verify Platelet Product – The centrifuge spun down in the middle of the procedure or the system suspected possible WBC or RBC contamination. Testing should be performed to check the red cell content of the platelet production per institutional guidelines. See Section 5.1 for additional scenarios where product testing may be required.

Low Platelet Yield – If the platelet product collected is less than a usable platelet dose, the screen will display $< 1.5 \times 10^{11}$, as shown in the previous image.

Section 5.3 Alerts

The operator plays an essential role in the safe operation of the AmiCORE Apheresis System. Alerts are provided when conditions exist that may affect donor safety, product quality, or donor comfort.

During an alert, the AmiCORE Apheresis System stops the pumps and closes the donor isolation clamps. The AmiCORE Apheresis System provides an Alert screen with instructions on how to resolve the alert. In addition, an audible tone sounds.



Help Button

Alert icons appear on the left side of the Alert screen and provide the operator with information regarding the alert, possible causes, and possible resolutions. The AmiCORE Apheresis System displays an instrument graphic in the center of the Alert screen. The troubleshooting areas displayed on the graphic in orange show possible locations of alert origin. For additional information about the alert and alert icons, touch the *Help button*. For a list with the descriptions of all alert icons, refer to the “Glossary of Graphics” at the end of this manual.

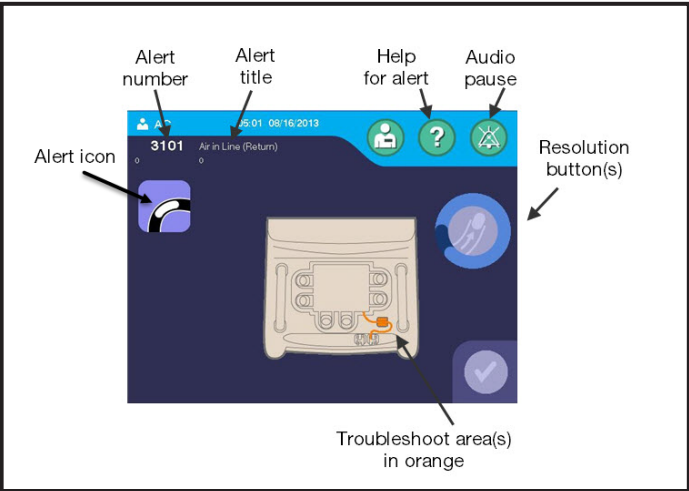
Alerts can be muted for two (2) minutes by pressing the *Audio Pause button*. Audible tones range from 60–120 dBA.

System Status	Screen Header	Audible Tone
Normal operating conditions	Blue	N/A
Procedure paused	Yellow	N/A
Extended paused	Yellow	BingBing
Alerts	Blue	Two tone chime
Attention	N/A	BaDing
Invalid entry	Highlighted Orange	BeDong
Final return	N/A	TaDa
Scanner input	N/A	Beep

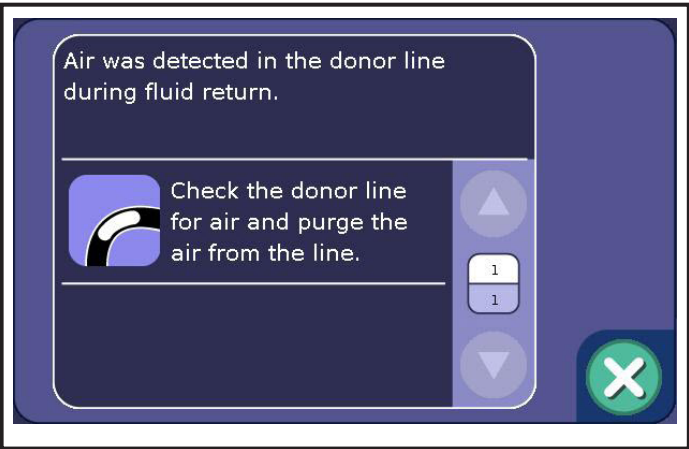
There are three general types of alerts:

Alert Type	Description/Additional Information
Recoverable	The operator should resolve the issue as indicated by the alert icons and touch the <i>Check button</i> once the issue is resolved. The procedure will continue as usual.
Non-Recoverable (End the procedure)	The operator must press the <i>STOP button</i> and choose how they want to end the procedure. The <i>Check button</i> on the Alert screen is disabled.

Non-Recoverable (Shut down)	The operator must press the <i>STOP button</i> . The only option available for this type of alert is to shut down the instrument. The <i>Check button</i> on the Alert screen is disabled.
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5.2 Typical Alert Screen



5.3 Typical Alert Help Screen

The following table provides a complete listing of Alerts that may occur while performing a platelet collection procedure.

Code	Description	Resolution
1002	Cannot Connect to Network The system could not connect to the DXT Relay server.	Check that the Ethernet cable is plugged in or the wireless network is functional.
1029	Fluid Detected During the initial instrument self-tests, the system detected fluid at the air detector.	If kit from the previous procedure is installed, remove the kit. Press the <i>STOP button</i> and shut off instrument. If previous kit was not on instrument when alert was triggered, air detector may be malfunctioning. Call a qualified service representative.
1030	Stop Button Error During the initial instrument self-tests, the <i>STOP button</i> test failed.	Power off instrument and call a qualified service representative.
1040	Calibration Failure The system detected a front scale error.	Touch the <i>Calibrate Scale button</i> to recalibrate the front scale. Ensure weight is hung properly on the front weigh scale. If alert persists, press <i>STOP button</i> . Power off the instrument and call a qualified service representative.
1041	Scale Disturbed The system was unable to collect a stable scale reading.	Check the scales and touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
1042	Weight on Scale The system detected an incorrect weight on the front scale.	Remove any weight from front scale. If no weight is on scale, touch the <i>Calibrate Scale button</i> and perform scale calibration. Touch the <i>Check button</i> to continue. If issue reoccurs, a hardware error is likely. Call a qualified service representative.
1043	Calibration Failure The system detected no weight change during front scale calibration. Touch the <i>Calibrate Scale button</i> to restart calibration.	Touch the <i>Calibrate Scale button</i> to redo scale calibration. Ensure that during scale calibration, the correct weights are hung on the front scale.
1044	Calibration Failure The system detected a back scale error.	Touch the <i>Calibrate Scale button</i> to recalibrate the back scale. Ensure weight is hung properly on the back weigh scale. If alert persists, press the <i>STOP button</i> . Power off the instrument and call a qualified service representative.
1045	Weight on Scale The system detected an incorrect weight on the back scale.	Remove any weight from back scale. If no weight is on scale, touch the <i>Calibrate Scale button</i> and perform scale calibration. Touch the <i>Check button</i> to continue. If issue reoccurs, a hardware error is likely. Call a qualified service representative.

Code	Description	Resolution
1046	Calibration Failure The system detected no weight change during back scale calibration.	Touch the <i>Calibrate Scale button</i> to redo scale calibration. Ensure that during scale calibration, the correct weights are hung on the back scale.
1101	Power Failure Instrument experienced power failure upon power on. Check that the instrument is plugged in.	Press the <i>STOP button</i> . Power off the instrument at both power switches. If issue persists, call a qualified service representative.
1102	Battery Charge Failed The system detected an error with the battery charge circuit.	Keep instrument plugged in and power switches ON to allow battery to charge. Touch the <i>Check button</i> to continue. Repeat to allow battery to charge. Call a qualified service representative if the issue persists.
2002	Kit Check Error The blood pump has pumped fluid without seeing a pressure change, indicating an occlusion problem with the Draw Line Clamp, Return Line Clamp, or Saline Clamp (inside the cassette).	Touch the <i>Check button</i> to continue. Ensure proper alignment of the cassette and tubing within clamps before reloading the cassette.
2003	Kit Check Error The saline clamp is not functioning or fluid has been introduced; probable Cassette load error.	Press the <i>STOP button</i> and end the procedure. Start over with new kit. Do not spike fluids until instructed to.
2004	Kit Check Error Blood pump is unable to generate sufficient pressure to perform saline prime; due to air leak or faulty saline solution connection.	Check all connections and tubes between saline container and blood pump for poor connections; fix connections and touch the <i>Check button</i> . If no connection issues exist, inspect for air leak. If an air leak is present, press the <i>STOP button</i> and end the procedure. Start over with new kit.
2005	Kit Check Error The blood pump has pumped fluid without seeing a change in pressure; the manual donor clamp is not properly occluding, the AC pump is not loaded correctly, or there is an air leak.	Check that donor line manual clamp is in fully closed position. Touch the <i>Check button</i> to continue. If issue persists, check that tube is properly oriented around AC pump and touch the <i>Check button</i> . If issue persists, air leak is likely the cause. Press the <i>STOP button</i> and end the procedure. Start over with new kit.
2007	Kit Check Error The system is unable to build pressure within the kit.	Touch the <i>Check button</i> to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.
2008	Kit Check Error The system is unable to build pressure within the kit.	Touch the <i>Check button</i> to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.

Code	Description	Resolution
2010	Leak Detected The system has detected a leak in the kit.	Press the <i>STOP button</i> and end the procedure. Unload kit and start again with new kit.
2011	Kit Check Error Draw line clamp or return line clamp is not loaded properly.	Touch the <i>Open Clamps button</i> to correctly install the draw and return lines. Touch the <i>Check button</i> to continue. Ensure proper alignment of the cassette and clamps before reloading the cassette.
2020	Leak Detected The system detected a leak in the kit during kit install check.	Press the <i>STOP button</i> and end the procedure. Unload kit and start over with a new kit.
2024	Install Kit Timeout The kit has been loaded on instrument for too long without confirmation of venipuncture.	A dry kit cannot be loaded on instrument for more than 8 hours and a wet kit cannot be loaded on instrument for more than 4 hours. Press the <i>STOP button</i> and end the procedure. Unload kit and start over with a new kit.
2025	Kit Check Error The system is unable to build pressure within the kit.	Touch the <i>Check button</i> and re-install kit. If issue persists, hardware error likely. Call a qualified service representative.
2026	Leak Detected The system detected a leak.	Press the <i>STOP button</i> and end the procedure. Unload kit and start over with a new kit.
2027	Plasma Container off Back Scale The system detected insufficient weight on the back scale.	Place the Plasma container on the back scale. Touch the <i>Check button</i> to continue.
2028	Extra Weight on Back Scale The system detected excessive weight on the back scale.	The Plasma container should be the only weight on the back scale. Remove any other weights and touch the <i>Check button</i> to continue.
2029	Containers Not on Front Scale The system detected insufficient weight on the front scale.	The In-Process and RBC containers should be the only weights on the front scale. Remove any other weights and touch the <i>Check button</i> to continue.
2030	Extra Weight on Front Scale The system detected excessive weight on the front scale.	Place only the In-Process and RBC containers on the front scale. Touch the <i>Check button</i> to continue.
2101	Pressure Out of Range During solutions prime, pressure in the disposable set was outside of the allowable range.	Inspect kit and lines for kinks. Once fixed, touch the <i>Check button</i> . If issue persists, hardware error likely. Call a qualified service representative.

Code	Description	Resolution
2102	Saline Prime Error The saline container is not connected properly or the air detector tubing is misloaded.	Ensure saline line is not kinked. Ensure the saline container has been spiked properly and is hanging on the solution pole on the right side. If there is fluid in the air detector tubing, the tubing should be removed and reinserted in order to be fully loaded in the air detector. Touch the <i>Check button</i> to continue
2103	AC Prime Error The AC container is not connected properly.	Spike the AC container, fix connection, or check for kinks. Touch the <i>Check button</i> to continue.
2104	Saline Container on Scale The saline container is on one of the scales.	Remove the saline container from the weigh scale and hang from solution pole on right side. Touch the <i>Check button</i> to continue.
2105	Saline Prime Error The draw and return lines are interchanged in clamps.	Touch the <i>Open Clamp button</i> . Reload tubing, switching so that inlet line is in its clamp and return line is in its clamp. Touch the <i>Check button</i> to close clamps and continue.
2106	AC Container Not on Scale AC container is not on the scale.	Place the AC container on the front scale. Touch the <i>Check button</i> .
2107	AC Prime Error Flow issue detected with AC container; the AC Container is not loaded properly, there is a kink in the AC line, or the AC container is empty.	Ensure AC container is loaded correctly. Inspect AC line for kinks. If AC level is low, touch the <i>Add Container button</i> to add a new AC container. Touch the <i>Check button</i> to continue.
2108	Saline Prime Error Flow issue detected with saline container; the saline container is not installed properly, there is a kink in the saline line, or the saline container is empty.	Ensure saline container is loaded correctly. Inspect saline line for kinks. If saline level is low, touch the <i>Add Container button</i> to add a new saline container. Touch the <i>Check button</i> to continue.
2109	Air in Line (Prime) Air detector has detected air where there should be liquid; a leak is present, the saline container is empty, or the appropriate line has become misaligned with the air detector.	Check that line is properly aligned with air detector. Check for saline in saline container and touch the <i>Add Container button</i> to replace if low. Inspect for leaks.
2110	Air Detector Failed Air detector has failed.	Hardware failure, call a qualified service representative.

Code	Description	Resolution
2111	Interface Detection Issue During prime, the system found the interface detector (optical sensor) was outside of the allowable range.	Touch the <i>Check button</i> to retry calibration. Check interface detector window and ramp for debris. Clean by soaking one edge of a lens cloth with isopropyl alcohol and wipe the interface detector window and ramp gently, in a circular motion, with the alcohol soaked edge of the lens cloth. Continue to wipe gently until the parts are completely clean. Call a qualified service representative if the issue persists.
2113	Leak Detected During prime, there was a leak in the kit or clamps are not working properly.	Press the STOP <i>button</i> and end the procedure. Unload kit and start over with a new kit.
2114	Incorrect Container on Front Scale An incorrect AC container weight was detected on the front scale.	Ensure that the volume of the AC container loaded is equivalent to the AC container volume displayed on the Hang AC screen/overlay. Touch the <i>Check button</i> to continue.
2115	Saline Prime Error There is an occlusion in the container lines or centrifuge lines.	Check for kinks or damaged tubing within the kit. Ensure that there is nothing disturbing the scale. Touch the <i>Check button</i> to continue.
2117	Platelet Clamp Closed The clamp on the platelet container line is closed and should be open.	Open the platelet line clamp if it is closed. Ensure the platelet container line is not kinked. Touch the <i>Check button</i> to continue.
2118	Plasma Container on Front Scale The system detected the plasma container on the wrong scale.	Place the plasma container on the back scale. Touch the <i>Check button</i> to continue.
2119	IP Container on Back Scale The system detected the in-process container on the wrong scale.	Place the in-process container on the front scale. Touch the <i>Check button</i> to continue.
2120	RBC Container on Back Scale The system detected the RBC container on the wrong scale.	Place the RBC container on the front scale. Touch the <i>Check button</i> to continue.
2121	Plasma Container Not on Scale The system detected the plasma container on the wrong scale.	Place the plasma container on the back scale. Touch the <i>Check button</i> to continue.
2122	IP Container Not on Scale The system detected the in-process container is not on the scale.	Place the in-process container on the front scale. Touch the <i>Check button</i> to continue.
2123	RBC Container Not on Scale The system detected the RBC container is not on the scale.	Place the RBC container on the front scale. Touch the <i>Check button</i> to continue.

Code	Description	Resolution
2124	Clamp Check Error The system was unable to relieve pressure in the kit.	Inspect kit and lines for kinks. Once fixed, touch the <i>Check button</i> . If issue persists, hardware error likely. Call a qualified service representative.
2125	Close Donor Clamp The system can not draw a negative pressure against the manual donor clamp. Verify this clamp is closed.	Check that the donor roller clamp is closed. Inspect for leaks. Touch the <i>Check button</i> to continue. If issue persists, a leak may be present in the kit.
2301	Pressure Out of Range During inter-procedure self-tests, one or more of the pressure sensors were not at atmospheric pressure.	Ensure nothing is resting on pressure sensors on the top panel prior to selecting a procedure. Check the sensors for debris and clean by using compressed air if necessary. If blue hydrophobic filters on donor pressure sensors are wet, replace them. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
2306	Weight on Scale During inter-procedure self-tests, a weight was detected on the front scale.	If a container is on the front scale, remove and touch the <i>Check button</i> . If no containers were on the scale, touch the <i>Calibrate Scale button</i> and follow instructions to calibrate the scale.
2307	Weight on Scale During inter-procedure self-tests, a weight was detected on the back scale.	If a container is on the back scale, remove and touch the <i>Check button</i> . If no containers were on the scale, touch the <i>Calibrate Scale button</i> and follow instructions to calibrate the scale.
2310	Cuff Pressure Out of Range During inter-procedure self-tests, the cuff pressure was outside of atmospheric range.	Check that cuff is properly deflated. Detach and vent any residual pressure. Touch the <i>Check button</i> to continue.
2312	Leak Detector Signal Error During inter-procedure self-tests, the leak detector signal was outside threshold.	Press the <i>STOP button</i> and touch shut down. Power off instrument and call a qualified service representative.
2313	Temperature Out of Range During inter-procedure self-tests, the temperature was outside of atmospheric range.	Ensure that the instrument is within specified operating temperatures. Move instrument from possible heat sources. Touch the <i>Check button</i> . Ensure instrument air filters are clean and unobstructed. Call a qualified service representative if the issue persists.

Code	Description	Resolution
3003	Interface Detector Blocked The interface detector signal was lost.	Check the interface detector window and ramp for debris or smudges. Clean by soaking one edge of a lens cloth with isopropyl alcohol and wipe the interface detector window and ramp gently in a circular motion, with the alcohol soaked edge of the lens cloth. Check the centrifuge pack and plasma container for lipemic plasma. If lipemic plasma is present, and the alert is retriggered, stop and end the procedure.
3004	Scale and Pumps Out of Sync There was a change in the weight on the scale that was inconsistent with the pump direction.	Inspect for kinks in lines. Inspect for pump misalignment. Fix appropriate issue and touch the <i>Check button</i> .
3005	Low Saline Container Volume The saline container volume is too low.	Touch the <i>Add Container button</i> to add new saline container. Touch the <i>Check button</i> to continue.
3006	Low AC Container Volume The AC container volume is too low.	Touch the <i>Add Container button</i> to add new AC container. Remove the old AC container before adding a new AC container. Touch the <i>Check button</i> to continue.
3007	Draw Volume Limit Exceeded The system reached the maximum draw volume limit.	Press <i>STOP button</i> and end procedure. Disconnect donor. Unload kit.
3008	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in the tubing lines. Touch the <i>Check button</i> to continue.
3009	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in the tubing lines. Touch the <i>Check button</i> to continue.
3010	Pressure Out of Range Pressure in the kit was out of range.	Check the kit for kinks in the tubing lines. Touch the <i>Check button</i> to continue.
3011	Container Volume Limit Exceeded The system reached the maximum container volume limit.	Press the <i>STOP button</i> and end procedure. Disconnect donor. Unload kit.
3012	Draw Volume Limit Exceeded The system reached the maximum draw volume limit.	Press the <i>STOP button</i> and end procedure. Disconnect donor. Unload kit.

Code	Description	Resolution
3101	Air in Line (Return) Air was detected in the donor line during fluid return.	The system will automatically perform an air purge. Check that the return line is properly inserted into the air detector. Following the air purge, check the return line for air and touch the <i>Check button</i> if it is enabled. If air remains in line or <i>Check button</i> is still disabled, touch the <i>Air Purge button</i> to perform a second air purge. Repeat if necessary. Once air has been cleared, touch the <i>Check button</i> to continue.
3102	Repeat Cent. Line Blockage A second consecutive centrifuge line occlusion was detected without purging the fistula of stagnant blood. Consider performing saline reinfusion.	Check the kit for kinks in the tubing lines. Touch the <i>Check button</i> to continue. When in collection, touch the <i>Pause button</i> and return saline to the donor in order to flush the fistula.
3103	Centrifuge Line Blockage The system detected a subsequent attempt to clear a centrifuge line occlusion.	Check the kit for kinks in the tubing lines. Touch the <i>Check button</i> to continue.
3104	Blood Prime Error Volume of whole blood drawn during blood prime is more than max volume that can be drawn.	Press the <i>STOP button</i> and end the procedure. Disconnect donor. Unload kit. Donor must be deferred; do not start over with new kit.
3105	Air in Line (Draw) The system detected air in donor inlet line repeatedly during the blood prime.	Check venipuncture and line in air detector. Make sure manual clamp on donor line is open. Check for kinks on donor line. Check that the tubing is properly aligned in the air detector. Touch the <i>Check button</i> to continue.
3508	No Blood Flow Vein pressure fell below the minimum pressure during a draw phase.	Check the inlet line for kinks or occlusions. Check venous access and position of the donor's arm. Instruct donor to squeeze when the cuff inflates. Check that the cuff on the inlet line is inflated to the appropriate pressure. Touch the <i>Decrease Value button</i> to decrease the inlet rate. Check that the donor roller clamp is open. Touch the <i>Check button</i> to continue. If issue persists, consider lowering the inlet rate further.
3601	High Pressure Return Vein pressure exceeded the maximum allowed pressure during a return phase.	Check the donor's return venipuncture site for signs of infiltration. Check the return line for kinks or occlusions. Touch the <i>Decrease Value button</i> to decrease the return rate. Touch the <i>Check button</i> to continue. If the issue persists, consider lowering the return rate further.
4002	Plasma Collection Error Insufficient plasma was collected.	Touch the <i>Check button</i> to continue.

Code	Description	Resolution
4003	Weight on Scale The system has detected an increase in the weight on the scale indicating that the operator has hung the centrifuge pack on one of the scales.	Remove centrifuge pack from scale and touch the <i>Check button</i> to continue.
4004	Weight on Scale The system has detected an increase in the weight on the scale indicating that the operator has hung the platelet container on one of the scales.	Remove platelet container from scale and touch the <i>Check button</i> to continue.
4005	Product Transfer Error The system detected a high pressure in the centrifuge pack.	Unload the centrifuge pack from the centrifuge and hang on the right solution pole. If the pack is already unloaded, check the kit for kinks in the tubing lines.
4007	Product Transfer Error PC Pump and scale are out of sync during product transfer.	Inspect tubes for kinks. Check for an empty plasma container. Rectify and touch the <i>Check button</i> .
8001	Software Error A SW communication error occurred.	Press STOP <i>button</i> and end procedure. Disconnect donor. Unload kit. Shut down the instrument and call a qualified service representative.
9001	Software Error The system was unable to capture the procedure record.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10001	Scale Disturbed The scales sensed a large, sudden change in weight.	Check that nothing is disturbing or moving the containers on the scale. Steady the containers. Touch the <i>Check button</i> to continue. If problem persists, evaluate surroundings for sources of disturbance (construction, etc.).
10002	Power Failure The system detected a loss of AC power.	Plug the instrument's power cord into the outlet and touch the <i>Check button</i> to continue. If instrument power is lost for 5 minutes, the instrument will shutdown.
10003	Incorrect Weight on Scale The weight on the scale is incorrect. Ensure that the correct containers are on the scale.	Check that the correct containers are on the scale. Check the scale for any disturbances.
10004	Air Detector Failed The system detected a fault within the air detector circuit.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.

Code	Description	Resolution
10005	Pump Speed Error The measured pump speed did not match the commanded pump speed.	Check pumps for obstructions. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10006	Pump Speed Error The pump head is failing to turn when a non-zero rate has been commanded.	Check pumps for obstructions. Ensure pumps are all the same height. If a pump is higher than the others, reinstall the pump as described in the Chapter 6 section entitled: "Cleaning and Disinfecting the Instrument." Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10007	Pump Direction Error The pump head is not turning in the commanded direction.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
10008	Pump Speed Error The AC pump is turning faster than the Blood pump.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
10010	Pump Speed Error The AC pump is turning during a return cycle.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
10011	Cassette Error Cassette had been unloaded following a power failure.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
10012	Pump Volume Error Pump head volume is not consistent with pump motor volume.	Check pumps for obstructions. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10013	Auto Recovery Failed Instrument pump speed auto-recovery retries exceeded.	Touch the <i>Check button</i> to continue. If the issue persists, call a qualified service representative.
10014	Clamp Error The clamps did not respond in time.	Touch the <i>Check button</i> to continue. If issue persists, call a qualified service representative.
10015	Pump Volume Error The system detected inconsistent pump volume.	Press the STOP <i>button</i> and end procedure. Disconnect donor. Unload kit.
10110	Centrifuge Speed Error The centrifuge speed was slower than expected.	Check for obstructions. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10111	Centrifuge Speed Error The centrifuge speed was faster than expected.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative if the issue persists.

Code	Description	Resolution
10115	Air Purge Malfunction Either the saline manual clamp is closed, the saline container is empty, or there is a problem with the saline line.	Inspect for kinks. Ensure manual saline clamp is open. Check that the tubing is properly aligned in the air detector. Check saline container level. Touch the <i>Add Container button</i> to add a replacement container. Touch the <i>Check button</i> to continue.
10116	Blood Prime Error The system detected insufficient flow from the donor.	Check the inlet line for kinks or occlusions. Check venous access and position of the donor's arm. Instruct donor to squeeze when the cuff inflates. Check that the cuff on the inlet line is inflated to the appropriate pressure. Check that the donor roller clamp is open. Touch the <i>Check button</i> to continue.
10117	Centrifuge Door Not Closed System has detected that the centrifuge door is not closed.	Close centrifuge door and touch the <i>Check button</i> to continue. If door is already closed, open it and close it again. Touch the <i>Check button</i> to continue.
10118	Centrifuge Door Error Centrifuge door did not successfully lock before timeout was reached.	Open and re-shut centrifuge door. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10120	Pump Speed Error A pump speed issue was detected.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10121	Clamp Over Current The system detected a pump error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10122	Pump Over current The system detected a pump error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10123	Pump Fault The system detected a pump error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10124	Centrifuge Fault The system detected a centrifuge error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10125	Centrifuge Speed Error The system detected a centrifuge speed error.	When the centrifuge has stopped, check the door lock for any obstructions. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.
10126	Pressure Out of Range Pressure in the disposable set was outside the expected range.	Check the kit for kinks in the tubing lines. Check that the platelet line clamp is open. Touch the <i>Check button</i> to continue.

Code	Description	Resolution
11002	Return Pathway Error The system detected that fluid was returned through the inlet line.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11003	Draw Pathway Error The system detected that fluid was drawn through the return line.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11004	Blood Prime Error Insufficient volume was drawn into kit during blood prime. Possible presence of air.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11005	Power Failure The system has detected a power failure.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off instrument and call a qualified service representative.
11006	Hardware Error Commanded safe state was not achieved.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument.
11007	AC Ratio Error The system detected an incorrect measured AC ratio above allowed maximum.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11008	Temperature Out of Range Exceeded centrifuge temperature limit for one of the centrifuge temperature sensors.	Check that nothing is blocking the air inlet filter or the air outlet paths. Remove any obstructions. Ensure that the instrument is within specified operating temperatures. Move instrument from possible heat sources. Touch the <i>Check button</i> . If issue persists, press the STOP <i>button</i> and end the procedure. Disconnect donor.
11010	Cuff Error Cuff pressure has gone above the maximum pressure limit.	Touch the <i>Check button</i> to continue. If the issue persists, contact a qualified service representative.
11011	Cassette Error Cassette unloaded while the donor was connected.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.
11012	AC Ratio Error The system detected a measured AC ratio below the safety minimum.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11013	Cuff Error The pressure cuff was inflated during a return cycle.	Check the cuff tubing connection. Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.

Code	Description	Resolution
11014	AC Ratio Error AC ratio error was detected during draw.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11015	Air Purge Malfunction Insufficient air purge volume pumped.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11017	Hardware Error A software error preventing HW safe state was detected.	Disconnect donor. Power off the instrument.
11018	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Call a qualified service representative.
11020	Air Detector Failed The system detected a fault within the air detector circuit.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.
11021	Pressure Out of Range The system detected a fault with a pressure sensor.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.
11022	Centrifuge Leak Detected The system detected a leak in the centrifuge.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11023	Leak Detector Signal Error The system detected a fault with the leak detector.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.
11024	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11025	Return Pathway Error The system detected that fluid was pumped into the saline container.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11026	Temperature Out of Range Both temperature sensors exceeded centrifuge temperature limit.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11027	Blood Prime Error Saline clamp open while return line is primed.	Press the STOP <i>button</i> and end the procedure. Disconnect donor.
11028	Hardware Error The inlet or return line clamp is open during an alert while the donor is connected.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument.

Code	Description	Resolution
11029	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative
11030	Centrifuge Speed Error The system detected an error with the centrifuge door lock or door lock sensor.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Call a qualified service representative.
11031	Software Error The system detected a software error.	Press STOP <i>button</i> and end procedure. Disconnect donor. Shut down the instrument and call a qualified service representative.
11032	Centrifuge Speed Error The centrifuge spin time reached the maximum limit.	Press STOP <i>button</i> and end procedure. Disconnect donor. Shut down the instrument and call a qualified service representative.
11033	Cassette Error The system detected a cassette motor error.	Press STOP <i>button</i> and end procedure. Disconnect donor. Unload kit. Call a qualified service representative if the issue persists.
11070	Cuff Error The system detected a cuff inflation error.	Cuff will vent pressure. Touch the <i>Check button</i> to continue. If alert occurred before venipuncture, perform venipuncture to avoid second occurrence of alert.
14011	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Contact a qualified service representative.
14012	Software Error The system detected a software safety check failure.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Contact a qualified service representative.
14013	Software Error The system detected a software safety check failure.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Contact a qualified service representative.
15001	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Contact a qualified service representative.
15002	Software Error The system detected a software error.	Press the STOP <i>button</i> and end the procedure. Disconnect donor. Power off the instrument. Contact a qualified service representative.
15003	Software Error The system detected a software error.	Press STOP <i>button</i> and end procedure. Disconnect donor. Shut down the instrument and call a qualified service representative.
15004	Software Error The system detected a software error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.

Code	Description	Resolution
15005	Software Error The system detected a software error.	Touch the <i>Check button</i> to continue. Call a qualified service representative if the issue persists.

Section 5.4 Unloading the Kit During Special Circumstances

This section describes instructions on how to unload a kit during special circumstances. For example, the instrument is shut down due to a non-recoverable alert, or a power outage occurs during a procedure. Also, it is possible to unload the kit if the operator accidentally touches the *Check button* after a procedure was completed, but before the kit was unloaded.

Non-Recoverable Alert Occurs or Kit Left On After Procedure Ends

If the operator shut down the instrument during a procedure because of a non-recoverable alert, or accidentally touched the *Check button* after a procedure ended, but before unloading the kit, perform the following steps.

1. Power cycle the instrument.



Note:

If the operator touches the *Check button* after a procedure was completed, but before the kit is unloaded, the instrument does not have to be power cycled.

2. On the Home screen, touch the *Information button*. In the *Instrument Settings tab*, touch the *Unload Cassette button*. The cassette base rises, and the inlet and return clamps open.
3. Remove the kit from the instrument.

Unloading a Kit Due to a Power Outage



Warning:

Do not actuate the centrifuge door lock override and open the centrifuge door while the centrifuge is spinning.

1. Press the blue CASSETTE BYPASS *button* on the back of the instrument within 5 minutes of losing power to raise the cassette.

2. The blue CASSETTE BYPASS *button* is located on the back of the instrument next to the pressure cuff connection site. This button is only active while the instrument is on battery power.
3. Unload the pump tubing and the tubing in the inlet and return clamps.
4. Insert a small tool, such as a 2 mm hex tool (or Allen wrench) or narrow screwdriver, into the circular slot underneath the centrifuge door latch (centrifuge door lock override). Press forward until the centrifuge door clicks unlocked.
5. While the tool is still in the slot, open the centrifuge door. Unload the centrifuge pack and umbilicus.

Section 5.5 (Optional) Manual Reinfusion

The purpose of this section is to provide instructions for performing manual reinfusion in the event the instrument was unable to perform this function automatically. This procedure should be used if automated reinfusion did not occur or cannot be performed during the procedure.

If an alert ended the procedure mid-collection and the option to perform reinfusion was not provided, the operator should not perform manual reinfusion. The residual fluids in the kit may be unsafe to return to the donor due to infusion of air or anticoagulant.

1. Close all the clamps on the kit including the donor line roller clamp. Remember to close the slide clamps on the platelet storage containers.
2. Place a hemostat on the return line below the blue slide clamp, and place another hemostat on the purple draw line.
3. Place one hemostat on each of the tubes coming from the top-left of the cassette, except the red-striped AC line. These are the in-process container line, the RBC container line, and the plasma container line.
4. Place a hemostat on each of the following lines as it emerges from the cassette:
 - a. Left-most line (PRP out)
 - b. Middle line (PPP)
 - c. Right-most line (PRP in)

5. At the junction where the draw, return and AC lines connect near the donor line roller clamp, separate the AC line from the draw and return lines. Seal the AC line near the junction.
6. Remove the in-process container from the red hanger on the front left scale and hang it over the side of the instrument, making sure the tubing is not kinked.



Note:

If manual product transfer will be performed after manual reinfusion, the operator should seal the platelet storage containers and plasma container before unloading the cassette. Follow the instructions in the section entitled: “Manual Product Transfer.”

7. Removal of the kit from the instrument can occur in one of the following three ways. Proceed to step 11 after the cassette is unloaded.
 - Removing the kit when prompted at the end of the procedure, as in step 8.
 - Touching the *Unload Cassette button* available outside of a procedure through the *Information button* by following the instructions in step 9.
 - Pressing the CASSETTE BYPASS *button* on the back of the instrument during a power loss scenario by following the instructions in step 10.
8. These instructions apply when the instrument instructs the operator to remove the kit at the end of the procedure.
 - a. DO NOT seal the saline line.
 - b. Open the centrifuge door and remove the centrifuge pack.
 - c. Close the centrifuge door and place the centrifuge pack on the cassette nest.
 - d. Go to step 11.
9. This scenario is entered if an alert required the operator to shut down the instrument immediately.
 - a. Power cycle the instrument and follow the instructions for the STOP *button* test and alert signals test.
 - b. On the Home screen, touch the *Information button*, touch the *Instrument Settings tab*, and touch the *Unload Cassette button*. The instrument will unload the cassette.

- c. Open the centrifuge door and remove the centrifuge pack.
 - d. Close the centrifuge door and place the centrifuge pack on the cassette nest.
 - e. Go to step 11.
10. These instructions apply to a power fail scenario:
- a. Press the CASSETTE BYPASS *button* on the back of the instrument. The instrument will raise the cassette but not unload the pump tubings.
 - b. Unload the pump tubings and unload the centrifuge pack by following the instructions in the section: "Unloading the Kit During Special Circumstances."
11. Move the hemostat on the return line from below the blue return clamp to below the air detector. Open the roller clamp on the return line to establish a saline drip to the donor.



Warning:

Continuously monitor the donor line for air bubbles during manual reinfusion of disposable contents or saline and immediately clamp the line if air is observed.

12. Hang the separation chamber and the RBC container on the front left scale hook. Remove hemostats from the RBC container and in-process container. Allow both the separation chamber and the RBC container to drain into the in-process container.
13. When the RBC container and separation chamber are empty, place a hemostat on the RBC container.
14. Place a hemostat on the saline line below the saline filter and move the hemostat from below the air detector to below the blue return clamp.
15. Place the in-process container on the front left scale hook.
16. Remove the clamp from the return line below the blue return clamp and drain the container into the donor line by gravity.



Warning:

Continuously monitor the donor line for air bubbles during manual reinfusion of disposable contents or saline and immediately clamp the line if air is detected.

17. When the in-process container has finished draining, close the roller clamp on the donor line and disconnect the donor.

Section 5.6 (Optional) Manual Product Transfer

This section describes how to perform manual product transfer for a single needle platelet collection procedure. Manual product transfer should only be used if automated platelet transfer did not occur or cannot be performed. The platelet product should be counted for WBC if manual product transfer is performed.

1. Close all clamps on the kit.
2. Disconnect the donor from the instrument.
3. Before unloading the cassette, seal all five centrifuge lines close to the cassette.
4. Seal near the cassette and remove the plasma container and platelet storage container from the kit. If the kit has two platelet storage containers, seal below the junction of the two platelet storage containers, close to the cassette.
5. Remove the kit. Follow the instructions in the section entitled: "To Remove the Disposable Kit from the Instrument."
6. Place a hemostat on the plasma container and close the slide clamp on each platelet storage container.
7. Sterile-connect the plasma container to the middle centrifuge pack line (PPP line) per institutional SOPs.
8. Sterile-connect the platelet storage container line to the right-most line (PRP in) per institutional SOPs.



Note: The location of the centrifuge lines is defined by how the centrifuge lines are attached to the cassette.

9. Hang the plasma container on the front right (white) hook and place the centrifuge pack on the cassette base.
10. Remove the hemostat from the plasma container. Plasma should now flow into the collection chamber of the centrifuge pack.
11. When the collection chamber is half full, place a hemostat on the plasma container.
12. Resuspend the platelets as directed in the Chapter 4 section entitled: *To Resuspend the Platelets*.
13. Determine the minimum volume of storage plasma for the platelet product by following the instructions in the Appendix section entitled:

“Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC.”

14. Remove the hemostat from the plasma container. Plasma should now flow into the collection chamber of the centrifuge pack.
15. When the centrifuge pack is full, place a hemostat on the plasma container and place the centrifuge pack on the front right hook.
16. Open the clamp on one of the storage containers. Plasma should now flow from the centrifuge pack to the storage container.
17. Place the storage container on a scale. Tare the scale and monitor container weight to determine the product volume.
18. When the collection chamber is empty, clamp the storage container and remove the centrifuge pack from the hook.
19. Repeat steps 14–18 until the desired volume of plasma (as determined in step 13) has been transferred to the storage container.
20. To remove air from the platelet storage container, roll the filled storage container from the bottom up to push air towards the port.
21. Carefully open the clamp on the filled platelet storage container and purge the air into the collection chamber until the platelets reach the bottom of the ports.
22. Close the clamp on the filled storage container.
23. Make three (3) hermetic seals directly on the line leading to the collection chamber. Ensure that adequate line length remains associated with the storage container to permit subsequent sterile connection needs.
24. Cut between two (2) seals, leaving two (2) seals on the line leading to the storage container.

Section 5.7 Considerations for Connecting a New Needle Mid-Procedure

In the unlikely event that the needle must be replaced after a procedure has begun, the following information must be taken into consideration before proceeding with the instructions below.



Warning:

Modification of the closed or functionally closed system or replacement of the needle under open conditions negates any claims to a closed system, unless sterile connection methods are employed. If closed or functionally closed system conditions have been maintained, the platelet product may be stored for up to 5 days.



Warning:

Improper priming of the new needle may result in infusion of air into the donor.

1. Touch the *Pause button*, close the roller clamp on the donor line, clamp the fistula, and disconnect the donor according to institutional SOPs.
2. Sterile-connect a new needle (17 gauge thin-walled apheresis needle) per institutional SOPs. Ensure that the new tubing is approximately the same length as the original tubing.
3. To prime the new needle, touch the *Return Saline button* on the screen. Open the roller clamp on the donor line and prime the line.
4. While the Administering Saline overlay is still displayed, close the donor line roller clamp.
5. Perform venipuncture per your institutional SOPs.
6. Touch the *Cancel button* on the Administering Saline overlay to stop saline flow. Open the roller clamp on the donor line. The instrument draws anticoagulated blood to re-prime the donor line.
7. Touch the *Resume Procedure button* to proceed with the procedure.

Section 5.8 Bacterial Sampling

1. Sampling should occur after the product transfer has been completed and the platelet storage container has been sealed and removed from the kit.
2. Sterile-connect the bacterial sampling setup to the platelet storage container tubing per institutional SOPs.
3. Bacterial testing should be completed per the manufacturer's instructions.

Chapter 6 - Maintenance and Cleaning

Section 6.1 Instrument Installation

Installation of the AmiCORE instrument must be performed by authorized service personnel, following the instructions provided in the AmiCORE Service Manual.

Section 6.2 Instrument Relocation



Caution:

The instrument must be used and stored in the proper operating environment (temperature, humidity, altitude, and surface incline requirements).



Caution:

When unplugging the power cord from the mains supply outlet, grasp the power cord at the plug. Do not pull the power cord wire.

When moving or relocating an instrument within a center, follow these instructions:

- Remove all disposable sets, solutions, and collection containers from the instrument.
- Turn the instrument off and disconnect the power cord from the instrument. Disconnect the ethernet cable, if applicable.
- Unlock the wheels before attempting to move or relocate the instrument.
- Rotate the weight scale and solution hook assemblies backward, until they click into place.
- Always use both blue knobs when moving or relocating the instrument.

- To prevent over balancing, do not push on instrument surfaces that are labeled “no pushing”.
- Do not move the instrument over large thresholds, objects on the floor, or other obstacles that may cause the instrument to tip over.
- Re-lock the wheels once the instrument is placed in the desired location.
- Rotate the weight scale and solution hook assemblies forward, until they click into position.

Section 6.3 Instrument Routine and Preventive Maintenance

Non-service personnel can replace only the following parts:

- Air filter
- Pump head
- Power cord
- Pressure cuff



Warning:

Do not perform maintenance tasks while a procedure is in progress.



Caution:

Use only replacement components and accessories authorized by the instrument manufacturer.



Caution:

Only authorized service personnel should remove any of the instrument panels.

Routine Maintenance

Routine maintenance and cleaning should be performed by authorized service personnel and documented according to institutional SOPs. If any discrepancies are found during routine maintenance, service may be required. Contact authorized service personnel or your local service representative as necessary to make repairs to the instrument.

Weekly

Clean the interface detector system.

Monthly

1. Clean the touch screen.
2. Clean centrifuge compartment walls.
3. Clean the cassette base.

Every Six Months

Replace the air filter.

Periodic Cleaning

Clean the pump housing and pump rotors periodically, using a mild detergent. See steps for these tasks later in this chapter.

Per Institutional SOPs

Perform the scale QC check. If the check indicates that scale calibration is necessary, perform the scale calibration.

Preventive Maintenance

Preventive maintenance items must be performed by authorized service personnel at least every twelve months. See the AmiCORE Service Manual for detailed instructions.

Replace the Air Inlet Filter

1. Remove the air inlet cover (a Phillips screwdriver is required). The cover is located on the front panel, below the storage compartment.
2. Remove the dirty air filter from the cover and replace with a clean air filter.
3. Ensure the air filter is properly aligned within the air inlet cover and then tighten the screw into place.

Cleaning the Interface Detector System

The interface detector system should be cleaned once every seven $\pm$ two days, or if it becomes dirty or smudged. Proper protective clothing should be worn to prevent contact with biohazardous materials and cleaning solutions. The interface detector system consists of the interface detector, the window, and the ramp.

To Clean the Interface Detector and Lens

1. Open the centrifuge door and pull back the zero omega arm.
2. The interface detector lens is located on the centrifuge compartment at the 11 o'clock position. It may be necessary to rotate the centrifuge to access the lens.

3. Soak one edge of a lens cloth with 70% isopropyl alcohol and wipe the interface detector lens gently, in a circular motion, with the alcohol soaked edge of the lens cloth. Continue to wipe gently until the lens is completely clean. Alternatively, an OPTICPAD CT811 optical surface cleaner may be used to clean the lens. (OPTICPAD CT811 is available from Advantus Corporation, Jacksonville, FL 32223, USA, to order go to Harmonycr.com or call 1-800-899-1255).



Caution:

Do not use any paper-based products on the interface detector lens, window, or ramp. Paper-based lens cloths can cause scratches which may potentially interfere with the separation characteristics of the AmiCORE Apheresis System.

4. Do not leave excess fluid on the lens. The interface detector lens should look clean, dry, and unscratched.
5. Continue by cleaning the window and ramp.

To Clean the Window and Ramp

1. Depress the gray and blue spool release buttons and remove the spool from the spool holder. Be careful to place the spool in a safe place.
2. Locate the window on the outside of the spool holder.
3. Locate the ramp on the inside of the spool holder. The ramp is a clear piece of plastic attached to the window.
4. Soak one edge of a lens cloth with isopropyl alcohol and wipe the window and ramp gently, in a circular motion, with the alcohol soaked edge of the lens cloth. Continue to wipe gently until the window and ramp are completely clean. Alternatively, an OPTICPAD CT811 optical surface cleaner may be used to clean the window and ramp.



Caution:

Do not use any paper-based products on the interface detector lens, window, or ramp. Paper-based lens cloths can cause scratches which may potentially interfere with the separation characteristics of the AmiCORE Apheresis System.

5. Do not leave excess fluid on the window or ramp. The window and ramp should look clean, dry, and unscratched.
6. Install the spool back into the spool holder by compressing the gray and blue spool release buttons and push the spool back into the spool holder.
7. Close the centrifuge and the zero omega arm.

8. Close the centrifuge door.

Scale QC Check and Calibration

Scale QC Check

The scale QC check should be completed at the frequency dictated by institutional SOPs or if the instrument is moved to a new location. Checks may only be performed outside of a procedure. One 1000 g calibrated weight (or a combination of equivalent smaller weights) is required for the scale QC check. Follow prompts to check each scale.



Caution:

If the instrument has been moved or relocated, perform weigh scale checks before starting a procedure to ensure scale accuracy.



Scale Calibration
Button



Scale Pass Icon



Scale Retry
Button

1. Touch the *Information button*.
2. Touch the *Instrument Settings tab* and then touch the *Scale Calibration button*.
3. Remove weight from the front scale. As each check is completed successfully, a *Scale Pass icon* appears in the appropriate area for that weight. Otherwise, a *Scale Retry button* appears.
4. When prompted, hang a 1000 g weight on the front scale and allow enough time for the instrument to perform the check.



Note:

All weight must be centered directly under the scale and should not contact any other part of the instrument. Do not touch or move the scale during the check.

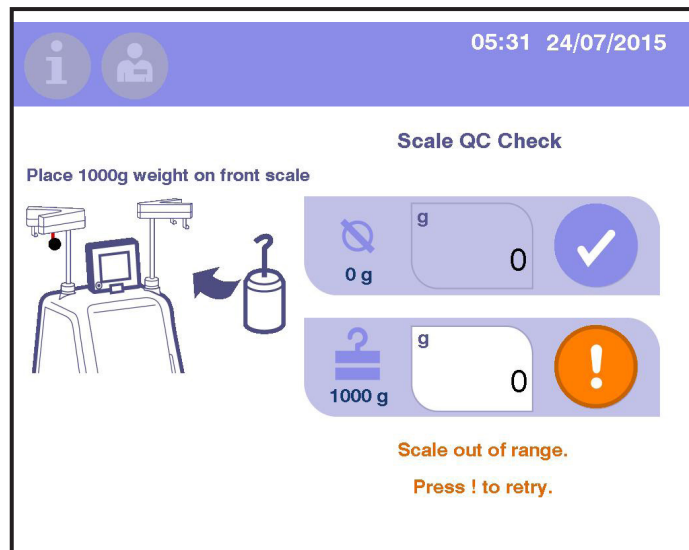


**6.1 Typical Scale QC Check Screen –
Remove Weight Prompt (from Front Scale)**

5. Once the front scale checks pass, remove the weight from the front scale.
6. Repeat steps 3–5 for the back scale.

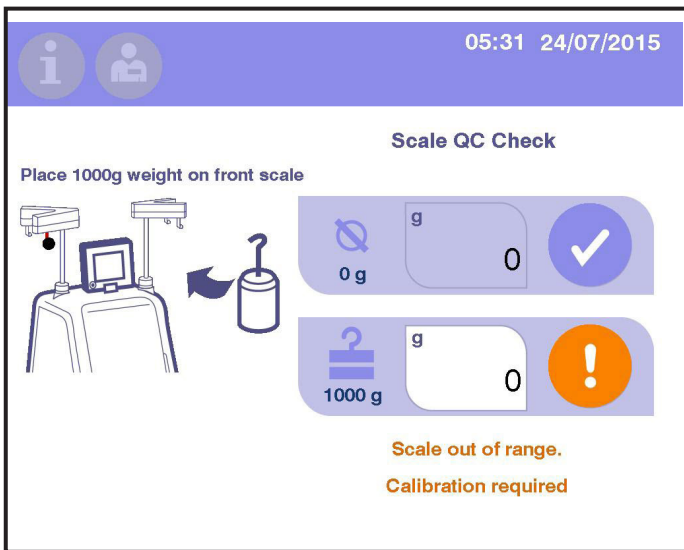
Once the scale QC check is successful, the touch screen displays the Home screen.

If a scale check does not pass, a *Scale Retry button* indicates which scale failed. Check that the weight on the scale matches the *Scale Check prompt*. Touch the *Scale Retry button* and follow the prompts.



6.2 Typical Scale Check Retry Screen

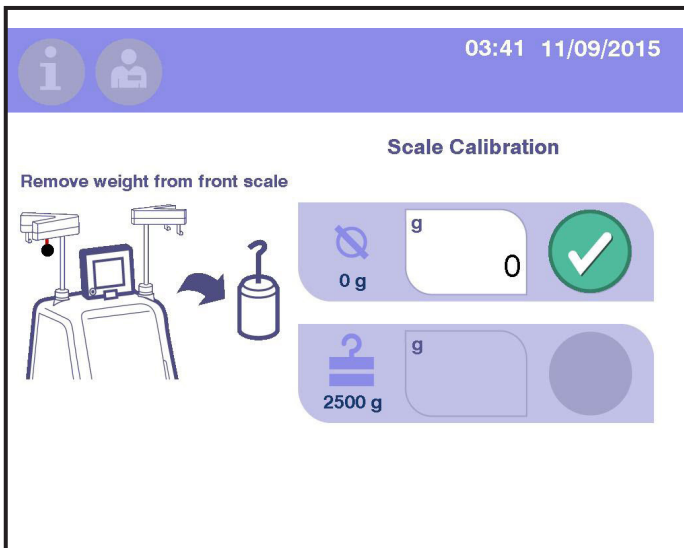
If a scale check does not pass for a second time, a *Scale Fail icon* indicates which scale check failed. The touch screen prompts the operator to perform a scale calibration.



6.3 Typical Scale Check Failed Screen

Scale Calibration

Scale calibration must be performed after a scale check failure or when the *Calibrate Scale button* is touched on a Scale Alert screen. One 2500 g calibrated weight (or a combination of equivalent smaller weights) is required for scale calibration.



**6.4 Typical Scale Calibration Screen –
Remove Weight Prompt (from Front Scale)**

1. Remove all weight from the scale indicated by the prompt.
2. Touch the *Check button* to verify that all weight has been removed from the scale. A *Scale Pass icon* is displayed when the check passes.

3. Hang a 2500 g weight on the scale indicated by the prompt.
4. Touch the *Check button* to verify that the 2500 g weight has been added to the scale. A *Scale Pass icon* is displayed when the check passes.

Once the scale has been calibrated, the Scale Check screen displays the *Remove Weight prompt*, indicating the scale needs to go through the scale QC check.

Section 6.4 Cleaning and Disinfecting the Instrument

As with any equipment used in the processing of human blood, there is the possibility of contamination of the equipment with potentially hazardous and infectious substances. It should be assumed that any equipment used is contaminated and the appropriate precautions observed. The instrument manufacture recommends to first clean and then disinfect components.

Components listed in this section may be cleaned and disinfected on an as-needed basis and recorded according to institutional SOPs.



Warning:

Clean and disinfect blood spills immediately. Treat all spills as potential biohazards. Precautions should be taken at all times to prevent exposure to and transmission of blood borne pathogens.



Warning:

Do not perform maintenance tasks while a procedure is in progress.



Caution:

Use only replacement components and accessories authorized by the instrument manufacturer.



Caution:

Power off the AmiCORE instrument and unplug from the power source before replacing the air filter and air inlet cover, disassembling pump assembly components, or replacing the power cord.



Caution:

Allow sanitizing agent to dry before installing the disposable set.



Caution:

Do not use solvents or abrasive cleaners on the instrument.

Suggested Equipment and Supplies

Equipment

- Rubber protective apron or disposable laboratory coat
- Protective gloves
- Splash resistant goggles

Miscellaneous Utensils

- Spray bottles for solutions, from 0.25 to 1 L, clearly labeled with contents
- Cloths or paper towels that are soft, absorbent, and lint-free, to be used for cleaning and drying
- Swabs, lint-free
- Waste containers marked for biohazardous materials
- Containers for soaking and rinsing pump heads

Instrument Manufacturer Recommended Disinfecting Agents

- Freka®-NOL manufactured by Fresenius Medical Care GmbH
- Hepacide Quat® II Virucidal Disinfectant EPA Reg. No. 5741-18 manufactured by Spartan Chemical Company, Inc.
- Sani-Cloth® Plus Super Germicidal Disposable Cloth EPA Reg. No. 9480-6 manufactured by PDI
- Sporidicin® Disinfecting Solution EPA Reg. No. 8383-3 manufactured by Contec, Inc.
- Virex® Tb Disinfectant Cleaner EPA Reg. 70627-2 manufactured by Johnson Diversey, Inc.
- A freshly prepared bleach solution (1 part bleach to 9 parts water) -or- 70% (v/v) aqueous isopropyl alcohol can be used.

Cleaning Solutions

- One of the following disinfectant solutions:
 - a. Freshly prepared bleach solution consisting of one part household bleach to nine parts water, -or-

b. Isopropyl alcohol, 70% (v/v) aqueous solution.

- Mild ammonia window cleaner (for cleaning exterior components)
- Mild soap/detergent



Note: For routine cleaning use a mild soap/detergent.



Note: Follow institutional SOPs when preparing cleaning solutions.

Rinsing Solutions

- Distilled water
- Tap water

Cleaning the Exterior Components



Note: Many conventional cleaning solutions do not disinfect surfaces.



Note: Treat exterior components as potential biohazards and disinfect the component after cleaning.

Perform the following steps:

1. Only use a damp lint-free cloth.
2. Only use cleaning solutions recommended from the instrument manufacturer. The instrument manufacturer recommends the following cleaning solutions for exterior components:
 - A mild detergent in tap water
 - A small amount of mild ammonia window cleaner
3. For detailed instructions on how to clean an instrument exterior component, refer to the desired component listed.

Disinfecting the Exterior Components



Note: Many disinfectants are not effective on dirt covered surfaces, so the instrument surface **must be** cleaned first. For detailed information on how to clean instrument exterior components, see “Cleaning the Exterior Components”.



Note: Treat exterior components as potential biohazards and clean the component before disinfecting. For

detailed cleaning instructions, see “Cleaning the Exterior Components”.

Perform the following steps:

1. Only use a damp lint-free cloth.
2. Only use recommended manufacturer disinfecting agents. Follow disinfectant application instructions on how to apply to cloth and surface.

Cleaning and Disinfecting the Air Detector

The air detector assembly should be free of debris to provide optimal transmission and reception of the ultrasonic signal. Clean the air detector with an appropriate sanitizing agent. Wipe the sensor dry with a lint-free swab or cloth.



Warning:

The air detector housing and tubing channel must be dry for proper detection of air.



Caution:

Allow sanitizing agent to dry before installing the disposable set.

Cleaning and Disinfecting the Touch Screen

Clean the touch screen with a small amount of mild ammonia window cleaner sprayed onto a lint-free cloth. Do not spray cleaner directly on the screen, as damage may result. Allow the screen to air dry.



Caution:

Do not use bleach to disinfect the spool holder, clamps, optical sensor, interface detector window, or pump heads. Using bleach on these parts may cause damage. Use only 70% isopropyl alcohol to disinfect these parts.

Cleaning and Disinfecting the Clamps

1. Spray cleaning solution carefully into the clamps. Use lint-free cloths or towels to keep excessive fluid from seeping in and around the cassette base or into the interior of the instrument.



Caution:

Do not use bleach to disinfect the spool holder, clamps, optical sensor, interface detector window, or pump heads. Using bleach on these parts may cause damage. Use only 70% isopropyl alcohol to disinfect these parts.

2. Wipe with lint-free swabs if needed.
3. Rinse with distilled or tap water.

4. Dry thoroughly with a lint-free cloth or towel.

Cleaning and Disinfecting the Cassette Base



Caution: Spraying fluids on the cassette base will cause corrosion of mechanical parts. Corroded parts will have to be replaced.

If minor cleaning is needed:

1. Wet a lint-free cloth or towel lightly with cleaning solution.
2. Wipe the base gently.
3. Wet a lint-free cloth lightly with distilled or tap water.
4. Wipe the base gently.
5. Dry gently with a lint-free cloth or towel.
6. If needed, remove the cassette base to gain access to the cassette plate assembly. Repeat steps 1–5, replace base with cassette plate assembly.



Caution: The AmiCORE cassette plate assembly contains four pressure sensors. Two pressure sensors have a black rubber surface and two have a blue vent. Do not press on the black rubber surface, or damage to the pressure sensors will occur. If the blue vents become wet, they must be replaced.

Cleaning and Disinfecting the Pumps

Removing the Pumps

Grasp the top of the pump head, push it in, and turn it counterclockwise to remove it from the pump shaft.

Cleaning and Disinfecting a Pump

1. Fill a container with enough 70% isopropyl alcohol or mild soap/detergent to cover the pump head.



Caution: Do not use bleach to disinfect the spool holder, clamps, optical sensor, interface detector window, or pump heads. Using bleach on these parts may cause damage. Use only 70% isopropyl alcohol to disinfect these parts.

2. Allow the pump head to soak for at least ten minutes.

3. Rinse the pump head thoroughly with warm tap water and dry with a lint-free cloth or towel. Let the pump head air-dry or use canned air to dry it.
4. Repeat steps 1–3 as many times as required to thoroughly clean and dry the pump head.
5. Repeat the cleaning process if the rocker arms will not move freely. If cleaning has been repeated and the arms still will not move, contact a qualified service representative for assistance.

Cleaning and Disinfecting the Pump Head Recess

1. Remove any fluid or debris accumulated in the pump head recess. In severe cases, alternating between alcohol wipes and paper products may be required.
2. Disinfect the pump head recess using cloths or towels to keep excessive fluid from seeping around the cassette base or into the interior of the instrument.

Reinstalling the Pump Head

Align the pump head with the pump shaft. Push the pump head in and turn it clockwise until it stops. Release the pump head.

Cleaning and Disinfecting the Centrifuge

The brown strip at the back of the centrifuge compartment is the leak detector. When it detects fluid, the centrifuge stops and all other parts of the AmiCORE Apheresis System are put into a safe condition. It must be dried thoroughly after cleaning in order to begin a new procedure.

1. Turn the instrument's power switch to the OFF position and unplug the AmiCORE Apheresis System from the power source.
2. Open the centrifuge door.



Note:

If the door is locked, locate the small opening at the bottom of the centrifuge door, near the bottom of the handle assembly. Using a small screw driver or Allen wrench, push forward until the locking pin releases from the door lock assembly.

3. Spray cleaning solution onto a lint-free cloth or towel and wipe blood from surfaces.
4. Spray the centrifuge compartment sparingly with disinfecting solution. Bleach may be used in the centrifuge compartment, but make sure it does not come in contact with the interface detector lens, spool holder, clamps, optical sensor, interface detector window, or pump heads.



Caution:

Do not use bleach to disinfect the spool holder, clamps, optical sensor, interface detector window, or pump heads. Using bleach on these parts may cause damage. Use only 70% isopropyl alcohol to disinfect these parts.



Caution:

Be careful when cleaning the centrifuge compartment, so you do not damage the interface detector lens and leak detector.

5. Allow disinfecting solutions to remain in contact with surfaces for at least ten minutes.
6. Thoroughly rinse parts with water. Bleach solution may cause severe corrosion if parts are not rinsed thoroughly.

Cleaning and Disinfecting the Spool

The spool may be removed and cleaned with any of the cleaning solutions listed in this section.

1. Open the centrifuge door and unlock the zero omega arm.
2. Turn the centrifuge until the blue centrifuge release buttons are at the 12 o'clock position. Squeeze the centrifuge release buttons and open the centrifuge.
3. Depress the gray and blue spool release buttons and remove the spool from the spool holder. Be careful to place the spool in a safe place.
4. Spray cleaning solution carefully on the spool. Do not immerse the spool in cleaning solution. Use care when cleaning around the reflective tape on the spool.
5. Rinse with distilled or tap water. Use care when rinsing around the reflective tape.
6. Dry the spool using a lint free cloth or towel.
7. Continue by cleaning the spool holder and centrifuge compartment.

Cleaning and Disinfecting the Spool Holder

1. Clean the metal surfaces of the spool holder with soap and water.
2. Refer to instructions earlier in this manual to clean the window and ramp.

Section 6.5 Part Replacement



Warning:

Only authorized service representatives should access the inside of the instrument.



Warning:

Do not perform maintenance tasks while a procedure is in progress.



Caution:

Use only replacement components and accessories authorized by the instrument manufacturer.



Caution:

Shut off the AmiCORE instrument and unplug from the power source before replacing the air filter and air inlet cover, removing the pump heads, or replacing the power cord.



Caution:

When maintaining the instrument, use only replacement components, cables and accessories authorized by the instrument manufacturer and be sure to replace the cassette base and all shields, covers, and screws in their exact locations as described in the AmiCORE Apheresis System Service Manual. Failure to do so may result in increased emissions or decreased immunity of the instrument.



Caution:

If the instrument has been moved or relocated, perform weigh scale checks before starting an apheresis procedure to ensure scale accuracy.

The following instrument parts can be replaced by non-service personnel:

- Air filter and air inlet cover
- Pump head assembly components
- Power cord
- Pressure cuff

Air Filter and Air Inlet Cover

The air filter and air inlet cover can be removed and replaced per the instructions in the previous section: “Instrument Routine and Preventive Maintenance.”

Pump Head Assembly

The pump head can be removed and replaced as described in the previous section: “Cleaning and Disinfecting the Instrument.”

Power Cord

1. Shut the instrument off. For detailed instructions, see the Chapter 3 section entitled: “Powering Off the Instrument.”
2. Remove the power cord plug from the power source receptacle.



Caution:

When unplugging the power cord from the mains supply outlet, grasp the power cord at the plug. Do not pull the power cord wire.

3. Remove the power cord from the instrument receptacle.
4. Attach replacement power cord to instrument receptacle and plug the power cord plug into the power source.

Pressure Cuff

1. Locate pressure cuff fitting on the back lower right of the instrument. Remove fitting from Luer by turning counterclockwise.
2. Install replacement pressure cuff to Luer by turning fitting clockwise. Ensure that cuff connector is locked to the instrument Luer.

Section 6.6 Instrument and Parts Disposal

Instrument (Product) Disposal

According to EU Directive WEEE 2012/19/EC, for instrument (product) disposal, please ensure the following:

- Do not dispose of this product as unsorted municipal waste.
- Collect this product separately.
- Use the collection and return systems available to you.

Parts Disposal

Certain components of the instrument require appropriate disposal.

These include:

- Battery
- Electrical and electronic parts
- Disposable set

Battery Disposal



Caution:

The backup battery should be properly handled and disposed of to prevent exposure to harmful chemicals.

The AmiCORE Apheresis System contains a sealed, lead acid rechargeable battery. The battery can only be replaced by an authorized service technician. This type of battery should be recycled after it has reached the end of its useful life. An automotive store or a local waste agency may accept these batteries for recycling. These batteries should not be disposed of in regular, unsorted, municipal waste. Consult institutional SOPs or contact your local service representative if you have any questions about battery recycling.

Electrical and Electronic Parts Disposal

After replacement of an electrical or electronic part (such as a pump, clamp, or printed circuit board), the authorized service technician should determine if the replaced part is under warranty. If under warranty, the part should be returned to manufacturer. If not under warranty, the part should be taken to a local Recycling Center that accepts electrical or electronic parts.

Disposable Goods Set Disposal

After use, the disposable set should be disposed of in an appropriate biohazard material container according to local regulations.

Section 6.7 Product Revalidation Following a Service Call

Whenever the AmiCORE Apheresis System requires service, the qualified service representative will document the service performed on the Service Order and on the AmiCORE Field Repair Test Matrix Checklist. A copy of both documents will be provided to the responsible party at the facility. Based on the information below, the Repair Test Matrix Checklist will indicate whether a product revalidation is suggested or not.

Product revalidation is suggested when the following parts are replaced, as noted in the Service Order.

- Centrifuge Assembly
- Spool Holder (Bowl) or Spool Holder Ramp
- Interface Detector
- Interface Controller Board

Product revalidation may be suggested when the following development defaults are adjusted, as noted in the Service Order.

- Ramp Detection Default 1–3
- Setpoint

Section 6.8 **Service**

In the U.S., for service, or to identify authorized service personnel or your local service representative, call: 1-800-448-5299. Outside the U.S., refer to the contact information on the back cover of this manual. For future reference you may want to record the following information:

In (Country) _____

Call: _____

Tel.: _____

Fax: _____

When you contact us for service, we will require the following information from you:

- Instrument serial number
- Account number and name
- Description of the problem
- If applicable, all alert and data 1/data 2 codes. For detailed information about these codes and their location, see “Alerts” in Chapter 5.

Section 6.9 Warranty Statement

Ask your customer service representative for a copy of the specific written warranty information applicable to your region.

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Chapter 7 - System Specifications

Section 7.1 Physical Specifications

The instrument is made of painted structural foam panel, painted steel and aluminum in white, blue and black. The following table provides the approximate physical specifications for the AmiCORE Apheresis System.

	US Customary	Metric
Height (to top of solution pole)	57 in	145 cm
Width	28 in	71 cm
Depth	36 in	91 cm
Weight	245 lb	111 kg
Floor space required	1,008 in ²	6,499 cm ²

Section 7.2 Instrument Specifications

Characteristic	Performance
Pump flow rate accuracy	±8% or 1 mL/min, whichever is greater
Needle Gauge	17 gauge
Return Filter	Prevent particles (200 µm or larger) from being returned to the donor.
Centrifuge	Maximum Speed: 1640 rpm Accuracy: Within the range 300–1640 rpm, 1% or 5 rpm (whichever is greater)
Leak Detector	Capable of detecting ≥1 mL droplet of fluid in centrifuge.
Donor Pressure Sensors	Operating Range: -250–450 mmHg Alarm point: -250 mm Hg (Draw); 450 mm Hg (Return) Accuracy: ±50 mmHg

Characteristic	Performance
Centrifuge Pressure Sensors	Operating Range: 0–1250 mmHg Alarm point: 750 mmHg Accuracy: ± 100 mmHg (0–500 mmHg); ± 200 mmHg (500–1250 mmHg)
Display Housing	Display Only: 10.4 in (26.4 cm) diagonally Display with Housing: 14.7 in (37.3 cm) diagonally 1024 x 768 pixel resolution color Liquid crystal display (LCD) Touch screen contains the display and provides a tactile interface. Includes external STOP <i>button</i> .

Section 7.3 Audible Tone Specifications

Audible tones range from 60–120 dBA. General instrument operating tones range from 60–70 dBA. Alarm tones range from 70–120 dBA, with the loudest tone as the highest priority alarm.

Section 7.4 Recommended Operating/Storage and Shipping Requirements

The instrument manufacturer recommends the following operating, storage, and shipping conditions:

Operating/Storage

	US Customary	Metric
Temperature	59–86°F	15–30°C
Humidity (non-condensing)	10–90% RH	
Altitude	0–8,000 ft	0–2,438 m
Incline	0 ±2°	

Shipping

	US Customary	Metric
Temperature	0–140°F	-18–60°C
Humidity (non-condensing)	0–90% RH	
Altitude	0–40,000 ft	0–12,192 m



Warning:

Do not use the AmiCORE Apheresis System in the presence of explosive gases (e.g., anaesthetics).



Caution:

The instrument must be used and stored in the proper operating environment (temperature, humidity, altitude, and surface incline requirements).



Note:

The instrument should be plugged in and powered ON at least once every six months for 24 hours to maintain battery charge should the instrument be stored for extended periods of time. Not doing so will cause the battery to become functionally inoperative and no longer able to hold an electrical charge, which will necessitate battery replacement.



Note:

Instrument is not rendered unsafe by spillage over the top of the instrument. Spill test per IEC 60601-1 3rd edition Section 11.6.3.

Section 7.5 Electrical Specifications

Electrical Rating

100–240 VAC, 50/60 Hz

Power Consumption

800 VA

Fuses

T10AL, 250 V



Note:

Mains power quality should be that of a typical commercial or hospital environment.

Power Cord

Three wire (13 A), 10 ft (3.05 m), with IEC receptacle.

Battery Specifications

A 12-volt rechargeable sealed lead acid (SLA) battery backup is used to maintain computer memory and display for approximately 5 minutes after a mains power loss.

A 3-volt non-rechargeable Lithium battery is included in the touch screen SBC module. The battery can be used for backup power for the real time clock, and has a life expectancy of 7–10 years.

Contact your authorized service personnel or local service representative for service and maintenance of these components.

Section 7.6 Standards Compliance

IEC 60601-1:2012: Medical electrical equipment - Part 1: General requirements for basic safety and essential performance.

Classification:

Type of Protection Against Electrical Shock: Class I, Internally Powered

Degree of Protection Against Electrical Shock: Type BF Applied Part

Equipment Not Suitable for Use in the Presence of a Flammable Anaesthetic Mixture With Air or With Oxygen or Nitrous Oxide

Mode of Operation: Continuous

Applied Parts: Needle, fistula tubing, disposable kit, and pressure cuff

IEC-60601-1-2:2007: Medical electrical equipment–Part 1–2: General requirements for basic safety and essential performance–Collateral standard: Electromagnetic compatibility–Requirements and tests.

Product Disposal

WEEE 2012/19/EC



For product disposal, please ensure the following:

- Do not dispose of this product as unsorted municipal waste.
- Collect this product separately.
- Use the collection and return systems available to you.

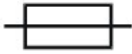
For more information on return, recovery, or recycling of this product, please contact your authorized service personnel or local service representative.

Section 7.7 Universal Markings

The following table provides a list of universal markings that may be seen on the AmiCORE instrument or carton, along with their meanings. This list does not include unique disposable set or instrument icons, prompts, or buttons displayed on the instrument's housing, disposable set, or touch screen. For a complete list of icons, prompts, and buttons used with the AmiCORE Apheresis System, see the "Glossary of Graphics."

Symbol	Meaning
	Fragile
	Caution
	Keep Dry

Symbol	Meaning
	This end up.
	Product Code
	Temperature Limit
	Humidity Limits
	Serial Number
	ON (Power)
	OFF (Power)
	Date of Manufacture
	Manufacturer
	Manufacturing Facility

Symbol	Meaning
	Fuse
	Type BF Equipment
	Underwriters Laboratories (UL) Classification Mark
	Recyclable Carton
	Product Disposal WEEE 2012/19/EC
	USB Port
	Ethernet Port
	WiFi
	This marking reflects compliance with the Medical Devices Directive 93/42/EEC issued by the Council of European Communities. The product was classified as a class IIb product.

Symbol	Meaning
	General Warning Sign
	Refer to instruction manual; follow instructions for use.
	Do not stack.
	RoHS 50 Year Environmentally Friendly Use Period
	RoHS 2011/65/EU Conformance
	RoHS 2011/65/EU Conformance

Section 7.8 Electromagnetic Compatibility

The following information is provided in accordance with IEC 60601-1-2:2007.

The AmiCORE system has been designed to withstand the effects of EMI (electromagnetic interference) and is in compliance with the applicable standards in Electromagnetic Compatibility (IEC 60601-1-2).



Caution:

Equipment connections to the instrument's communication ports must comply with all appropriate UL/CSA/IEC standards for the

equipment type in order to avoid interference issues between the system and other devices.



Caution:

Instrument performance may be affected by external electromagnetic fields generated by non-EMC compliant medical equipment and other electromagnetic sources (such as small hand-held radio transceivers).



Caution:

When maintaining the instrument, use only replacement components, cables, and accessories authorized by the instrument manufacturer and be sure to replace all shields, covers, screws, and gaskets in their exact locations as described in the AmiCORE Service Manual. Failure to do so may result in increased emissions or decreased immunity of the instrument.



Note:

Equipment connected to the ethernet port must be compliant to IEC 60950-1.

To reduce the risk of EMI, follow these recommendations:

- Be aware that modifying the instrument or adding accessories or components, not specifically authorized by the instrument manufacturer, may make the instrument more susceptible to interference from radio waves.
- The following cables, accessories, and transducers have been approved for use with the AmiCORE system and comply with current EMC standards:
 1. Power Cord (approved by the instrument manufacturer)
 2. Ethernet Cable (approved by the instrument manufacturer)
 3. Guidance and Manufacturer's Declaration – Electromagnetic Emissions

The AmiCORE system is intended for use in the electromagnetic environment specified below. The user of the AmiCORE system should assure that it is used in such an environment.

Emissions Test	Compliance	Electromagnetic Environment – Guidance
RF emissions CISPR 11	Group 1	The AmiCORE system uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class B	The AmiCORE system is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Class A	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

Guidance and Manufacturer's Declaration – Electromagnetic Immunity

The AmiCORE system is intended for use in the electromagnetic environment specified below. The user of the AmiCORE system should assure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
Electrostatic Discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	±2 kV for power supply lines ±1 kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV differential mode ±2 kV common mode	±1 kV differential mode ±2 kV common mode	Mains power quality should be that of a typical commercial or hospital environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
Voltage dips, short interruptions, and voltage variations on power supply input lines. IEC 61000-4-11	<5% UT (>95% dip in UT) for 0.5 cycle	<5% UT (>95% dip in UT) for 0.5 cycle	Mains power quality should be that of a typical commercial or hospital environment. If the user of the AmiCORE system requires continued operation during power mains interruptions, it is recommended that the AmiCORE instrument be powered from an uninterrupted power supply.
	40% UT (60% dip in UT) for 5 cycles	40% UT (60% dip in UT) for 5 cycles	
	70% UT (30% dip in UT) for 25 cycles	70% UT (30% dip in UT) for 25 cycles	
	<5% UT (>95% dip in UT) for 5 seconds	<5% UT (>95% dip in UT) for 5 seconds	
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
 Note: UT is the AC mains voltage prior to application of the test level.			

Guidance and Manufacturer's Declaration – Electromagnetic Immunity – For Equipment and Systems that are Not Life-Supporting

The AmiCORE system is intended for use in the electromagnetic environment specified below. The user of AmiCORE system should assure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms	Portable and mobile RF communications equipment should be used no closer to any part of the AmiCORE system, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance: $d = [3.5 / 3] \sqrt{P}$
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2.5 GHz	3 V/m	$d = [3.5 / 3] \sqrt{P}$ 80 MHz to 800 MHz $d = [7 / 3] \sqrt{P}$ 800 MHz to 2.5 GHz where: P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey ^a , should be less than the compliance level in each frequency range ^b . Interference may occur in the vicinity of equipment marked with the following symbol: 
 Note: At 80 MHz and 800 MHz, the higher frequency applies.			
 Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
^a Field strengths from fixed transmitters—such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast—cannot be predicted with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the AmiCORE system is used exceeds the applicable RF compliance level above, the AmiCORE system should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the AmiCORE system.			
^b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.			

Recommended Separation Distance Between Portable and Mobile RF Communications Equipment and AmiCORE – For Equipment and Systems that are Not Life-Supporting

The AmiCORE system is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The user of the AmiCORE system can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the AmiCORE system as recommended below, according to the maximum output power of the communications equipment.

Rated Maximum Output Power of Transmitter	Separation Distance According to Frequency of Transmitter (m)		
	150 kHz to 80 MHz $d = [3.5/3] \sqrt{P}$	80 MHz to 800 MHz $d = [3.5/3] \sqrt{P}$	800 MHz to 2.5 GHz $d = [7/3] \sqrt{P}$
W			
0.01	0.12	0.12	0.24
0.1	0.37	0.37	0.74
1	1.17	1.17	2.4
10	3.69	3.69	7.4
100	11.7	11.7	24
For transmitters rated at a maximum output power not listed above, the recommended separation distance (d) in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
 Note: At 80 MHz and 800 MHz, the higher frequency applies.			
 Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

Section 7.9 US Federal Communications Commission (FCC) Notice

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference; and
2. This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.



Caution:

Exposure to Radio Frequency Radiation. The equipment contains a transmitter with FCC ID: R68MTCHDRCT. The antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm (0 feet 7.87 inches) from all persons and must not be co-located or operated in conjunction with any other antenna or transmitter. End-users must follow the instructions from this document for transmitter operating conditions to satisfy Radio Frequency exposure compliance.

Section 7.10 Protocols and Standards

This wireless functionality references and uses the following industry protocols and standards:

- 802.11b and 802.11g are wireless networking standards developed by the Institute of Electrical and Electronics Engineers (IEEE). 802.11b/g apply to wireless LANs and work in the 2.4 GHz band and operate at a maximum raw data rate of 11Mbps for 802.11b and 54 Mbps for 802.11g. 802.11g devices are backwards compatible with 802.11b devices. Refer to ISO/ IEC 8802-11:2005 standards for local and metropolitan area networks for more information.

- 802.11i (WiFi Protected Access) is a security standard for wireless networks. Refer to IEEE 802.11i-2004 for more information. Your device will be configured to 802.11i-PSK with AES-CCMP encryption or equivalent.
- TCP/IP (Transmission Control Protocol/Internet Protocol) is a standard data transport protocol used for the Internet and other similar networks. Refer to RFC 1122 for more information.

Section 7.11 Electronic Records

The AmiCORE instrument can be configured to export data to be used to create an electronic record. In this context, an “electronic record” is data stored for official quality or regulatory purposes. To create an electronic record, the operator enters all information related to the donor and procedure, as required by the center’s SOPs.



Note: Compliance with USFDA’s 21 CFR Part 11 for Electronic Records is only achievable when integrated at a system-wide level. The data from the AmiCORE instrument on its own is not fully 21 CFR Part 11 compliant.



Note: Consult with your center’s RA/QA Representative for details on your center’s specific adaptation of manufacturing and electronic records.

When a procedure is completed, the data is automatically transmitted to the data management system for reporting.



Note: Required information according to your center’s SOPs must be entered to qualify as an electronic record.

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Estimated Apheresis Kit Plasma Volumes

These estimated plasma volumes include the contents of the separation and collection containers, but not the contents of plasma containers, and whole blood and red cell containers. Whole blood and red cell containers are in single needle kits only.

Single Needle Kit = 205 – RBC volume; includes AC

Estimated Red Blood Cell (RBC) Volumes after Reinfusion

These are the approximate volumes of RBCs, at a 100% hematocrit, remaining in the apheresis kit after the successful completion of reinfusion.

$$\text{RBC Volume} = \frac{[\text{AC Ratio} \times \text{Donor Hct} \div (1 + \text{AC Ratio})] \times 52.0 + \text{Donor Hct} \times 1.8}{1}$$

Estimated Apheresis Kit Plasma Volumes after Reinfusion

The estimated residual plasma volume, excluding AC, remaining in the AmiCORE apheresis kit after a successful reinfusion and product transfer is calculated with the equation below.

The AC Fraction in Plasma may be estimated using the table in Appendix A.4. The estimated absolute volume of plasma remaining in the apheresis kit may be calculated using the following equation:

$\text{Absolute Volume of Residual Plasma (mL)} = 145.6 \times [1 - (\text{AC Ratio} \times \text{Donor Hct}) \div (1 + \text{AC Ratio})] + 1.8 \times (1 - \text{Donor Hct})$
--

Estimated Total Plasma Loss

The estimated total plasma loss after a successful reinfusion and product transfer can be calculated using the following equation:

$\text{Total Plasma Loss (mL)} = \text{Storage Fluid (mL)} + \text{Residual Plasma (mL)}$

If unable to reinfuse during a procedure, the estimated plasma loss can be calculated as follows:

$\text{Total Plasma Loss (mL)} = \text{Storage Fluid (mL)} + \text{Estimated Single Needle Kit Plasma Volume (mL)} + \text{Max Cycle Vol (mL)} \times (1 - \text{Donor Hct}) \times \text{AC Ratio} \div (1 + \text{AC Ratio})$

The AC Fraction in Plasma may be estimated using the table in the Appendix section entitled: “Formula for AC in Plasma.” The absolute total plasma volume loss can be calculated using the following equation:

Absolute Total Plasma Loss (mL) = Total Plasma Loss – (Total Plasma Loss × AC Fraction in Plasma)
--

Section A.2 AmiCORE Equipment and Materials

- AmiCORE Apheresis System
- AmiCORE Apheresis Kit
- AmiCORE Platelet Spool Holder
- AmiCORE Platelet Spool



Caution: The correct spool and spool holder must be used to achieve intended results.

- Anticoagulant Citrate Dextrose Formula A (ACD-A) 500 or 1000 mL
- 0.9% Sodium Chloride Injection (saline) 500 to 1000 mL
- Product Sample Pack
- Platelet Storage Container (if applicable)
- Pressure Cuff
- Bar Code Scanner
- USB Drive



Note: Actual product availability may depend upon the regulatory status in the region.

Contact your local customer service representative for the list of approved apheresis kits and additional materials.

Section A.3 Calculations for Product Yields and Collection Efficiencies

The total number of platelets collected is a function of the product volume and the cell concentration in the product. Calculate the estimated collection efficiency by dividing the total cells collected by the total estimated cells processed. To calculate the total estimated cells processed, multiply the mean cell count of donor pre- and post-counts/microliter (μL) by the total volume of blood processed less AC.

Platelet Products

To determine platelet product yield and collection efficiency, you need the following counts and volumes:

- Donor pre- and post-platelet counts
- Product platelet count
- Volume of the product
- Blood volume processed
- Anticoagulant volume used

Product Yields

Use the following equation to calculate the total number of platelets in the final product:

$\text{Platelet Yield} = \text{Product Volume (mL)} \times \text{Product Count (platelets}/\mu\text{L)} \times \text{Conversion Factor (1000 } \mu\text{L/mL)}$

Example:

Product Volume = 190 mL

Product Count = 2250×10^3 platelets/ μL

Platelet Yield = $190 \text{ mL} \times (2250 \times 10^3 \text{ platelets}/\mu\text{L}) \times 1000 \mu\text{L/mL}$
= 4.3×10^{11} platelets



Note:

If platelet product sample is placed in a pediatric 2 mL EDTA tube, a dilution correction factor of 1.02 for the 0.04 mL EDTA solution would result in a more accurate count and is independent of other dilution corrections.



Note:

The weight of a 450 mL pack container is approximately 28 grams. The weight of a 600 mL

pack container is approximately 32 grams. The weight of an 800 mL pack container is approximately 39 grams. One platelet storage container with sample pack, associated tubing, and clamp weighs approximately 37 grams. See apheresis kit product insert for details.



Note: The density of plasma is approximately 1.027 g/mL.

Collection Efficiency

Take donor pre- and post-counts/ μL . Then use the following equations to calculate collection efficiency:

$\text{Total Platelets} = \left[\left(\frac{\text{Pre-count}/\mu\text{L} + \text{Post-count}/\mu\text{L}}{2} \right) \times \text{Total Blood Volume Processed}^* (\text{mL}) \times \text{Conversion Factor} (1000 \mu\text{L}/\text{mL}) \right]$
$^*\text{Total Blood Volume Processed} = \text{Blood Volume Processed (mL)} - \text{Anticoagulant (mL)}$

Example:

Pre-count	= 250,000 platelets/ μL
Post-count	= 149,000 platelets/ μL
Total Blood Volume Processed	= 5100 mL – 450 mL = 4650 mL
Total Platelets Processed	= $\left[\left(\frac{250,000 \text{ plts}/\mu\text{L} + 149,000 \text{ plts}/\mu\text{L}}{2} \right) \times 4650 \text{ mL} \times 1000 \mu\text{L}/\text{mL} \right]$
	= 9.3×10^{11} platelets

To calculate the collection efficiency:

$\text{Collection Efficiency} = (\text{Platelet Yield} \div \text{Total Platelets Processed}) \times 100$

Example:

Platelet Yield	= 6.6×10^{11} platelets
Total Platelets Processed	= 9.3×10^{11} platelets
Collection Efficiency	= $\left[\frac{6.6 \times 10^{11}}{9.3 \times 10^{11}} \right] \times 100$
	= 71%

Note that 0.55×10^{11} is the minimum standard for platelet units made from whole blood. To determine the number of equivalent standard units collected, use the following equation:

$\text{Number of Standard Units} = \text{Platelet Yield} \div (0.55 \times 10^{11})$
--

Section A.4 Formula for AC in Plasma

The volume of AC in plasma product and storage plasma can be estimated using the following equation:

Multiply the volume of plasma by the AC fraction that corresponds to the AC Ratio and donor hematocrit to calculate the AC in that volume of plasma.

$$\text{Volume of AC in Plasma} = [\text{AC Fraction}] \times \text{Volume of Plasma}$$

The AC Fraction may be found on the AC Fraction in Plasma table found on the following page. It was calculated using the following equation:

$$\text{AC Fraction} = (1 \div \text{AC Ratio}) \div [(1 - \text{Donor Hct}) + 1 \div \text{AC Ratio}]$$

To calculate the volume of plasma in the plasma container use the following equation:

$$\text{Volume of Plasma} = [\text{Plasma Product Weight (g)} - \text{Tare Weight of Plasma Container (g)}] \div \text{Density Factor (g/mL)}$$



Note: The approximate density of plasma is 1.027 grams/mL.



Note: The tare weight of a 450 mL pack container is approximately 28 grams. The tare weight of the plasma container is approximately 39 grams.

Example:

Donor Hematocrit	=	48% (or .48)
AC Ratio	=	10:1
Plasma Product Weight	=	250 g
AC Fraction (from table)	=	0.161
Volume of Plasma	=	$[250 - 39] \div 1.027 = 205 \text{ mL}$
Volume of AC in Plasma	=	$0.161 \times 205 = 33.0 \text{ mL}$

		AC Ratio							
Donor Hematocrit		8	9	10	11	12	13	14	15
	0.20	0.135	0.122	0.111	0.102	0.094	0.088	0.082	0.077
	0.21	0.137	0.123	0.112	0.103	0.095	0.089	0.083	0.078
	0.22	0.138	0.125	0.114	0.104	0.097	0.090	0.084	0.079
	0.23	0.140	0.126	0.115	0.106	0.098	0.091	0.085	0.080
	0.24	0.141	0.128	0.116	0.107	0.099	0.092	0.086	0.081
	0.25	0.143	0.129	0.118	0.108	0.100	0.093	0.087	0.082
	0.26	0.145	0.131	0.119	0.109	0.101	0.094	0.088	0.083
	0.27	0.146	0.132	0.120	0.111	0.102	0.095	0.089	0.084
	0.28	0.148	0.134	0.122	0.112	0.104	0.097	0.090	0.085
	0.29	0.150	0.135	0.123	0.114	0.105	0.098	0.091	0.086
	0.30	0.152	0.137	0.125	0.115	0.106	0.099	0.093	0.087
	0.31	0.153	0.139	0.127	0.116	0.108	0.100	0.094	0.088
	0.32	0.155	0.140	0.128	0.118	0.109	0.102	0.095	0.089
	0.33	0.157	0.142	0.130	0.119	0.111	0.103	0.096	0.090
	0.34	0.159	0.144	0.132	0.121	0.112	0.104	0.098	0.092
	0.35	0.161	0.146	0.133	0.123	0.114	0.106	0.099	0.093
	0.36	0.163	0.148	0.135	0.124	0.115	0.107	0.100	0.094
	0.37	0.166	0.150	0.137	0.126	0.117	0.109	0.102	0.096
	0.38	0.168	0.152	0.139	0.128	0.118	0.110	0.103	0.097
	0.39	0.170	0.154	0.141	0.130	0.120	0.112	0.105	0.099
	0.40	0.172	0.156	0.143	0.132	0.122	0.114	0.106	0.100
	0.41	0.175	0.158	0.145	0.134	0.124	0.115	0.108	0.102
	0.42	0.177	0.161	0.147	0.136	0.126	0.117	0.110	0.103
	0.43	0.180	0.163	0.149	0.138	0.128	0.119	0.111	0.105
	0.44	0.182	0.166	0.152	0.140	0.130	0.121	0.113	0.106
	0.45	0.185	0.168	0.154	0.142	0.132	0.123	0.115	0.108
	0.46	0.188	0.171	0.156	0.144	0.134	0.125	0.117	0.110
	0.47	0.191	0.173	0.159	0.146	0.136	0.127	0.119	0.112
	0.48	0.194	0.176	0.161	0.149	0.138	0.129	0.121	0.114
	0.49	0.197	0.179	0.164	0.151	0.140	0.131	0.123	0.116
	0.50	0.200	0.182	0.167	0.154	0.143	0.133	0.125	0.118
	0.51	0.203	0.185	0.169	0.156	0.145	0.136	0.127	0.120
	0.52	0.207	0.188	0.172	0.159	0.148	0.138	0.130	0.122
	0.53	0.210	0.191	0.175	0.162	0.151	0.141	0.132	0.124
	0.54	0.214	0.195	0.179	0.165	0.153	0.143	0.134	0.127
	0.55	0.217	0.198	0.182	0.168	0.156	0.146	0.137	0.129



Note:

To convert hemoglobin to hematocrit for this table, multiply the hemoglobin value by 0.03.

Section A.5 Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC

A maximum of 4.7×10^{11} platelets in the appropriate storage fluid volume can be stored in a single platelet storage container (or 9.4×10^{11} platelets in 2 containers). The Estimator recommends storage in two containers when the expected collection is more than 4.7×10^{11} platelets. If the platelet yield exceeds 9.4×10^{11} , a third storage container is required.

**Note:**

The administrator should only choose a storage duration of 7 Days if approved by the appropriate regulatory authorities for the region.

Number of Platelets ($\times 10^{11}$)	5 Day Storage*		7 Day Storage*	
	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.2	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.5	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.2	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.5
1.5	96	102	128	137
2.0	121	132	162	176
2.1	126	138	168	184
2.2	130	143	174	191
2.3	135	148	180	198
2.4	139	153	185	204
2.5	143	158	191	211
2.6	147	163	196	217
2.7	151	168	201	224
2.8	154	173	205	230
2.9	158	177	210	236
3.0	161	181	214	241
3.1	164	186	219	247
3.2	167	190	222	252
3.3	170	194	226	257
3.4	173	197	230	262
3.5	175	201	233	267
3.6	178	204	236	271
3.7	180	208	239	276
3.8	182	211	241	280
3.9	184	214	244	284
4.0	186	217	246	287
4.1	187	219	248	290
4.2	189	222	250	294
4.3	190	224	251	296
4.4	191	226	252	299
4.5	192	228	253	301
4.6	192	230	254	303
4.7	193	231	255	305
4.8	278	306	370	408
4.9	286	316	382	422
5.0	286	316	382	422
5.1	294	326	392	434
5.2	294	326	392	434

Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC

Number of Platelets ($\times 10^{11}$)	5 Day Storage*		7 Day Storage*	
	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.2	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.5	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.2	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.5
5.3	302	336	402	448
5.4	302	336	402	448
5.5	308	346	410	460
5.6	308	346	410	460
5.7	316	354	420	472
5.8	316	354	420	472
5.9	322	362	428	482
6.0	322	362	428	482
6.1	328	372	438	494
6.2	328	372	438	494
6.3	334	380	444	504
6.4	334	380	444	504
6.5	340	388	452	514
6.6	340	388	452	514
6.7	346	394	460	524
6.8	346	394	460	524
6.9	350	402	466	534
7.0	350	402	466	534
7.1	356	408	472	542
7.2	356	408	472	542
7.3	360	416	478	552
7.4	360	416	478	552
7.5	364	422	482	560
7.6	364	422	482	560
7.7	368	428	488	568
7.8	368	428	488	568
7.9	372	434	492	574
8.0	372	434	492	574
8.1	374	438	496	580
8.2	374	438	496	580
8.3	378	444	500	588
8.4	378	444	500	588
8.5	380	448	502	592
8.6	380	448	502	592
8.7	382	452	504	598
8.8	382	452	504	598
8.9	384	456	506	602
9.0	384	456	506	602
9.1	384	460	508	606
9.2	384	460	508	606
9.3	386	462	510	610
9.4	386	462	510	610
9.5	501	570	666	756
9.6	501	570	666	756
9.7	510	582	678	771
9.8	510	582	678	771
9.9	510	582	678	771
10.0	519	591	690	786
10.1	519	591	690	786
10.2	519	591	690	786

Appendix A - Platelets

Number of Leukocytes Reduced Platelets and Minimum Storage Fluid Volumes, Including AC

Number of Platelets ($\times 10^{11}$)	5 Day Storage*		7 Day Storage*	
	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.2	Minimum Storage Fluid Volume (mL), Including AC, pH ≥ 6.5	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.2	Minimum Storage Fluid Volume (mL) Including ACD pH ≥ 6.5
10.3	525	603	699	801
10.4	525	603	699	801
10.5	525	603	699	801
10.6	534	612	708	813
10.7	534	612	708	813
10.8	534	612	708	813
10.9	540	624	717	828
11.0	540	624	717	828
11.1	540	624	717	828
11.2	546	633	723	840
11.3	546	633	723	840
11.4	546	633	723	840
11.5	552	642	732	852
11.6	552	642	732	852
11.7	552	642	732	852
11.8	558	651	738	861
11.9	558	651	738	861
12.0	558	651	738	861
12.1	561	657	744	870
12.2	561	657	744	870
12.3	561	657	744	870
12.4	567	666	750	882
12.5	567	666	750	882
12.6	567	666	750	882
12.7	570	672	753	888
12.8	570	672	753	888
12.9	570	672	753	888
13.0	573	678	756	897
13.1	573	678	756	897
13.2	573	678	756	897
13.3	576	684	759	903
13.4	576	684	759	903
13.5	576	684	759	903
13.6	576	690	762	909
13.7	576	690	762	909
13.8	576	690	762	909
13.9	579	693	765	915
14.0	579	693	765	915
14.1	579	693	765	915

*Day of collection is counted as Day 0.



Note:

Per the Code of Federal Regulations, platelets shall be suspended in sufficient storage fluid so that the pH determined at the temperature of storage shall be 6.2 or greater in the units tested at the end of the allowable storage interval.

Section A.6 Normal Blood Volume Tables

Predicted Normal Blood Volumes in Men*

Weight kg	lb	Height (m)	1.52	1.58	1.63	1.68	1.73	1.78	1.83	1.88
		(in)	60	62	64	66	68	70	72	74
45.4	100		3365	3500	3643	3795	3957	4129	4311	4503
49.9	110		3512	3646	3789	3941	4103	4275	4457	4649
54.5	120		3658	3792	3935	4088	4250	4422	4603	4796
59.0	130		3804	3938	4082	4234	4396	4658	4750	4942
63.5	140		3951	4085	4228	4380	4542	4714	4896	5088
68.0	150		4097	4231	4374	4527	4689	4860	5042	5235
72.5	160		4243	4377	4521	4673	4835	5007	5189	5381
77.0	170		4389	4524	4667	4819	4981	5153	5335	5527
81.6	180		4536	4670	4813	4971	5128	5299	5481	5673
86.2	190		4682	4816	4959	5112	5274	5446	5627	5820
90.7	200		4828	4963	5106	5258	5420	5592	5774	5966
95.3	210		4975	5109	5252	5405	5566	5738	5920	6112
99.8	220		5121	5255	5398	5551	5713	5885	6066	6295
103.4	230		5267	5402	5545	5697	5859	6031	6213	6405
108.9	240		5414	5548	5692	5843	6005	6177	6359	6551
113.4	250		5560	5694	5837	5990	6152	6323	6505	6698
118.0	260		5706	5840	5984	6136	6298	6470	6652	6844
122.5	270		5852	5987	6130	6282	6444	6616	6798	6990
127.0	280		5999	6133	6276	6429	6591	6762	6944	7136
131.6	290		6145	6279	6423	6575	6737	6909	7091	7283
136.1	300		6291	6426	6569	6721	6883	7055	7237	7429
140.6	310		6438	6572	6715	6868	7030	7201	7383	7575

$$PBV = (0.3669 \times H^3 + 0.03219 \times W + 0.6041) \times 1,000$$

H = Height in meters

W = Weight in kilograms

*Nadler, S.B., et.al.: Prediction of Blood Volume in Normal Human Adults.
Surgery Vol 51 (2): pp 224-232, 1962.

Appendix A - Platelets
Normal Blood Volume Tables

Predicted Normal Blood Volumes in Women*

Weight kg	lb	Height (m) (in)	1.52 60	1.58 62	1.63 64	1.68 66	1.73 68	1.78 70	1.83 72	1.88 74
36.2	80		2646	2776	2915	3036	3220	3387	3564	3750
40.8	90		2796	2927	3066	3214	3371	3537	3714	3901
45.4	100		2947	3077	3216	3364	3521	3688	3864	4052
49.9	110		3097	3227	3366	3514	3671	3838	4015	4201
54.5	120		3247	3378	3517	3665	3822	3989	4165	4352
59.0	130		3398	3528	3667	3815	3972	4139	4315	4502
63.5	140		3548	3678	3817	3965	4123	4289	4466	4652
68.0	150		3698	3829	3968	4116	4273	4440	4616	4803
72.5	160		3849	3979	4118	4266	4423	4590	4766	4953
77.0	170		3999	4129	4268	4416	4574	4740	4917	5103
81.6	180		4150	4280	4419	4567	4724	4891	5067	5254
86.2	190		4300	4430	4569	4717	4874	5041	5217	5404
90.7	200		4450	4581	4719	4867	5025	5191	5368	5554
95.3	210		4601	4731	4870	5018	5175	5342	5518	5705
99.8	220		4751	4881	5020	5168	5325	5492	5669	5855
103.4	230		4901	5032	5171	5318	5476	5642	5819	6005
108.9	240		5052	5182	5321	5469	5626	5793	5969	6156
113.4	250		5202	5332	5471	5619	5776	5943	6120	6306
118.0	260		5352	5483	5622	5770	5927	6093	6270	6457
122.5	270		5503	5633	5772	5920	6077	6244	6420	6607
127.0	280		5653	5783	5922	6070	6227	6394	6571	6757
131.6	290		5803	5934	6073	6221	6378	6544	6721	6907

$$PBV = (0.3561 \times H^3 + 0.03308 \times W + 0.1833) \times 1,000$$

H = Height in meters

W = Weight in kilograms

*Nadler, S.B., et.al.: Prediction of Blood Volume in Normal Human Adults.
Surgery Vol 51 (2): pp 224-232, 1962.

Glossary of Graphics

This glossary provides the names and meanings of the graphics for tabs, buttons, icons, and prompts that are displayed on the touch screen. Administrative settings control which graphics appear for your institution. The tables below describe each type of system graphics.

Tabs

Tabs enable the operator to view or enter data.

Main Data Entry Tabs

Tab	Name
	Donor ID
	Gender
	Height
	Hematocrit/Hemoglobin
	Pre-count
	Procedure ID
	Product Yield
	Weight

Information Tabs (Access with the Information Button)

Tab	Name
	Instrument Settings
	Procedure Information
	Procedure Settings

Fields

Field	Name
	Code Number
	Expiration Date
	Lot Number

Buttons

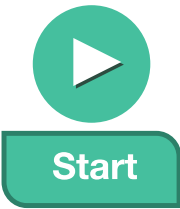
Buttons enable the operator to indicate that a task has been completed or command the instrument to proceed to another task or step.

Button	Name
	Add Container
	Admin Mode
	Audio Pause
	Calibrate Scale
	Cancel
	Cassette Unload
	Check
	Citrate Infusion Rate

Button	Name
 	Clear
	Cuff Pressure
	Data Entry
	Decrease Value
	Delete
	Demo Mode
	Draw Rate
	Events
	Expel Air
	Female

Button	Name
	Help
	Increase Value
	Information
	Keypad View
	Kit Information
	Male
	New Procedure
	No
	Numeric
	Open Clamp
	Operator ID

Button	Name
	Page Down
	Page Up
	Pause
	Preset View
	Resume from Stop
	Return Rate
	Return Saline
	Save
	Scale Retry
	Service View
	Shift (Lowercase)

Button	Name
	Shift (Uppercase)
	Shut Down
	Software Update
	Sound Level
	Special Characters
	Start
	System Info
	Tare Scale
	Terminate
	Transfer

Button	Name
	View Last Donation Procedure Summary
	Yes

Icons

Icons are symbols displayed on the touch screen, instrument, or disposable set that help the operator in understanding various functions or procedural events. Touching an icon will not result in any action being taken by the instrument. Icons are for informational use only.

Types of icons are:

- General Operating
- Alert

General Operating

These icons appear on the touch screen or Donor Display to provide additional information to the operator/donor regarding procedural events.

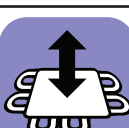
Icon	Name
	AC in Product
	AC Ratio
	AC Used
	Citrate Infusion Rate

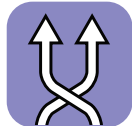
Icon	Name
	Collection Time
	Cycle Volume
	Do Not Perform Venipuncture
	Draw
	Plasma Loss
	Platelet Yield
	Post Count
	Processing
	RBC Loss
	Return

Icon	Name
	Saline Used
	Scale Fail
	Scale Pass
	Storage Volume
	Whole Blood Processed

Alert

Alert icons appear on the touch screen during alert situations. They provide the operator with information regarding the alert, possible causes, and possible resolutions. More information regarding alerts and troubleshooting is provided in Chapter 5.

Icon	Name
	AC Low
	Air Leak
	Call Service
	Cassette Load/Unload
	Centrifuge Occlusion
	Centrifuge Speed Error
	Clamp is Closed
	Close Door
	Disconnect Donor

Icon	Name
	Door Unlocked
	Draw and Return Lines are Interchanged
	Hardware Error
	Incorrect Container Placement
	Incorrect Weight
	Interface Detector
	Leaks
	Lipemia
	Network Error
	Open Door

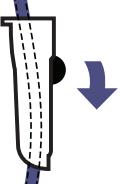
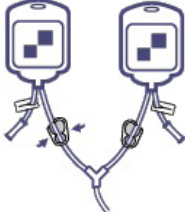
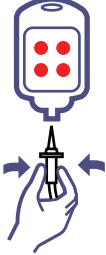
Icon	Name
	Power Loss
	Press STOP Button
	Pressure Cuff
	Saline Low
	Scale Disturbed
	Solution Connection
	Temperature High/Low
	Tubing (Air in Line)
	Tubing Damaged
	Tubing Kinked
	Tubing Misaligned (Air Detector)

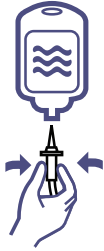
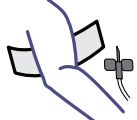
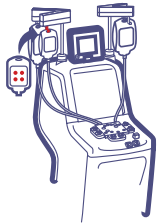
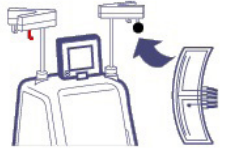
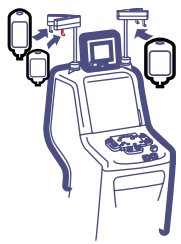
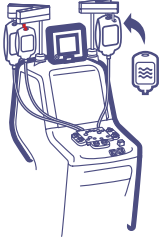
Glossary of Graphics

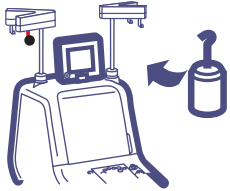
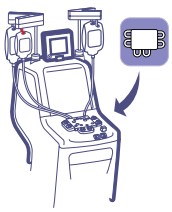
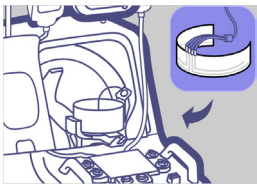
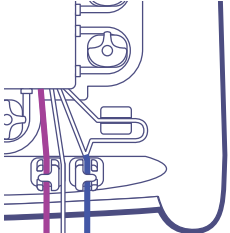
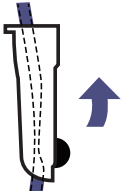
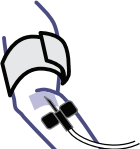
Icon	Name
 	Tubing Misaligned (Clamp)
	Tubing Misaligned (Pump)
	Turn Power OFF
	Unload Centrifuge Pack
	Venipuncture Site

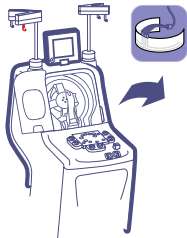
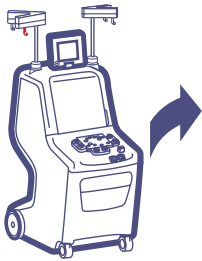
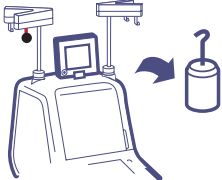
Prompts

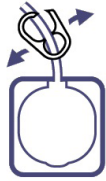
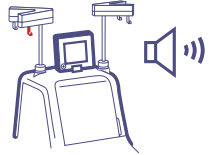
Prompts are touch screen graphics that indicate tasks for the operator to perform. Some screens contain multiple prompts that must all be completed before touching the *Check button*.

Prompt	Name
	Close Clamp
	Close Donor Line Roller Clamp
	Close One Platelet Clamp
	Close Slide Clamp
	Connect AC

Prompt	Name
	Connect Saline
	Disconnect Donor
	Expel Air
	Hang AC Container
	Hang Centrifuge Pack
	Hang Containers (Platelet and Plasma)
	Hang Saline Container

Prompt	Name
	Hang Weights
	Install Cassette
	Install Centrifuge Pack
	Install Lines
	Open Donor Line Roller Clamp
	Open Platelet Clamp
	Perform Venipuncture

Prompt	Name
	Power OFF
	Press STOP Button (Test)
	Remove Centrifuge Pack
	Remove Kit
	Remove Weights
	Resuspend Platelets
	Seal Tubing

Prompt	Name
	Take Samples
	Verify Audio Signal

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Fresenius Kabi AG
61346 Bad Homburg / Germany
Tel.: +49 (0) 61 72 / 686-0
www.fresenius-kabi.com



Plexus Manufacturing Sdn. Bhd.
Plot 87, Lebuhraya Kampung Jawa
11900 Bayan Lepas
Penang, Malaysia

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